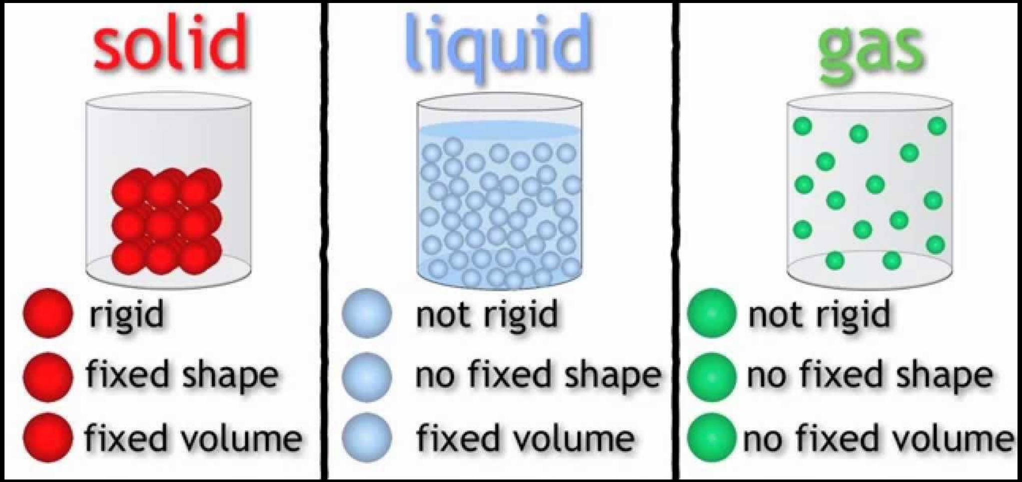
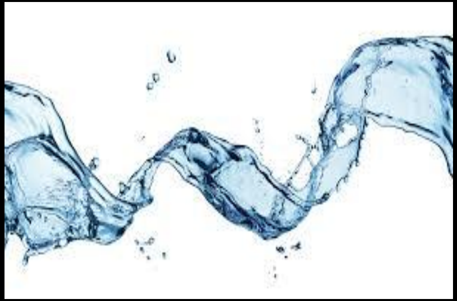
Solid State

States of Matter

Matter → 3 States



- Liquids & Gases → Fluids → Ability to flow

- Solids → Rigids →

- At given set of T & P, Most stable state of substance depends on -

1. Intermolecular forces

   Tend to keep the molecules (atoms and ions) closer

2. Thermal energy

   Tends to keep molecules (atoms and ions) apart

- At low T, thermal energy is low and intermolecular forces bring molecules(atoms and ions) very close. Hence, substance exists in solid state.

General Characteristics of Solids

- Structures of solids : Due to different arrangement of solid particles.
- Properties of solids : Due to different structures of solids

Get Modified → Structural Imperfection

Get Modified → Presence of impurities

The following are the characteristic properties of the solid state:

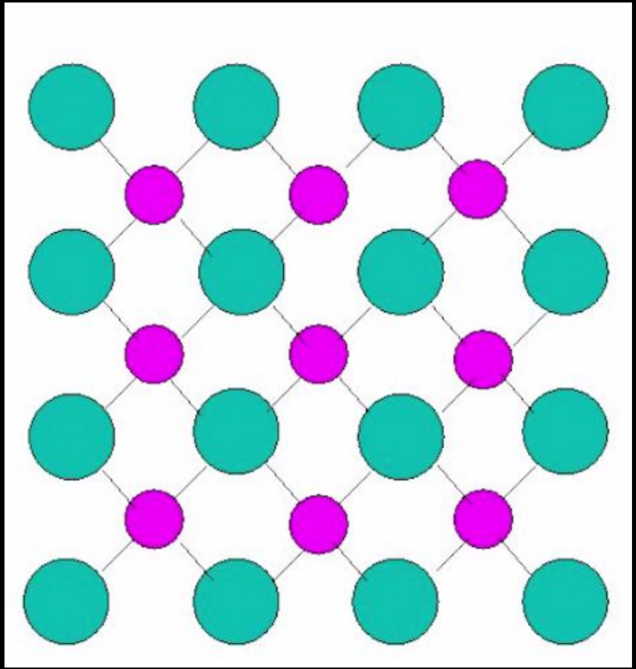
- They have definite mass, volume and shape.
- Intermolecular distances are short.
- Intermolecular forces are strong.
- They are incompressible and rigid.

Types of Solids :

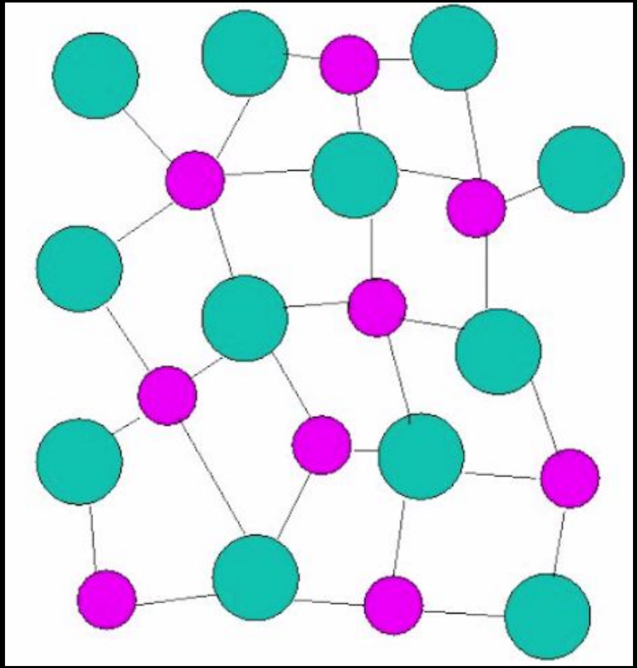
On the Basis of nature of order present in the arrangement of particles :

- Solids
  - Crystalline Solids
  - Amorphous Solids

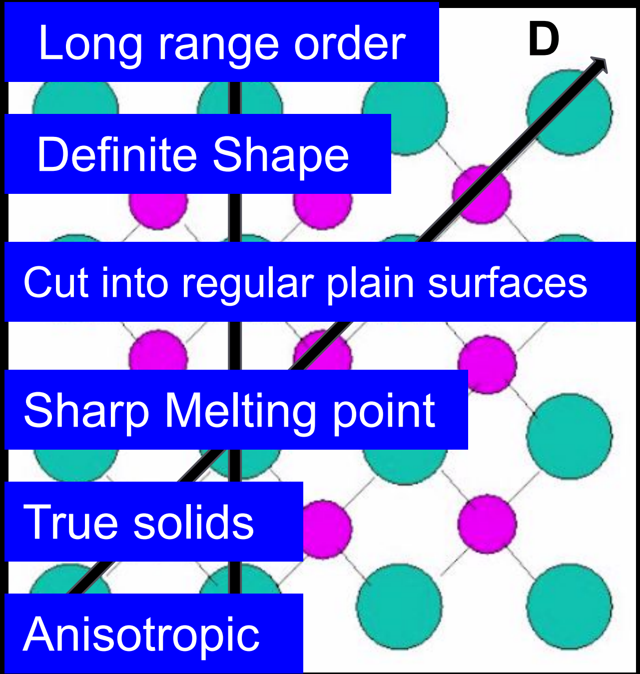
Crystalline Solids



Amorphous Solids

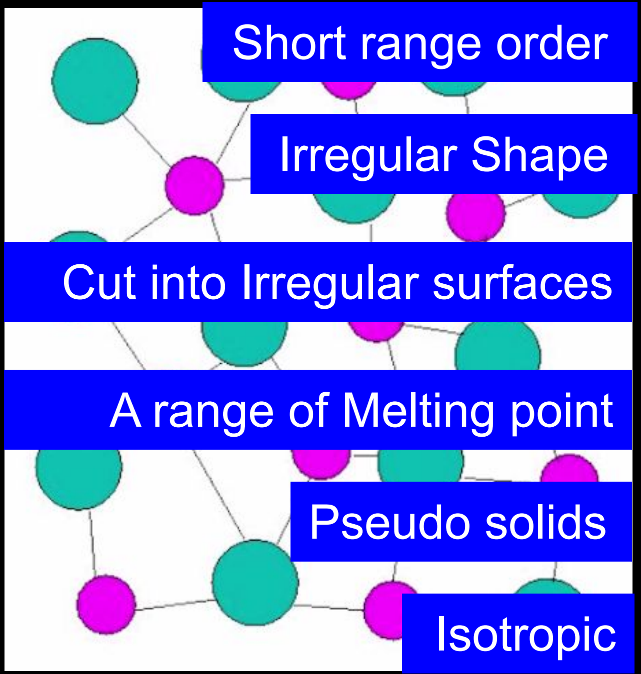


Crystalline Solids



- Long range order
- Definite Shape
- Cut into regular plain surfaces
- Sharp Melting point
- True solids
- Anisotropic

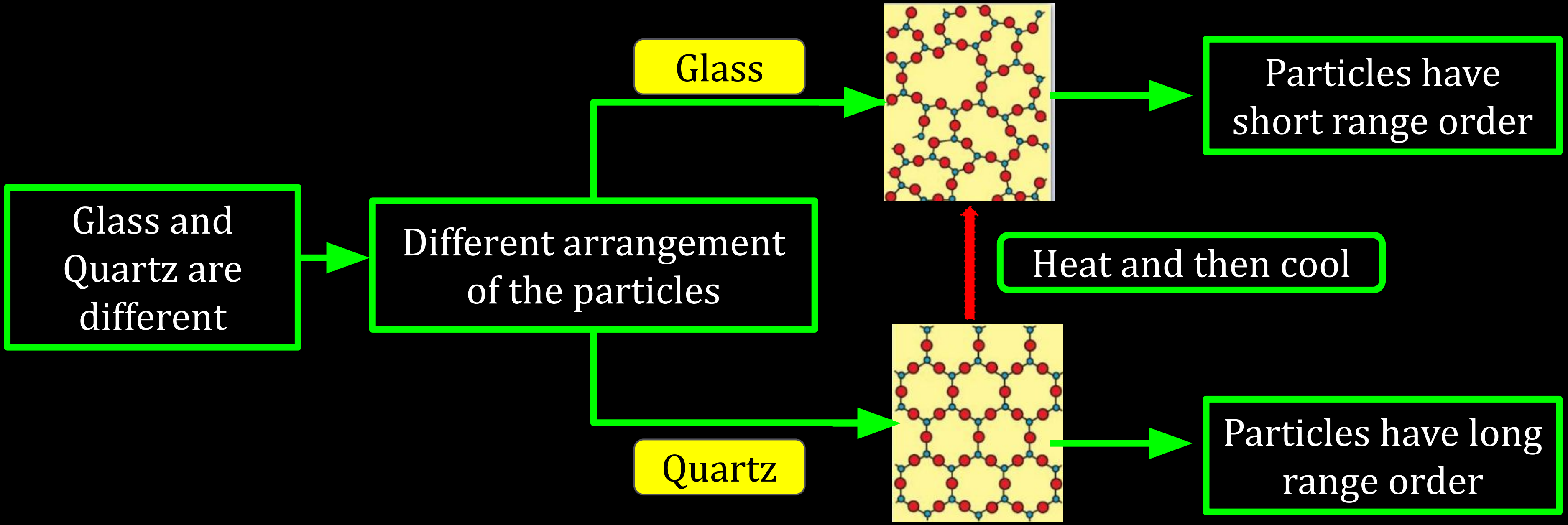
Amorphous Solids



- Short range order
- Irregular Shape
- Cut into Irregular surfaces
- A range of Melting point
- Pseudo solids
- Isotropic

Question : What makes a glass different from a solid such as quartz? Under what conditions could quartz be converted into glass?

Solution :



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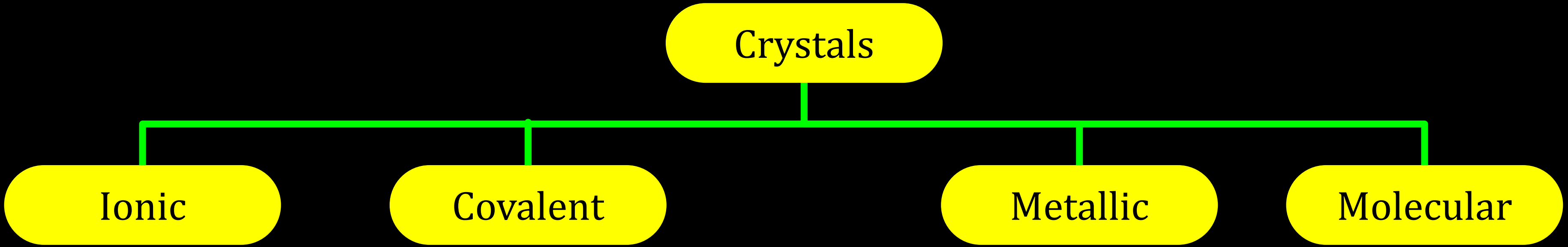
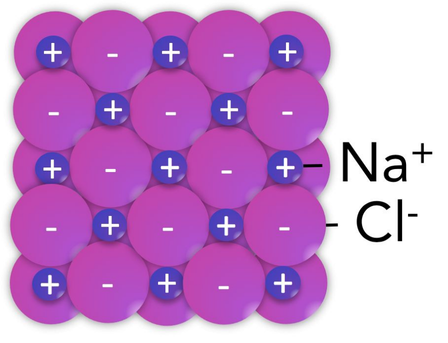

Crystals
Ionic
Covalent
Metallic
Molecular

Crystals

- Ionic
- Covalent
- Metallic
- Molecular
  - Non-polar
  - Polar
  - H - bonded

Ionic

- Made of Ions
- Electrostatic forces
- Hard and Brittle
- High melting and boiling points
- Electrical insulators in solid state
- Electrical conductors in molten and aqueous state.
- Examples are NaCl, CaF_2.

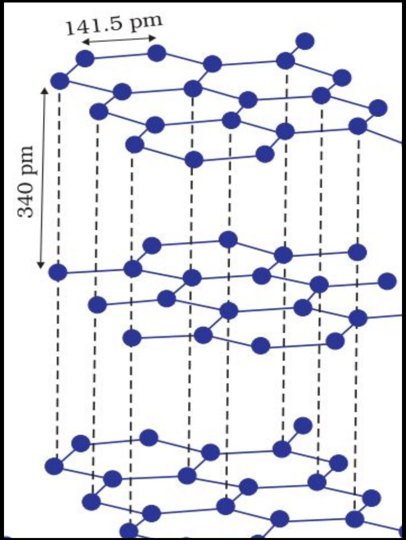


Crystals

- Ionic
- Covalent
- Metallic
- Molecular
  - Non-polar
  - Polar
  - H - bonded

Covalent

- Covalent bonds are present.
- Such solids are very hard and brittle.
- Extremely high melting points.
- They are insulators and do not conduct electricity.
- Examples are, diamond, silicon carbide and graphite.
- Exceptionally, Graphite is soft and is a conductor of electricity.



Crystals

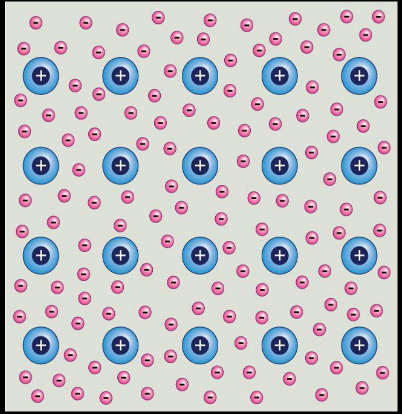
Ionic

Covalent

Metallic

Molecular

- Metallic bonds are present.
- Collection of positive ions surrounded by and held together by a sea of free electrons.
- High electrical and thermal conductivity due to free electrons.
- Lustrous, malleable and ductile in nature.
- Example are Fe, Cu, Ag etc.



Non-polar

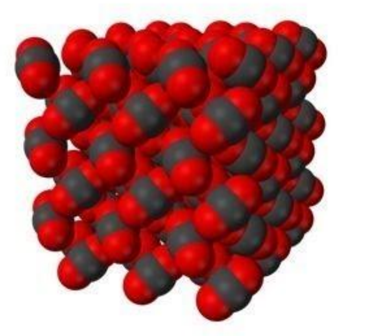
Polar

H - bonded

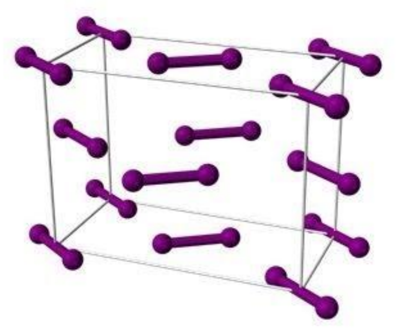
Crystals

- Ionic
- Covalent
- Metallic
- Molecular
  - Non-polar
  - Polar
  - H - bonded

- Molecules are the constituent particles of molecular solids.



Carbon di oxide
(CO_2)



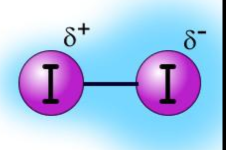
Iodine
(I_2)

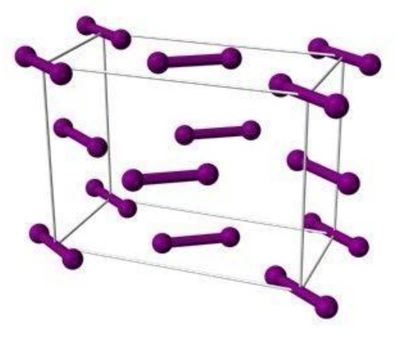
12C01.1 Solid State & its Classification

Crystals

- Ionic
- Covalent
- Metallic
- Molecular
  - Non-polar
  - Polar
  - H - bonded

- Contain either atoms like He and Ar or non polar molecules like H_2, Cl_2 and I_2.
- Weak dispersion or London forces are present.
- Soft and non-conductors of electricity.
- Low melting points
- liquid or gaseous state at room temp .





Iodine (I_2)

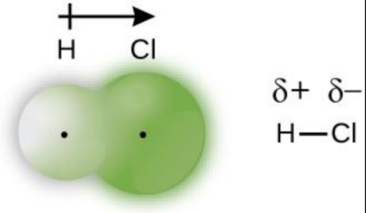
12C01.1 Solid State & its Classification

Crystals

Ionic | Covalent | Metallic | Molecular

Molecular: Non-polar | Polar | H - bonded

- Formed by molecules like HCl, SO_2 with Polar covalent bonds.
- Strong dipole-dipole interactions are present.
- Soft and non-conductors of electricity.
- M.P are higher than non polar molecular solids.
- Generally, gases or liquids at room temperature.



12C01.1 Solid State & its Classification

Crystals

Ionic

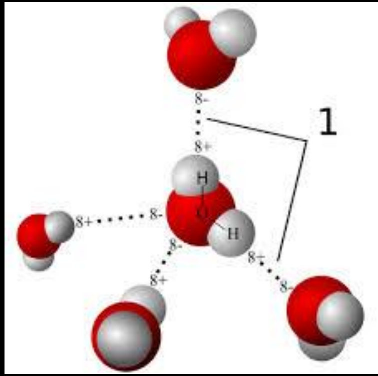
Covalent

Metallic

Molecular

- Non-polar
- Polar
- H - bonded

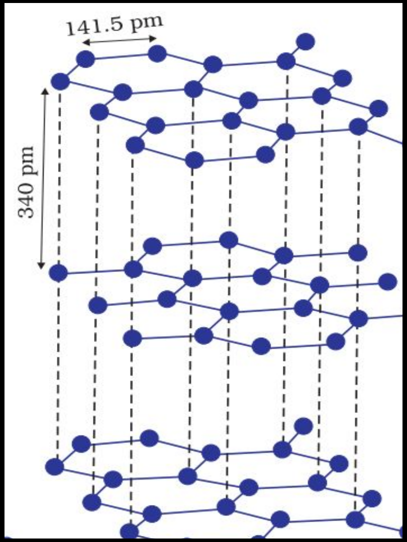
- Molecules of these solids contain polar covalent bonds between H and F, O or N atoms.
- Hydrogen bonds are present.
- Non-conductors of electricity.
- Volatile liquids or soft solids under room temp.
- Example is H_2O (ice).



Graphite - Exception of Covalent Or Network Solid

Exceptional properties of graphite are due to its typical structure.

- C- atoms are arranged in different layers.
- Each atom is covalently bonded to three of its neighbouring atoms in same layer.
- Fourth valence electron of each atom is present between different layers and is free to move about.
- Free electrons make it a good conductor.



Conceptest

Ready for Challenge

12C01.1 Solid State & its Classification

Question : Classify the following as ionic, metallic, molecular, covalent or amorphous.

(i) Tetra phosphorus decoxide (P_4O_{10}) (ii) Ammonium phosphate $(NH_4)_3PO_4$

(iii) SiC (iv) I_2 (v) P_4 (vi) Plastic (vii) Graphite (viii) Brass (ix) Rb (x) LiBr (xi) Si

Solution : ?

(NCERT Exercise Q. No. 1.3 Page no. 32)

Question : Classify the following as ionic, metallic, molecular, covalent or amorphous.

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Solution : ? (NCERT Exercise Q. No. 1.3 Page no. 32)

Pause the Video Time duration 2 minute

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Solution :

Ionic →

12C01.1 Solid State & its Classification

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Solution :

Ionic → (ii) Ammonium phosphate $(NH_4)_3PO_4$, (x) LiBr

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Solution :

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Metallic →

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Solution :

Ionic → (ii) Ammonium phosphate $(NH_4)_3PO_4$, (x) LiBr

Metallic → (viii) Brass, (ix) Rb

12C01.1 Solid State & its Classification

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Solution :

Ionic →	(ii) Ammonium phosphate $(NH_4)_3PO_4$, (x) LiBr
Metallic →	(viii) Brass, (ix) Rb
Molecular →	

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Solution :

Ionic →	(ii) Ammonium phosphate $(NH_4)_3PO_4$, (x) LiBr
Metallic →	(viii) Brass, (ix) Rb
Molecular →	(i) Tetra phosphorus decoxide (P_4O_{10}), (iv) I_2, (v) P_4

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Covalent →	

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Covalent →	(iii) SiC, (vii) Graphite, (xi) Si
Amorphous →	

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Covalent →	(iii) SiC, (vii) Graphite, (xi) Si
Amorphous →	(vi) Plastic

12C01.1 Solid State & its Classification

: Reference Questions :

Intext Questions : 1.1, 1.2,1.3 ,1.6, 1.7, 1.9 (NCERT)

Exercise Question : 1.1 ,1.9 (NCERT)

Work Book : Question 8

12C01.2

Crystal Structure & Unit Cell

12C01.2 Crystal Lattices and Unit Cells

- Crystal Lattice, Unit Cell and its Parameters

12C01.2 Crystal Lattices and Unit Cells

- Crystal Lattice, Unit Cell and its Parameters

- Types of Unit Cells

12C01.2 Crystal Lattices and Unit Cells

- Crystal Lattice, Unit Cell and its Parameters

- Types of Unit Cells

- Bravais Lattices

12C01.2 Crystal Lattices and Unit Cells

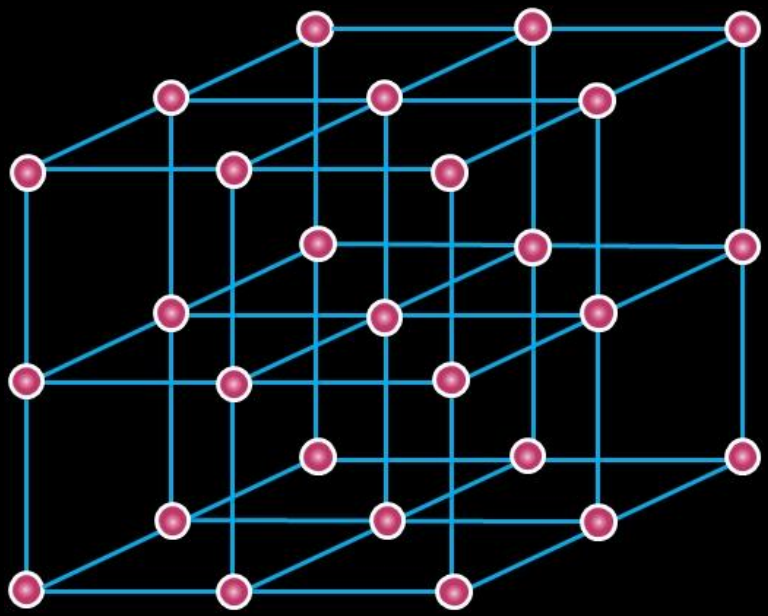
- Crystal Lattice, Unit Cell and its Parameters
- Types of Unit Cells
- Bravais Lattices
- Number of Atoms in a Cubic Unit Cell

CV 1

Crystal Lattice, Unit Cell and its Parameters

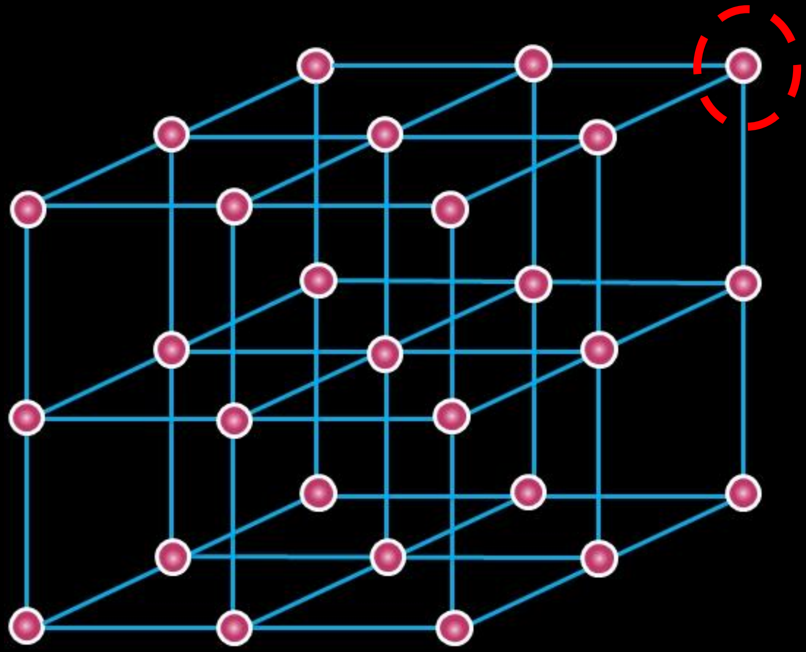
12C01.2 Crystal Lattices and Unit Cells

Crystal lattice or space lattice



12C01.2 Crystal Lattices and Unit Cells

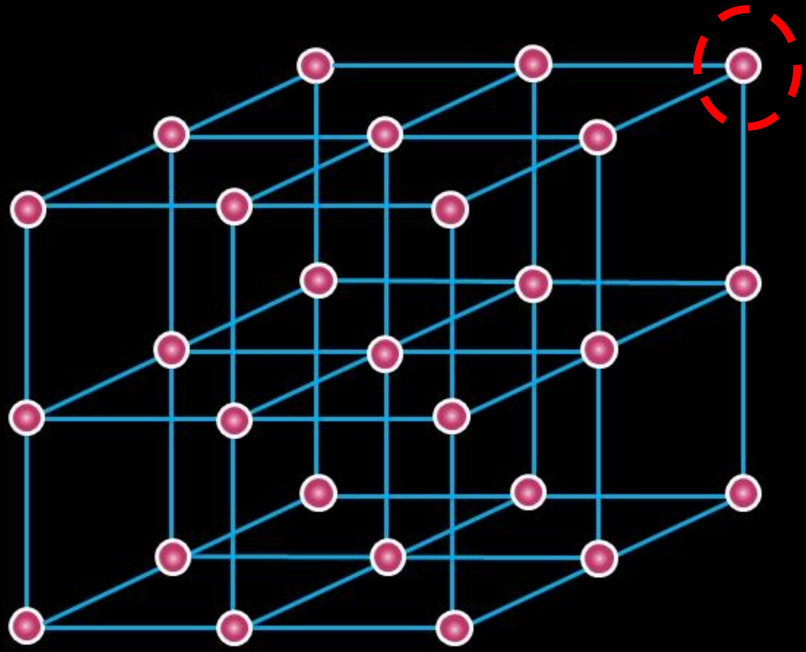
Crystal lattice or space lattice



Each point called Lattice Point

12C01.2 Crystal Lattices and Unit Cells

Crystal lattice or space lattice

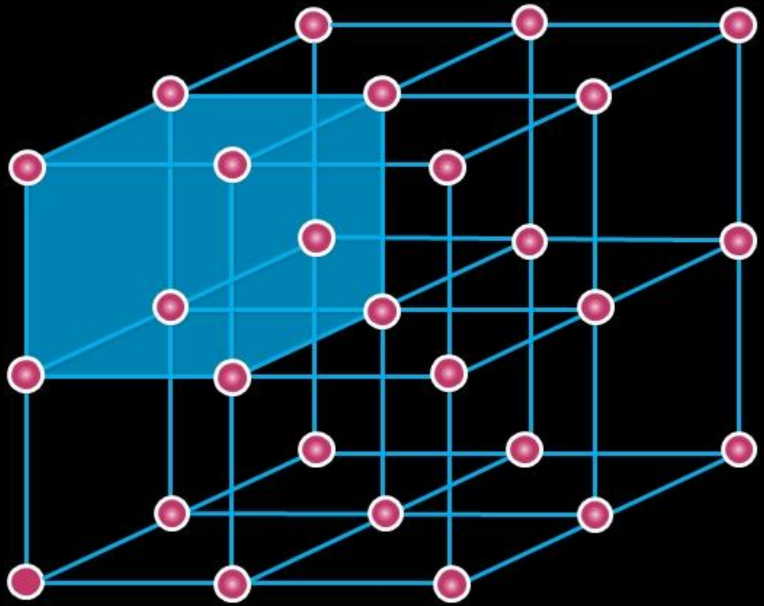


Each point called Lattice Point

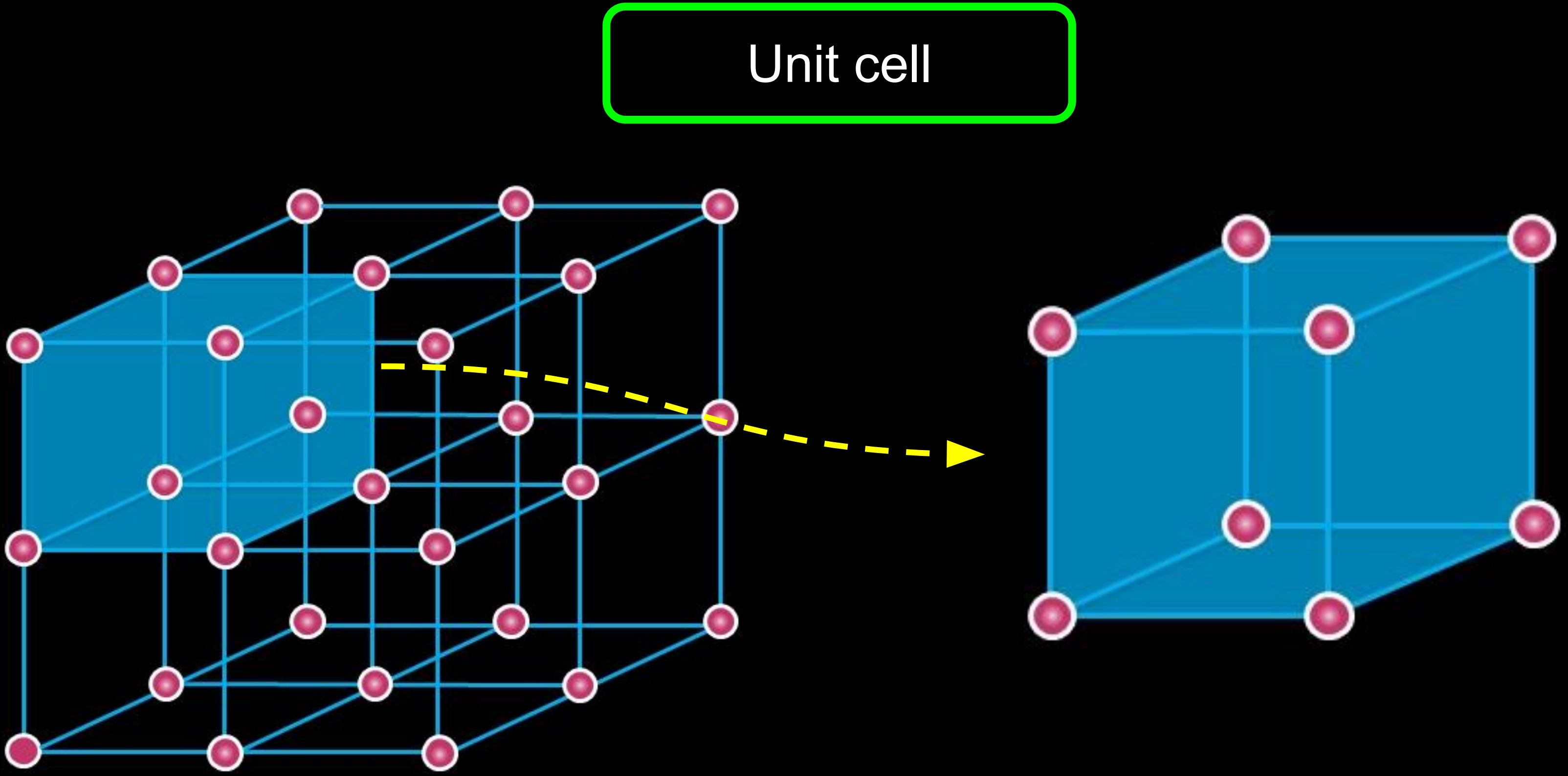
Atom, molecule or ion

12C01.2 Crystal Lattices and Unit Cells

Crystal lattice or space lattice

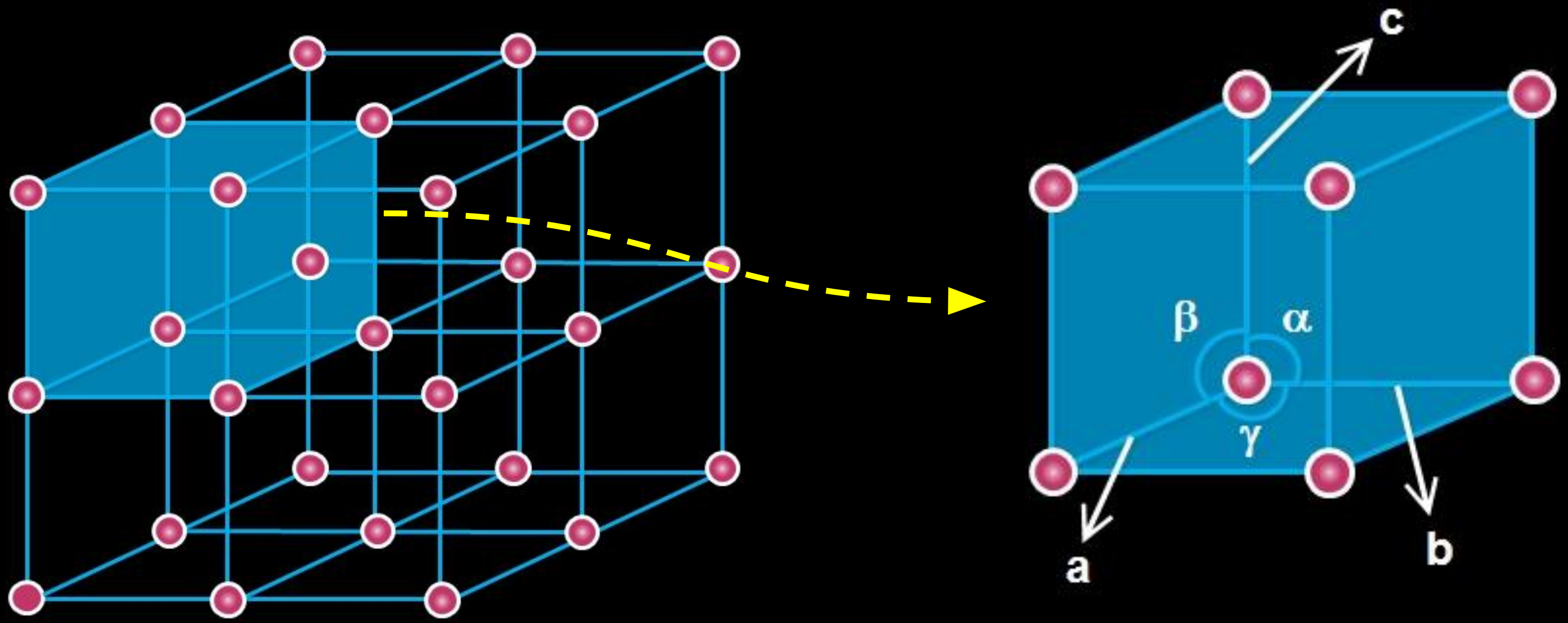


12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

Unit cell

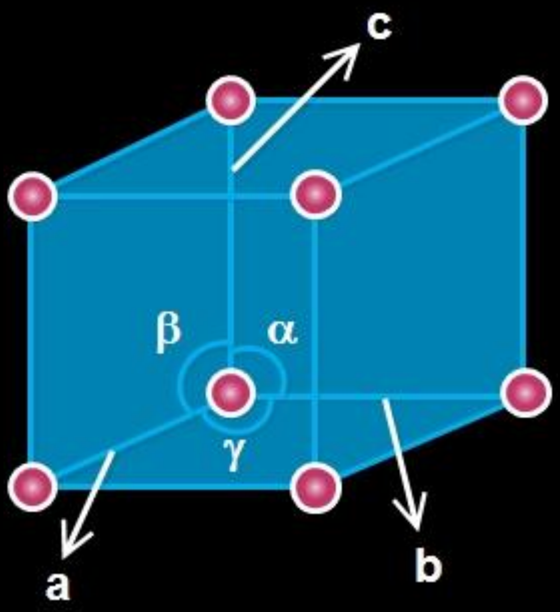


12C01.2 Crystal Lattices and Unit Cells

Edge lengths a, b and c.

Parameters of unit cell
- 6 parameters

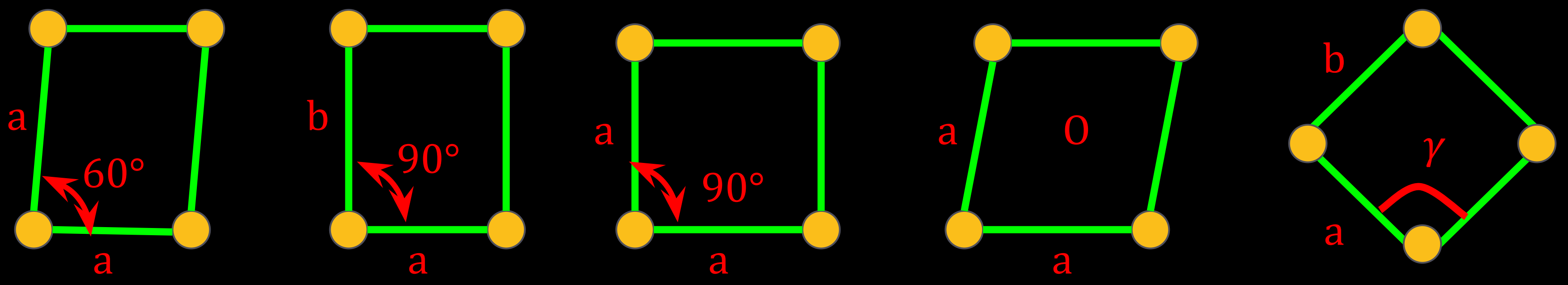
Angles between the edges :
α, β , γ



Unit Cells in 2-D :

In two dimensions a parallelogram with side of length ‘a’ and ‘b’ and an angle r between these sides is chosen as unit cell.

Possible unit cells in two dimensions



CV 2

Types of Unit Cells

12C01.2 Crystal Lattices and Unit Cells

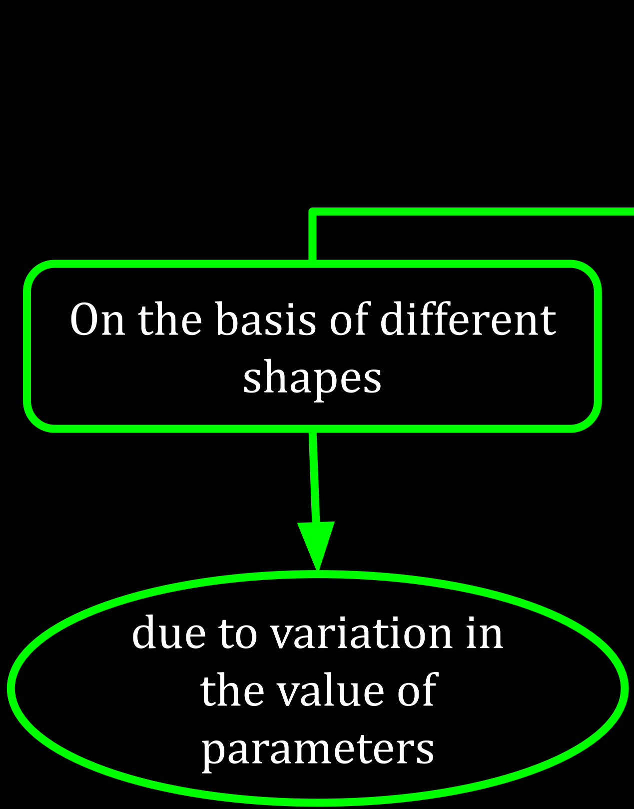
Types of unit cell

12C01.2 Crystal Lattices and Unit Cells

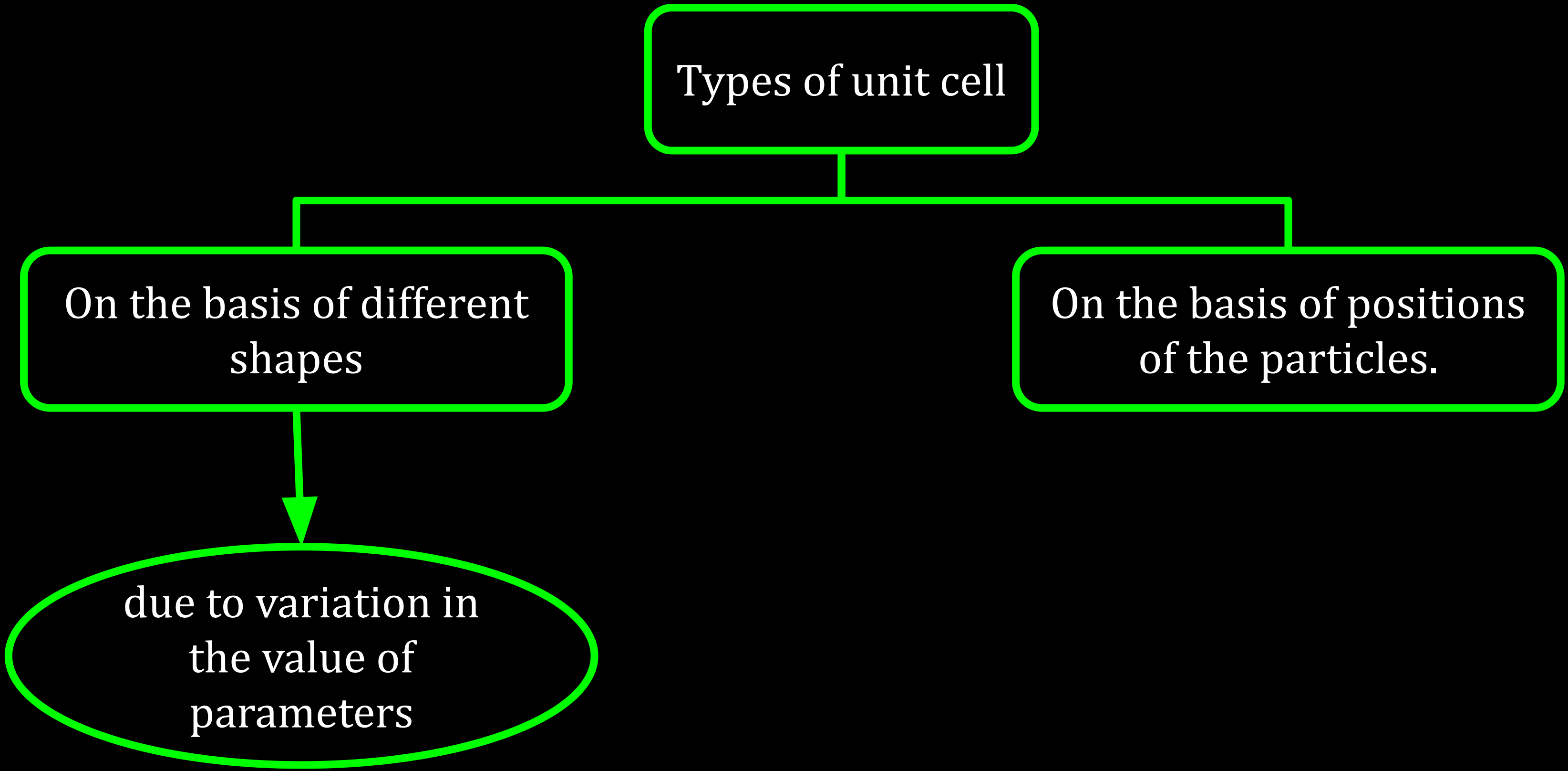
Types of unit cell

On the basis of different shapes

12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

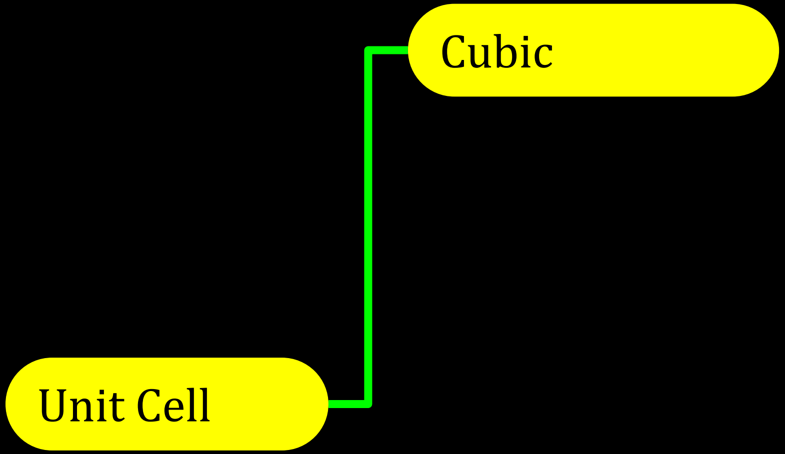
Types of unit cell

- On the basis of different shapes
  - due to variation in the value of parameters
- On the basis of positions of the particles.

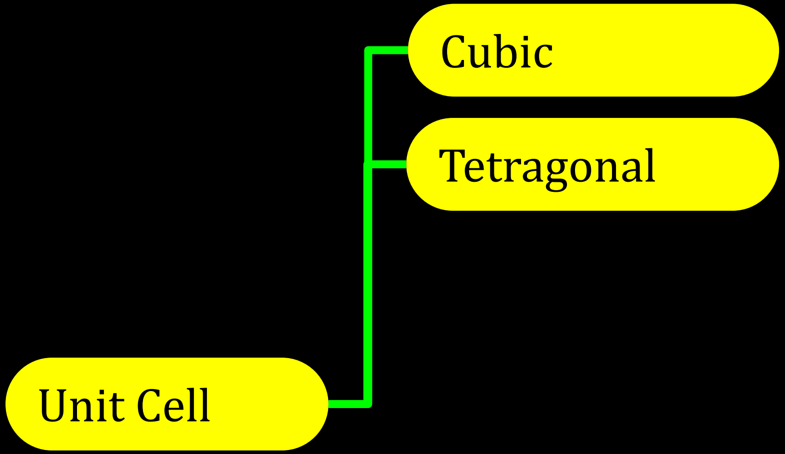
12C01.2 Crystal Lattices and Unit Cells

Unit Cell

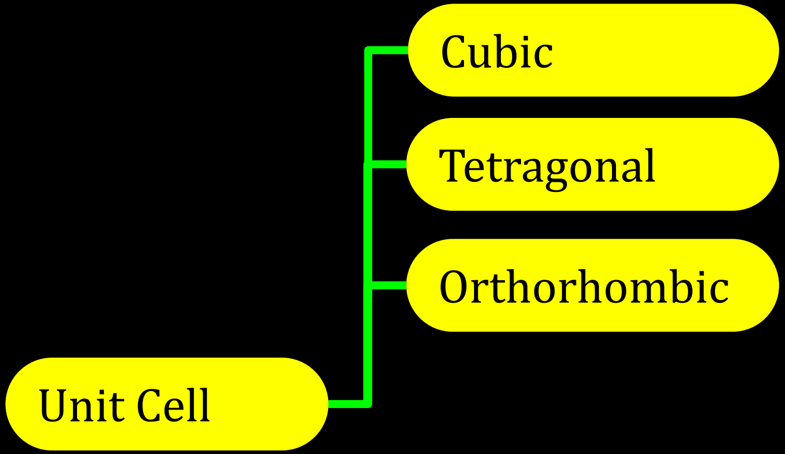
12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells



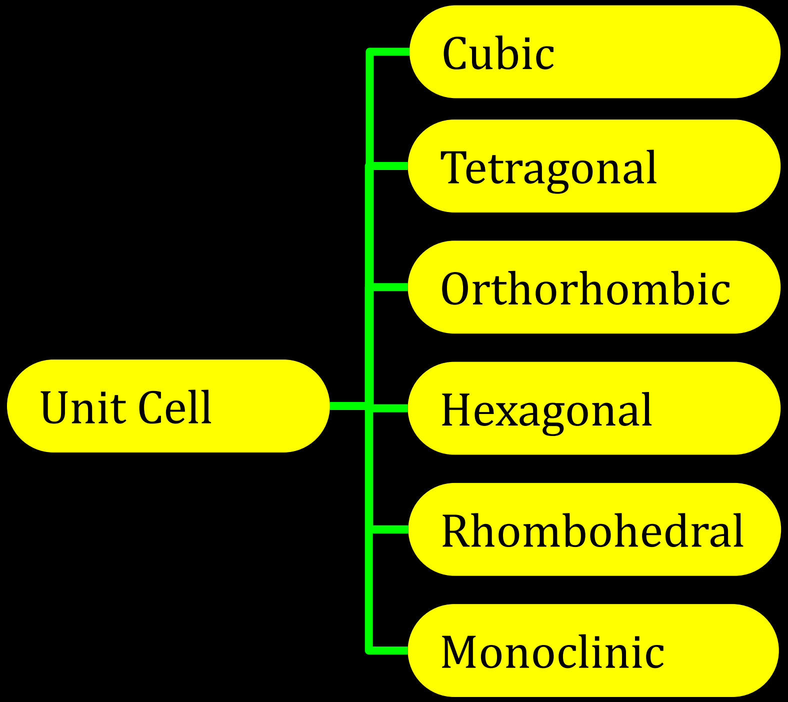
12C01.2 Crystal Lattices and Unit Cells

- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal

12C01.2 Crystal Lattices and Unit Cells

- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral

12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral
  - Monoclinic
  - Triclinic

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
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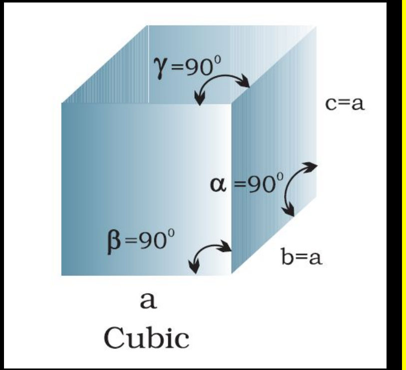


Edge lengths: $a = b = c$

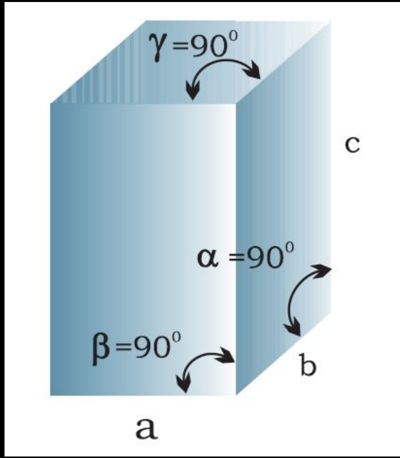
Axial angles:
$\alpha = \beta = \gamma = 90°$

Examples: NaCl, Zinc blende, Cu

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- **Tetragonal**
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



Edge lengths: $a = b \neq c$

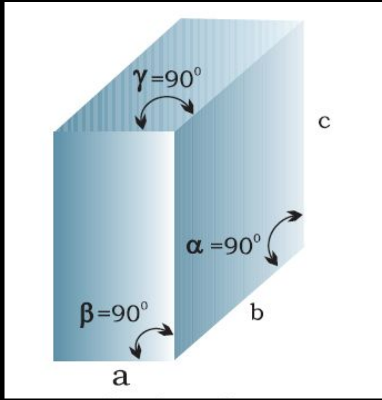
Axial angles:
$\alpha = \beta = \gamma = 90°$

Examples: White tin, SnO_2, TiO_2, $CaSO_4$

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



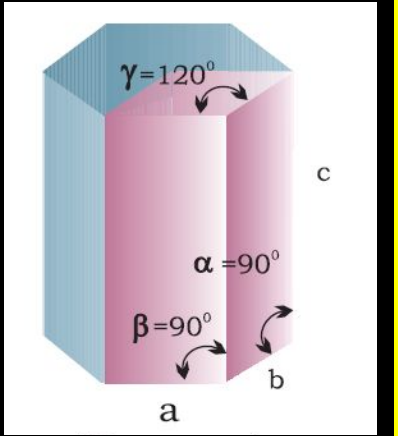
Edge lengths: $a \neq b \neq c$

Axial angles:
$\alpha = \beta = \gamma = 90°$

Examples: Rhombic sulphur, KNO_3, $BaSO_4$

12C01.2 Crystal Lattices and Unit Cells

Unit Cell
- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



Edge lengths: $a = b \neq c$

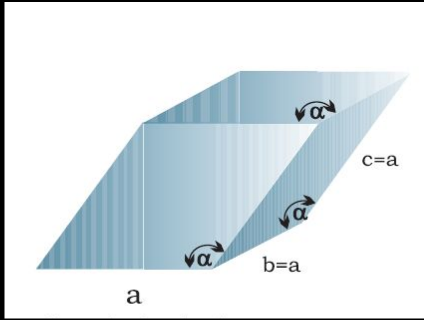
Axial angles:
$\alpha = \beta = 90°$
$\gamma = 120°$

Examples: Graphite, ZnO, CdS

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



Edge lengths: $a = b = c$

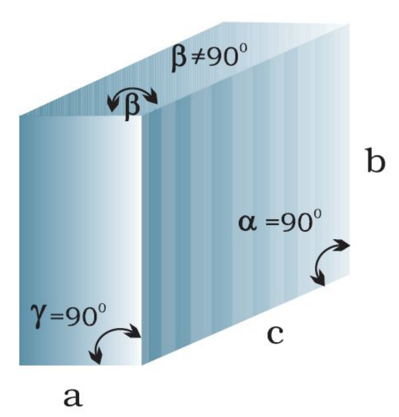
Axial angles:
$\alpha = \beta = \gamma \neq 90°$

Examples: Calcite ($CaCO_3$), HgS (cinnabar)

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



Edge lengths: $a \neq b \neq c$

Axial angles:

$\alpha = \gamma = 90°$

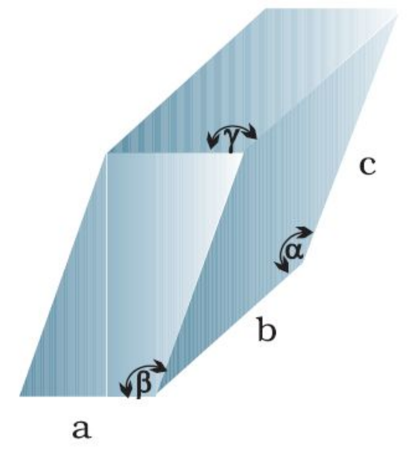
$\beta \neq 90°$

Examples: Monoclinic sulphur, $Na_2SO_4.10H_2O$

12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
- Triclinic



Edge lengths: $a \neq b \neq c$

Axial angles:
$\alpha \neq \beta \neq \gamma \neq 90°$

Examples: $K_2Cr_2O_7$, $CuSO_4 . 5H_2O$, H_3BO_3

12C01.2 Crystal Lattices and Unit Cells

- Types of unit cell
  - On the basis of different shapes
    - due to variation in the value of parameters
  - On the basis of positions of the particles.

12C01.2 Crystal Lattices and Unit Cells

Types of unit cell

On the basis of different shapes

due to variation in the value of parameters

On the basis of positions of the particles.

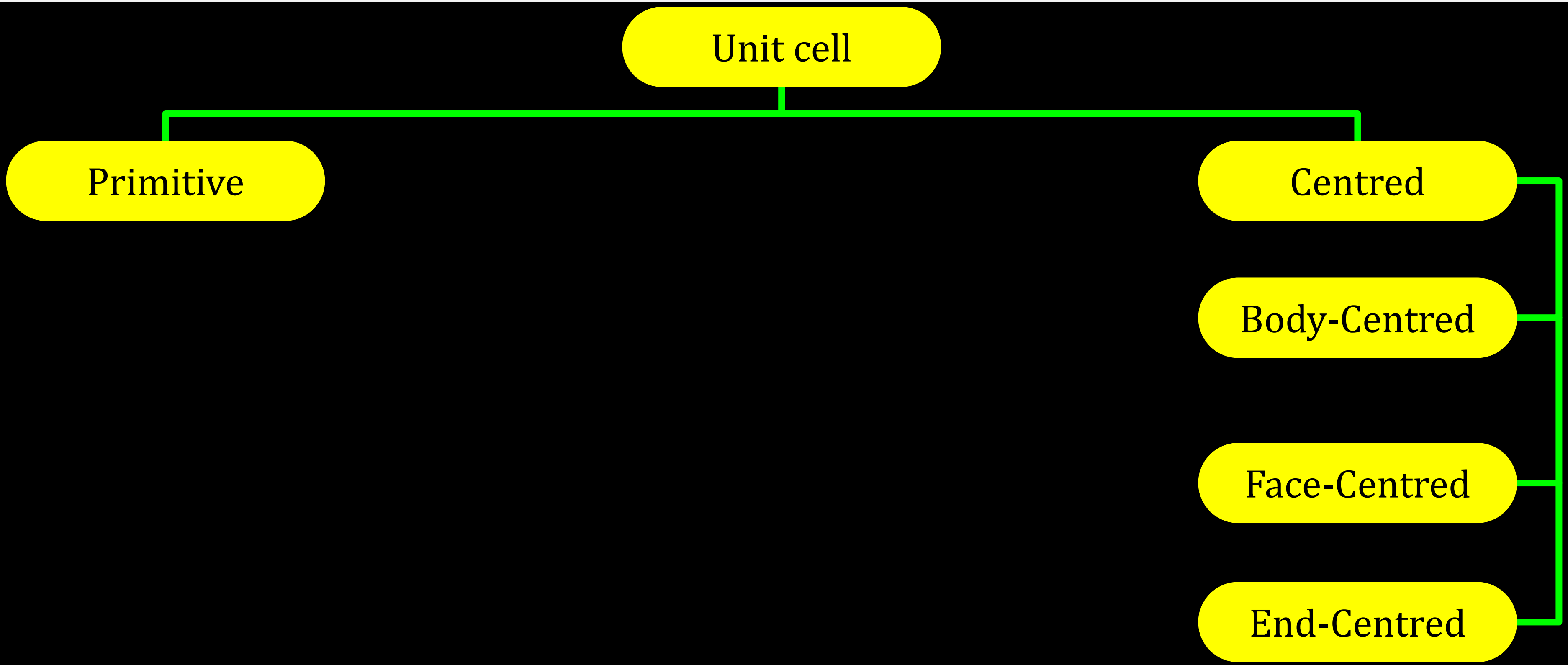
12C01.2 Crystal Lattices and Unit Cells

Unit cell

12C01.2 Crystal Lattices and Unit Cells

- Unit cell
  - Primitive
  - Centred

12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

Unit cell

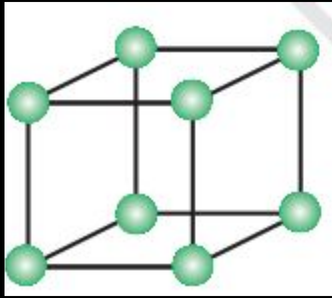
- Primitive
  - Constituent particles are present only on the corner positions of a unit cell.
- Centred
  - Body-Centred
  - Face-Centred
  - End-Centred

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

Constituent particles are present only on the corner positions of a unit cell.



Open structure

Centred

Body-Centred

Face-Centred

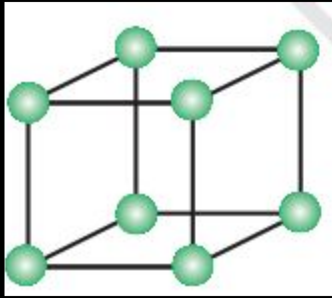
End-Centred

12C01.2 Crystal Lattices and Unit Cells

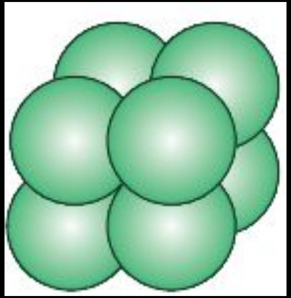
Unit cell

Primitive

Constituent particles are present only on the corner positions of a unit cell.



Open structure



Space-filling

Centred

Body-Centred

Face-Centred

End-Centred

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

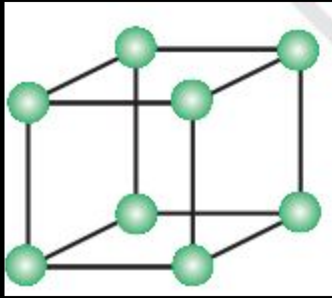
Centred

Body-Centred

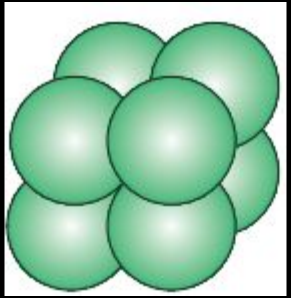
Face-Centred

End-Centred

Constituent particles are present only on the corner positions of a unit cell.



Open structure



Space-filling

Actual portions of atoms

12C01.2 Crystal Lattices and Unit Cells

Unit cell

- Primitive
- Centred
  - Body-Centred
  - Face-Centred
  - End-Centred

One or more particles present at positions other than corners in addition to those at corners

12C01.2 Crystal Lattices and Unit Cells

Unit cell

- Primitive
- Centred
  - Body-Centred
  - Face-Centred
  - End-Centred

Body-Centred: One constituent particle (atom, molecule or ion) at its body-centre besides ones that are at its corners.

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

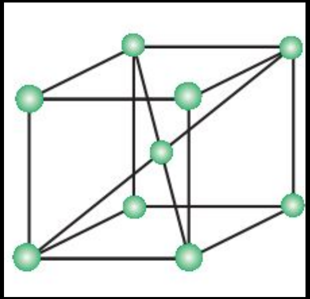
Centred

Body-Centred

Face-Centred

End-Centred

One constituent particle (atom, molecule or ion) at its body-centre besides ones that are at its corners.



Open structure

12C01.2 Crystal Lattices and Unit Cells

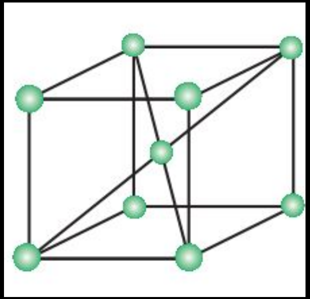
Unit cell

Primitive

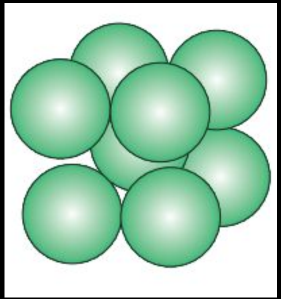
Centred

- Body-Centred
- Face-Centred
- End-Centred

One constituent particle (atom, molecule or ion) at its body-centre besides ones that are at its corners.



Open structure



Space-filling

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

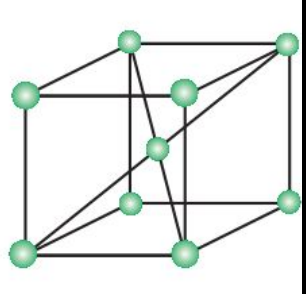
Centred

Body-Centred

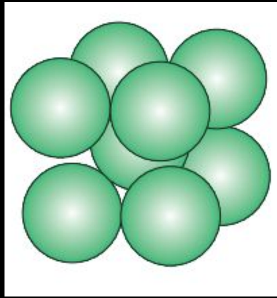
Face-Centred

End-Centred

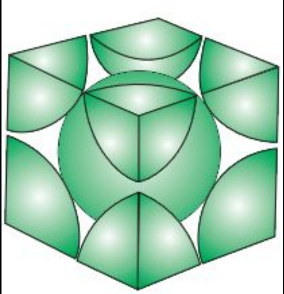
One constituent particle (atom, molecule or ion) at its body-centre besides ones that are at its corners.



Open structure

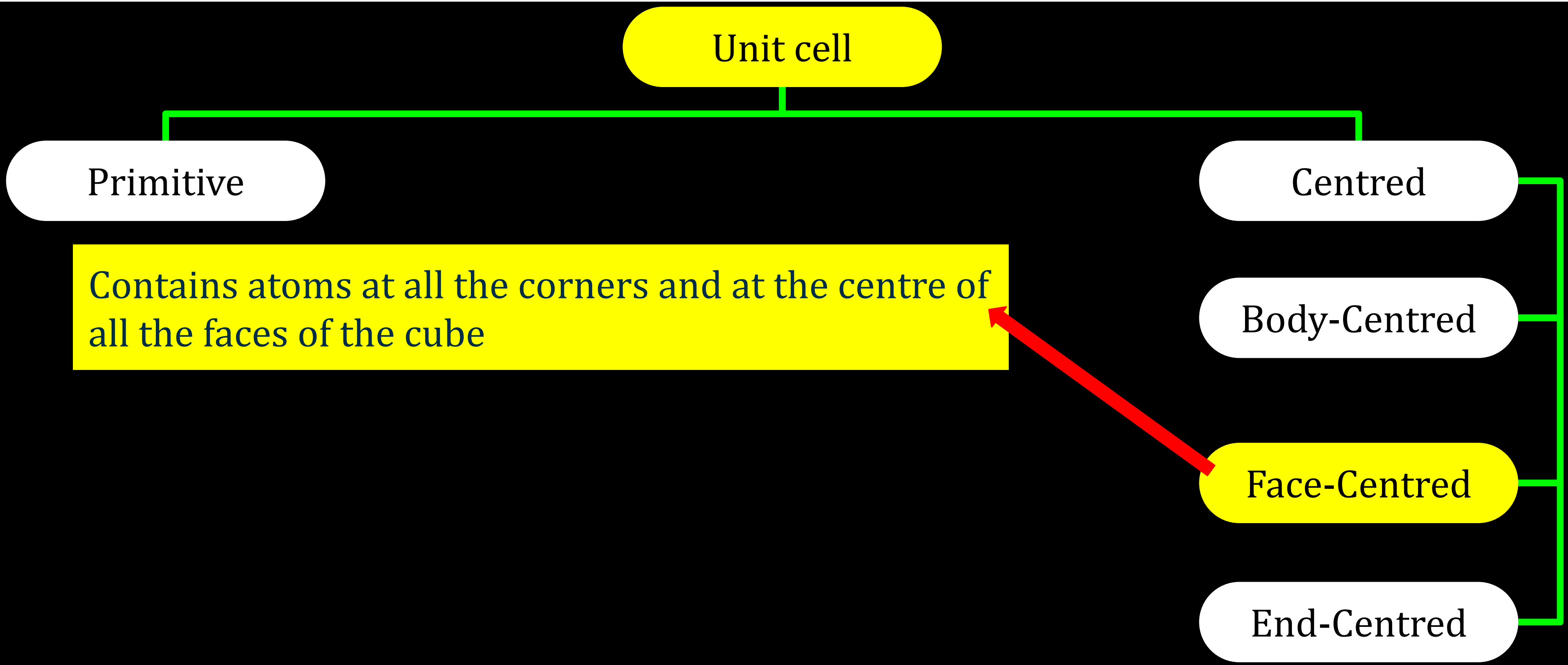


Space-filling



Actual portions of atoms

12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

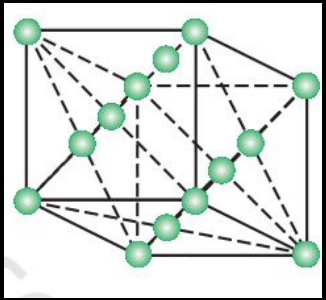
Centred

Body-Centred

Face-Centred

End-Centred

Contains atoms at all the corners and at the centre of all the faces of the cube



Open structure

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

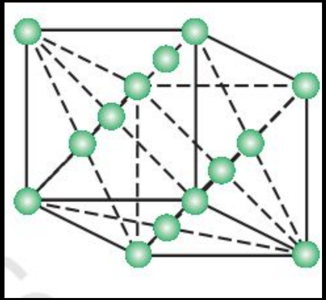
Centred

Body-Centred

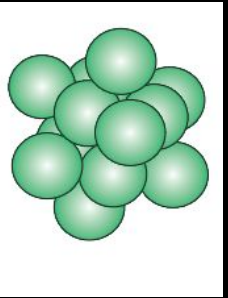
Face-Centred

End-Centred

Contains atoms at all the corners and at the centre of all the faces of the cube



Open structure



Space-filling

12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

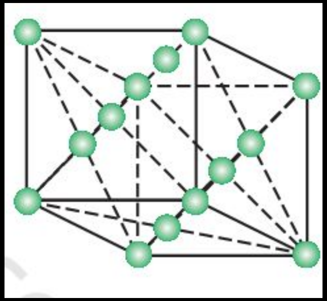
Centred

Body-Centred

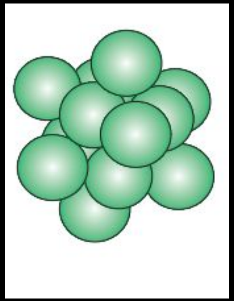
Face-Centred

End-Centred

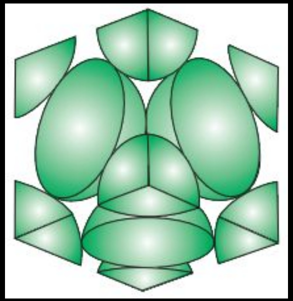
Contains atoms at all the corners and at the centre of all the faces of the cube



Open structure

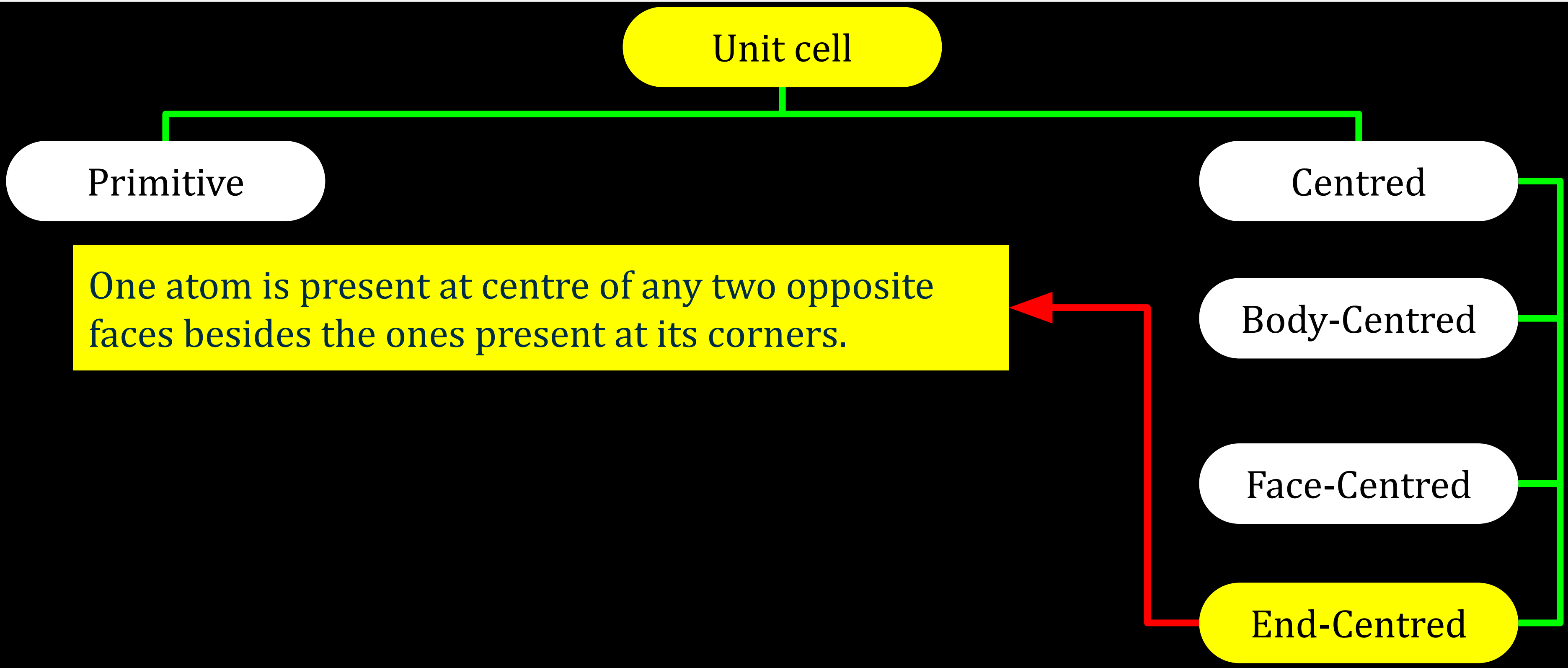


Space-filling



Actual portions of atoms

12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

Unit cell

Primitive

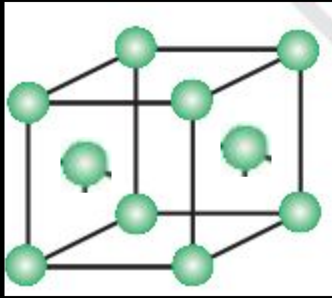
Centred

Body-Centred

Face-Centred

End-Centred

One atom is present at centre of any two opposite faces besides the ones present at its corners.



Open structure

Conceptest

Ready for Challenge

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?
(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution : ? (NCERT Exercise Q. No. 1.8 Page no. 33)

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?

(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution : ? (NCERT Exercise Q. No. 1.8 Page no. 33)

Pause the Video Time duration 2 minute

Question : How many lattice points are there in a unit cell of each of the given lattice?
(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution :

$\boxed{\text{Total no. of Lattice Points}}$

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?

(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution :

Total no. of Lattice Points

- Face-centred cubic
- Face-centred tetragonal
- Body-centred cubic

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?

(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution :

Total no. of Lattice Points

Face-centred cubic

Face-centred tetragonal

Body-centred cubic

8 from the corners
6 from the faces
Total = 14

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?

(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution :

Total no. of Lattice Points

- Face-centred cubic
  - 8 from the corners
    6 from the faces
    Total = 14
- Face-centred tetragonal
  - 8 from the corners
    6 from the faces
    Total = 14
- Body-centred cubic

12C01.2 Crystal Lattices and Unit Cells

Question : How many lattice points are there in a unit cell of each of the given lattice?

(i) Face-centred cubic (ii) Face-centred tetragonal (iii) Body-centred

Solution :

Total no. of Lattice Points

- Face-centred cubic
  - 8 from the corners
  - 6 from the faces
  - Total = 14
- Face-centred tetragonal
  - 8 from the corners
  - 6 from the faces
  - Total = 14
- Body-centred cubic
  - 8 from the corners
  - 1 from the centre
  - Total = 9

CV 3

Bravais Lattices

12C01.2 Crystal Lattices and Unit Cells

Bravais Lattices:

A French mathematician,
Bravais, showed

12C01.2 Crystal Lattices and Unit Cells

Bravais Lattices:

A French mathematician, Bravais, showed

↓

There are only 14 possible three dimensional lattices.

12C01.2 Crystal Lattices and Unit Cells

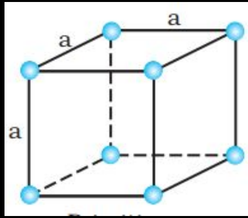
Unit Cell

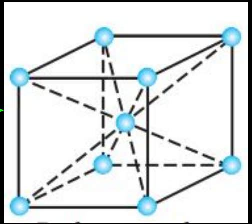
12C01.2 Crystal Lattices and Unit Cells

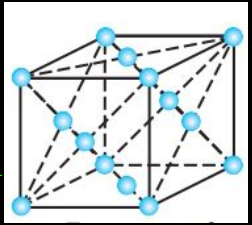
- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral
  - Monoclinic
  - Triclinic

12C01.2 Crystal Lattices and Unit Cells

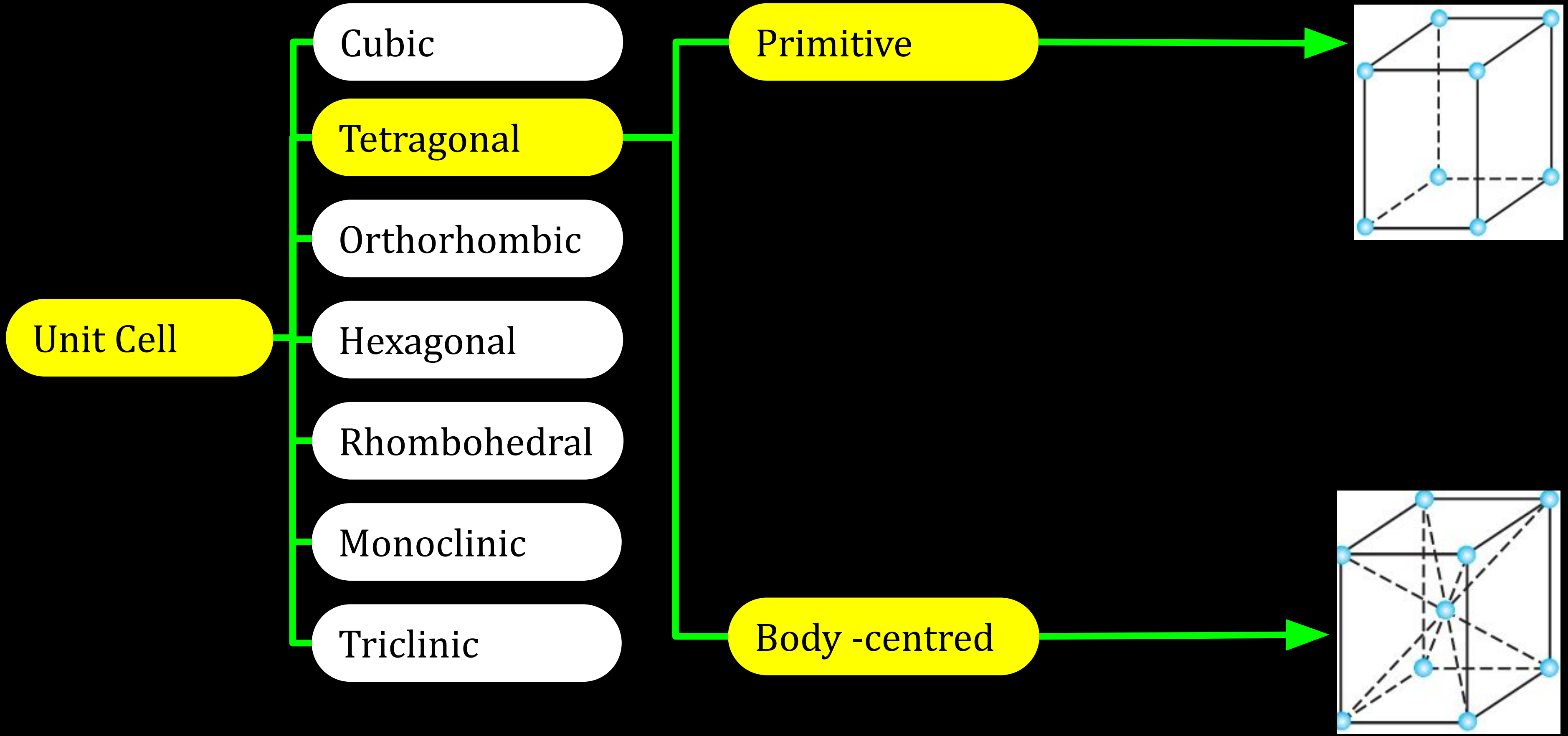
- Unit Cell
  - Cubic
    - Primitive
    - Body-centred
    - Face-centred
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral
  - Monoclinic
  - Triclinic



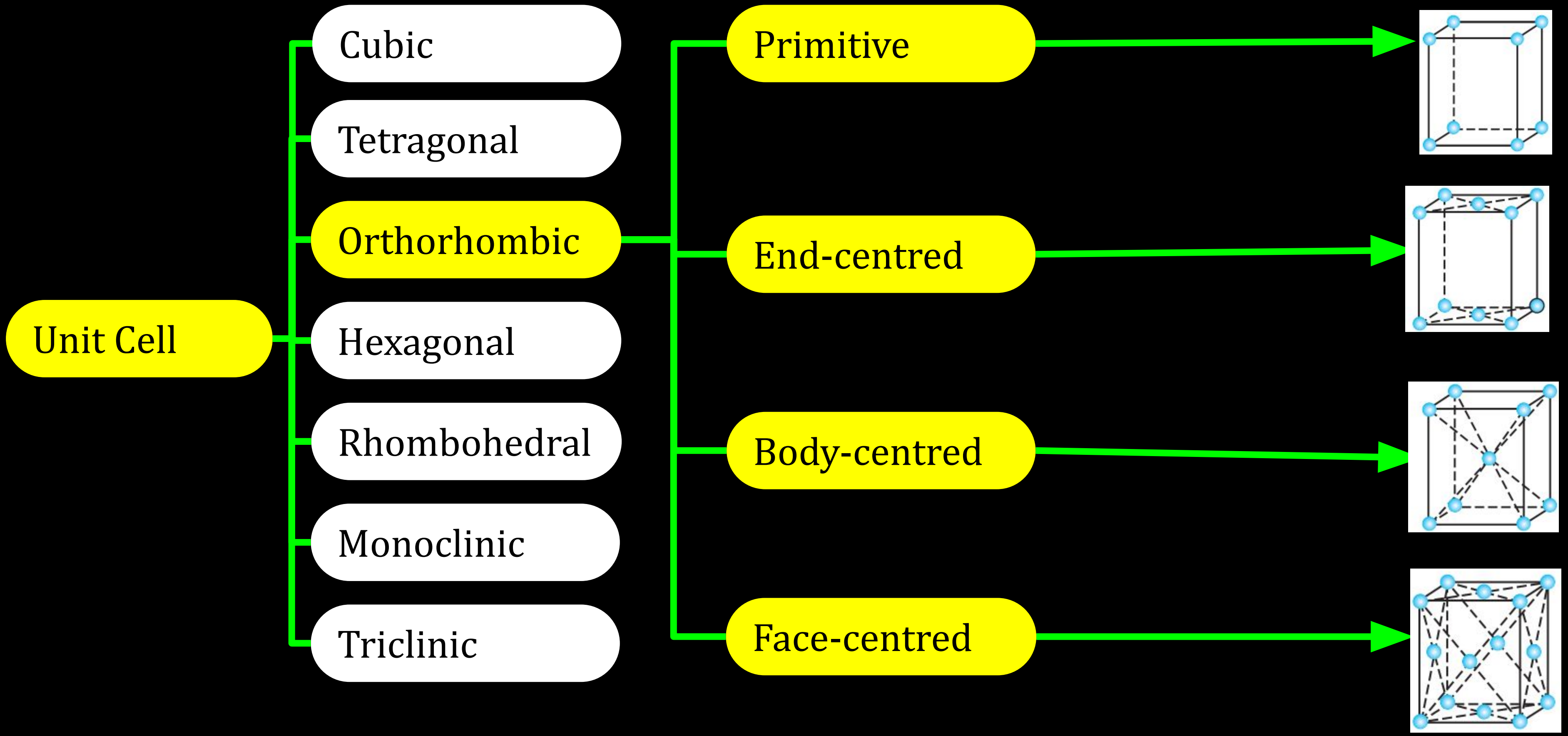




12C01.2 Crystal Lattices and Unit Cells



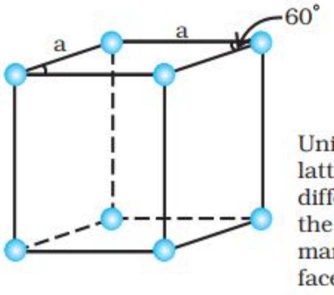
12C01.2 Crystal Lattices and Unit Cells



12C01.2 Crystal Lattices and Unit Cells

Unit Cell

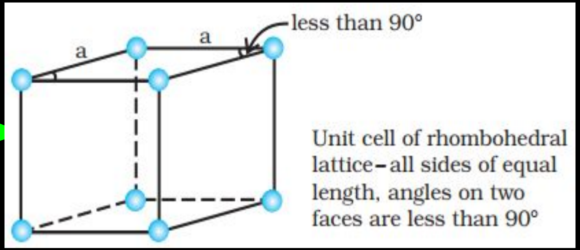
- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal → Primitive →
- Rhombohedral
- Monoclinic
- Triclinic



Unit cell of hexagonal lattice – one side different in length to the other two, the marked angles on two faces are 60°

12C01.2 Crystal Lattices and Unit Cells

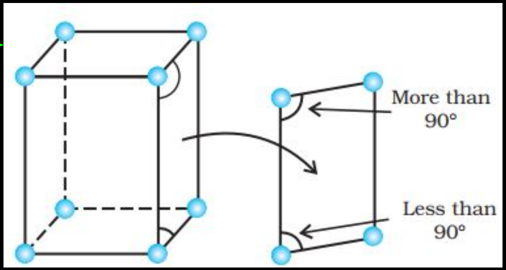
- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral
    - Primitive
  - Monoclinic
  - Triclinic



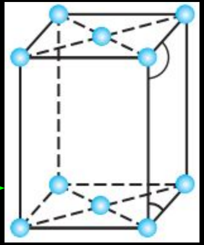
12C01.2 Crystal Lattices and Unit Cells

Unit Cell

- Cubic
- Tetragonal
- Orthorhombic
- Hexagonal
- Rhombohedral
- Monoclinic
  - Primitive



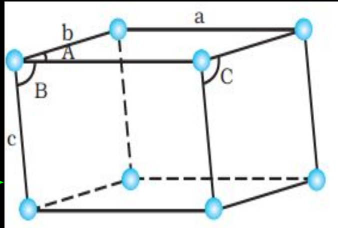
  - End-centred



- Triclinic

12C01.2 Crystal Lattices and Unit Cells

- Unit Cell
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Hexagonal
  - Rhombohedral
  - Monoclinic
  - Triclinic
    - Primitive



Unit cell of triclinic lattice– unequal sides a, b, c, A, B, C are unequal angles with none equal to 90°

CV 4

Number of Atoms in a Cubic Unit Cell

12C01.2 Crystal Lattices and Unit Cells

Number of Atoms in a Cubic Unit Cell

We shall consider three types of cubic unit cells and for simplicity assume that the constituent particle is an atom.

12C01.2 Crystal Lattices and Unit Cells

Insert : -

ALC C15.1.3 - (Hinglish)

Time Stamps are : -

1:53 - 3:38
3:42 - 4:55
4:59 - 6:15

Conceptest

Ready for Challenge

12C01.2 Crystal Lattices and Unit Cells

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution : ? (NCERT Exercise Q. no. 1.12 page no. 33)

12C01.2 Crystal Lattices and Unit Cells

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution : ? (NCERT Exercise Q. no. 1.12 page no. 33)

Pause the Video Time duration 2 minute

12C01.2 Crystal Lattices and Unit Cells

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

12C01.2 Crystal Lattices and Unit Cells

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

P present at the body-centre

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

P present at the body-centre

$\downarrow$

No. of P atoms = 1

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

P present at the body-centre

Q present at the corners

No. of P atoms = 1

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

P present at the body-centre

No. of P atoms = 1

Q present at the corners

No. of Q atoms = 8 * ⅛ = 1

12C01.2 Crystal Lattices and Unit Cells

Question : A cubic solid is made of two elements P and Q. Atoms of Q are at the corners of the cube and P at the body-centre. What is the formula of the compound?

Solution :

In a given cubic solid

P present at the body-centre

No. of P atoms = 1

Q present at the corners

No. of Q atoms = $8 * \frac{1}{8} = 1$

Formula of compound = PQ

12C01.2 Crystal Lattices and Unit Cells

: Reference Questions :

Intext Questions : 1.10, 1.11 ,1.12 ,1.13 (NCERT page no. 14)

Workbook Question : Question no. 14

12C01.3

Close Packed Structures

12C01.3 Close Packed Structures

- Close Packing in One Dimension

12C01.3 Close Packed Structures

- Close Packing in One Dimension

- Close Packing in Two Dimensions

12C01.3 Close Packed Structures

- Close Packing in One Dimension

- Close Packing in Two Dimensions

- Close Packing in Three Dimensions

12C01.3 Close Packed Structures

- Close Packing in One Dimension
- Close Packing in Two Dimensions
- Close Packing in Three Dimensions
- Formula of a Compound and Number of Voids filled

12C01.3 Close Packed Structures

- Close Packing in One Dimension
- Close Packing in Two Dimensions
- Close Packing in Three Dimensions
- Formula of a Compound and Number of Voids filled
- Locating Tetrahedral and Octahedral Voids

CV 1

Close Packing in 1D and 2D

Close Packed Structures

In solids

12C01.3 Close Packed Structures

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

In solids

12C01.3 Close Packed Structures

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

In solids

↓

3D structure in 3 steps

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

In solids

3D structure in 3 steps

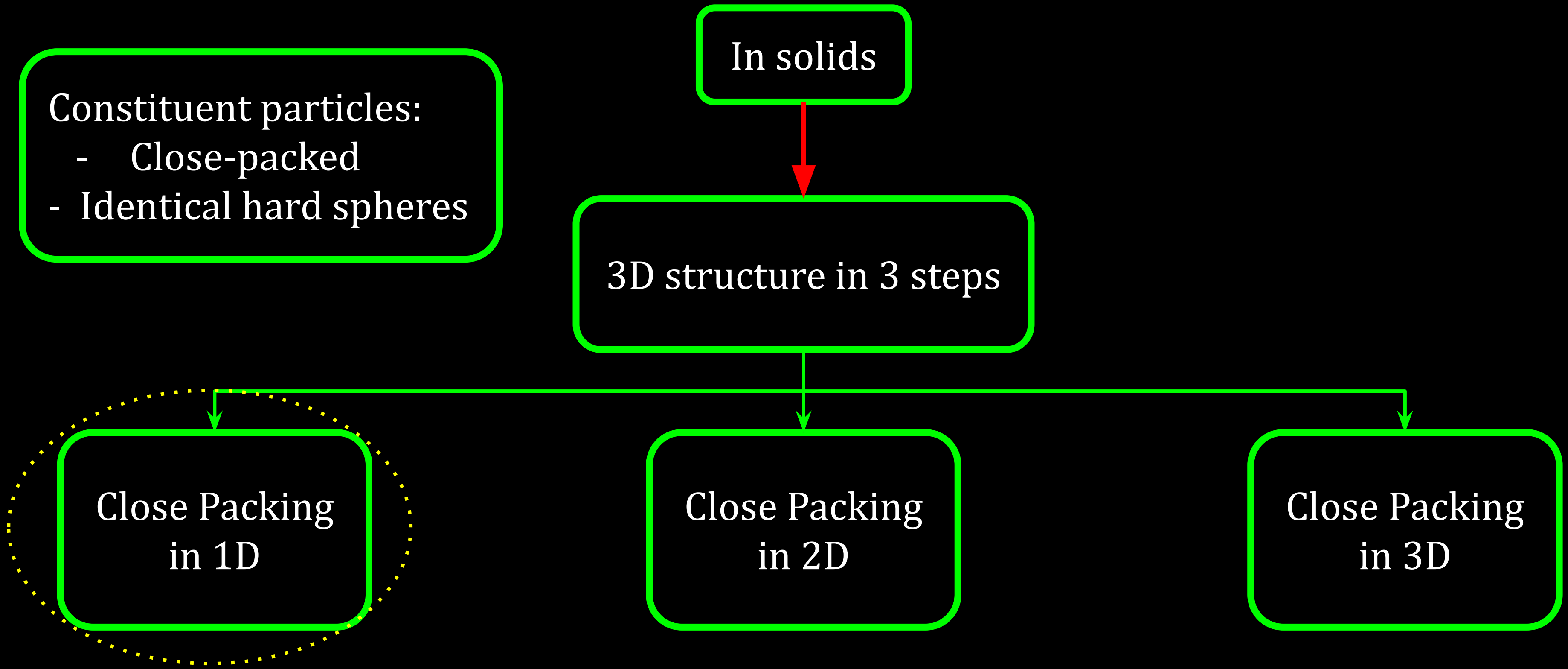
Close Packing in 1D

Close Packing in 2D

Close Packing in 3D

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

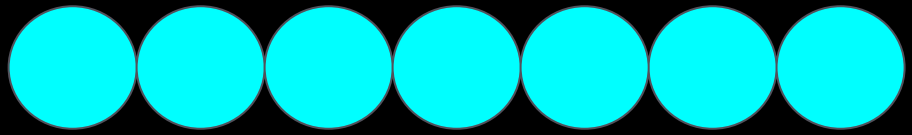


12C01.3 Close Packed Structures

Close Packing in 1D

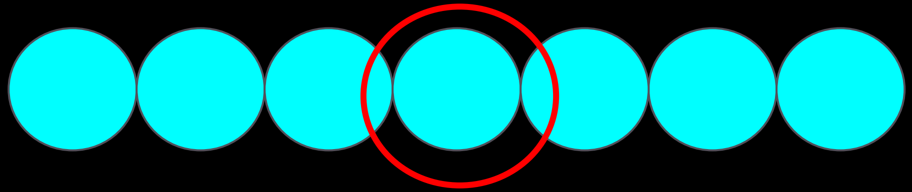
12C01.3 Close Packed Structures

Close Packing in 1D



12C01.3 Close Packed Structures

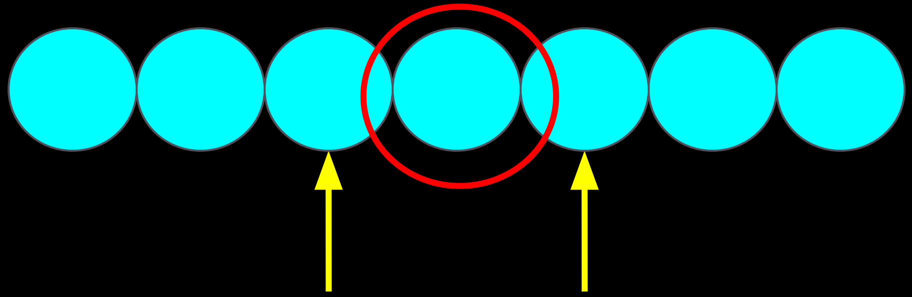
Close Packing in 1D



12C01.3 Close Packed Structures

Close Packing in 1D

Contact with 2 neighbours

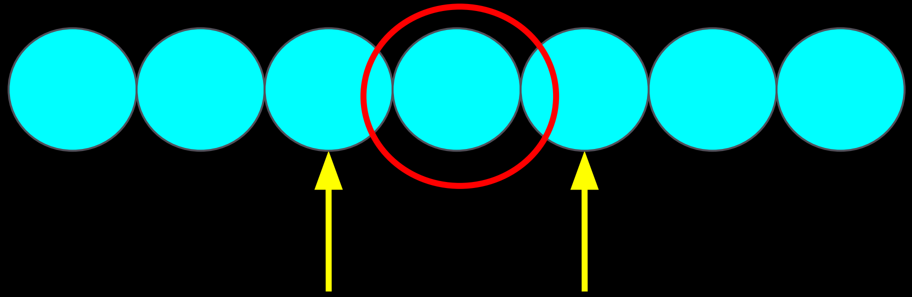


12C01.3 Close Packed Structures

Close Packing in 1D

Contact with 2 neighbours

Coordination no. = 2



12C01.3 Close Packed Structures

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

In solids

3D structure in 3 steps

Close Packing in 1D

Close Packing in 2D

Close Packing in 3D

12C01.3 Close Packed Structures

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres

In solids

3D structure in 3 steps

Close Packing in 1D

Close Packing in 2D

Close Packing in 3D

12C01.3 Close Packed Structures

Close Packing in 2D

12C01.3 Close Packed Structures

Close Packing in 2D

It can be generated by placing the rows of close packed spheres

12C01.3 Close Packed Structures

Close Packing in 2D

It can be generated by placing the rows of close packed spheres

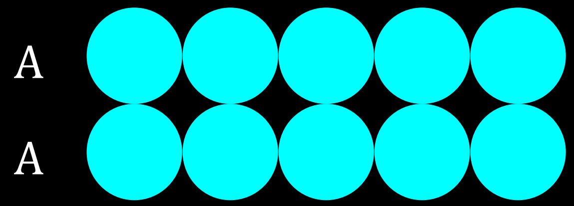
2 different ways

12C01.3 Close Packed Structures

Close Packing in 2D

It can be generated by placing the rows of close packed spheres

2 different ways

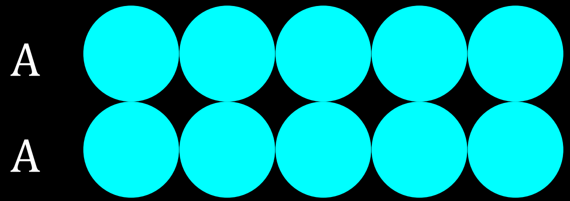


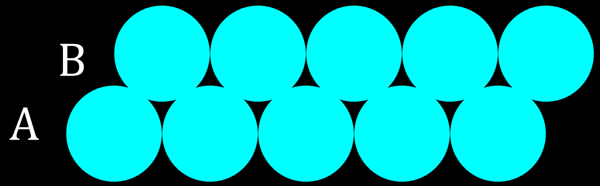
12C01.3 Close Packed Structures

Close Packing in 2D

It can be generated by placing the rows of close packed spheres

2 different ways



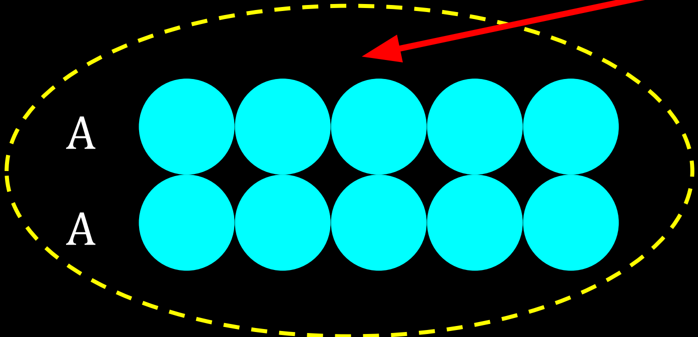


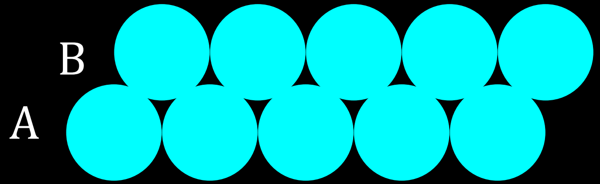
12C01.3 Close Packed Structures

Close Packing in 2D

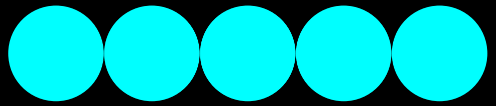
It can be generated by placing the rows of close packed spheres

2 different ways

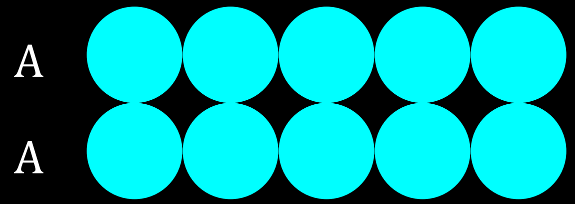




12C01.3 Close Packed Structures

A 

12C01.3 Close Packed Structures



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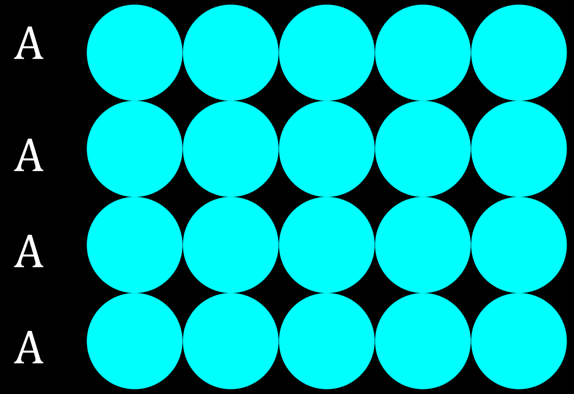
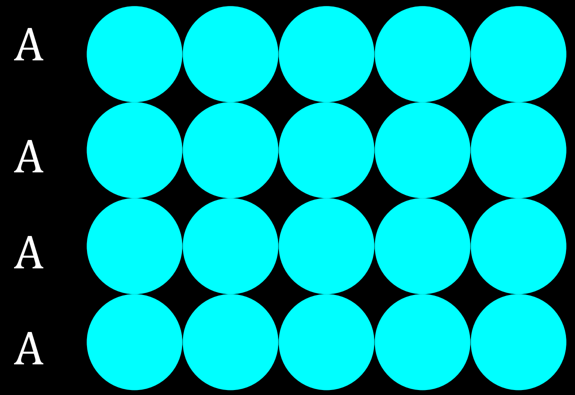


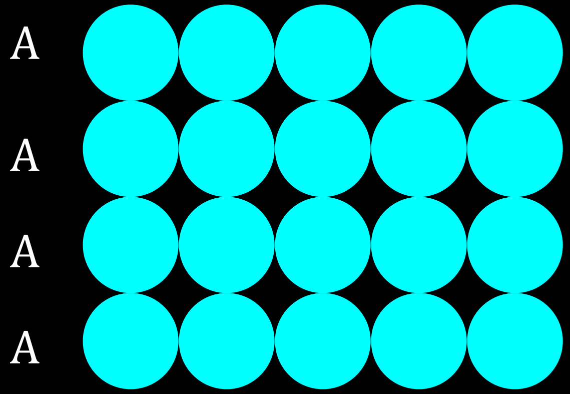
12C01.3 Close Packed Structures



AAA type of arrangement

12C01.3 Close Packed Structures

Spheres are aligned horizontally & vertically

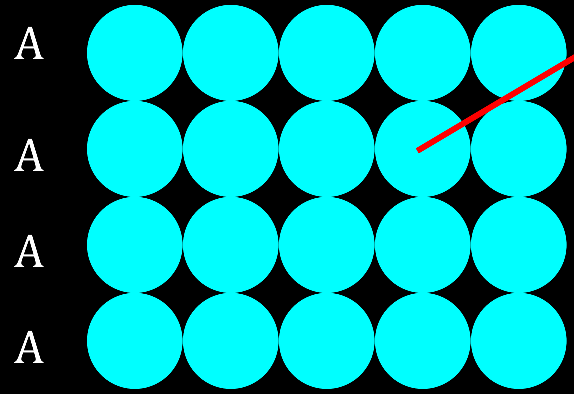


AAA type of arrangement

12C01.3 Close Packed Structures

Spheres are aligned horizontally & vertically

Contact with 4 neighbours. coordination no. is = 4

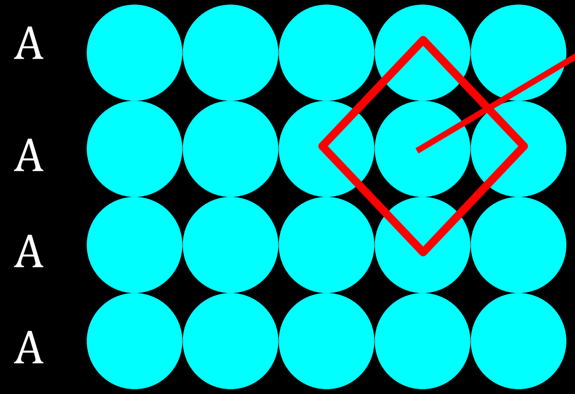


AAA type of arrangement

12C01.3 Close Packed Structures

Spheres are aligned horizontally & vertically

Contact with 4 neighbours.
coordination no. is = 4

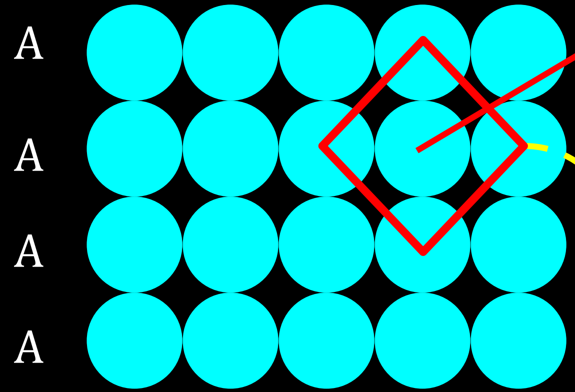


AAA type of arrangement

12C01.3 Close Packed Structures

Spheres are aligned horizontally & vertically

Contact with 4 neighbours. coordination no. is = 4



Square close packing in 2D

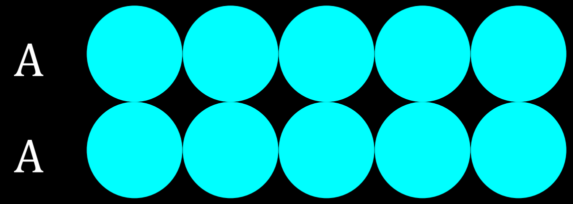
AAA type of arrangement

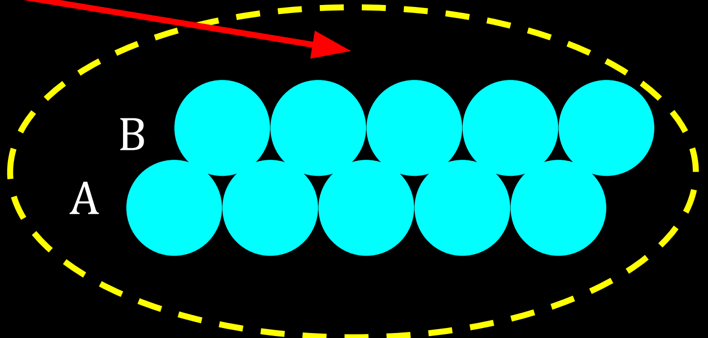
12C01.3 Close Packed Structures

Close Packing in 2D

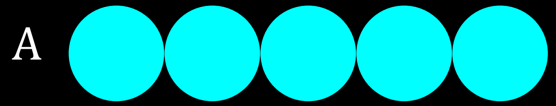
It can be generated by placing the rows of close packed spheres

2 different ways

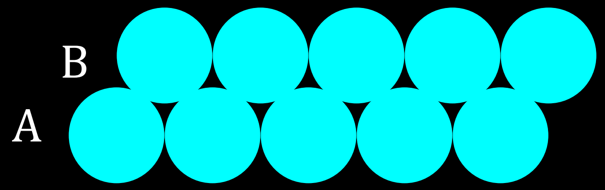




12C01.3 Close Packed Structures



12C01.3 Close Packed Structures



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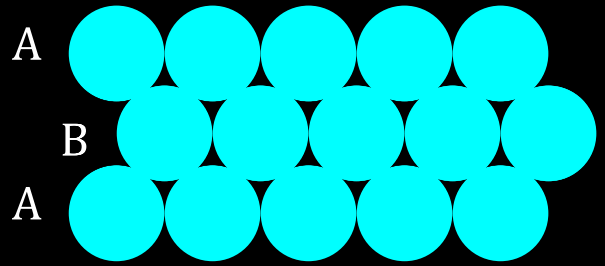
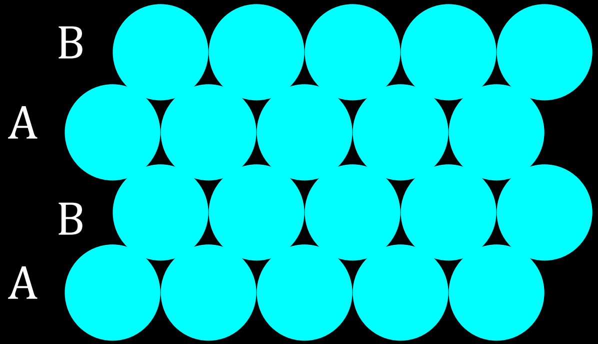


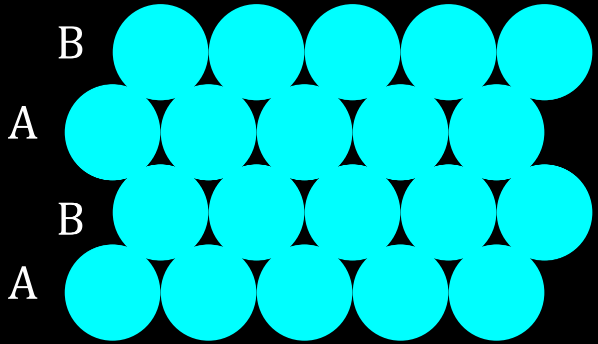
12C01.3 Close Packed Structures



ABAB type of arrangement

12C01.3 Close Packed Structures

Less free space.
More efficient Packing

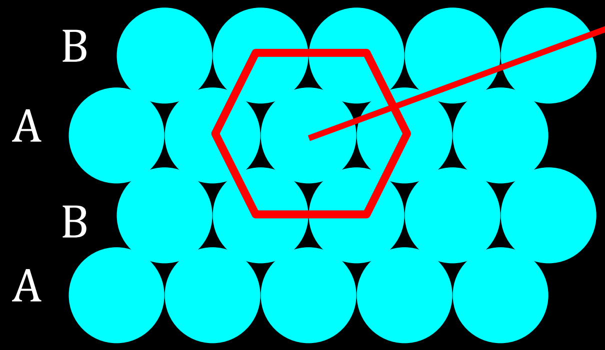


ABAB type of arrangement

12C01.3 Close Packed Structures

Less free space.
More efficient Packing

Contact with 6 neighbours.
coordination no. = 6

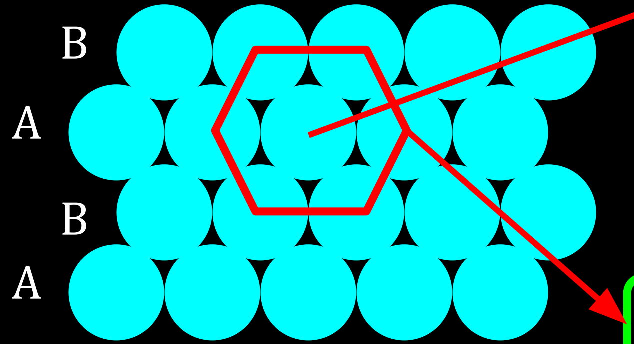


ABAB type of arrangement

12C01.3 Close Packed Structures

Less free space.
More efficient Packing

Contact with 6 neighbours.
coordination no. = 6



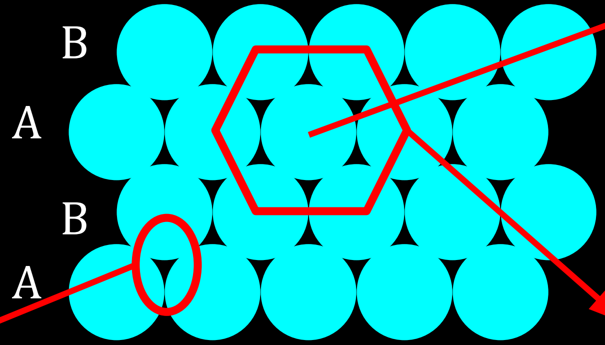
2-D hexagonal close packing

ABAB type of arrangement

12C01.3 Close Packed Structures

Less free space.
More efficient Packing

Contact with 6 neighbours.
coordination no. = 6



Triangular voids

2-D hexagonal close packing

ABAB type of arrangement

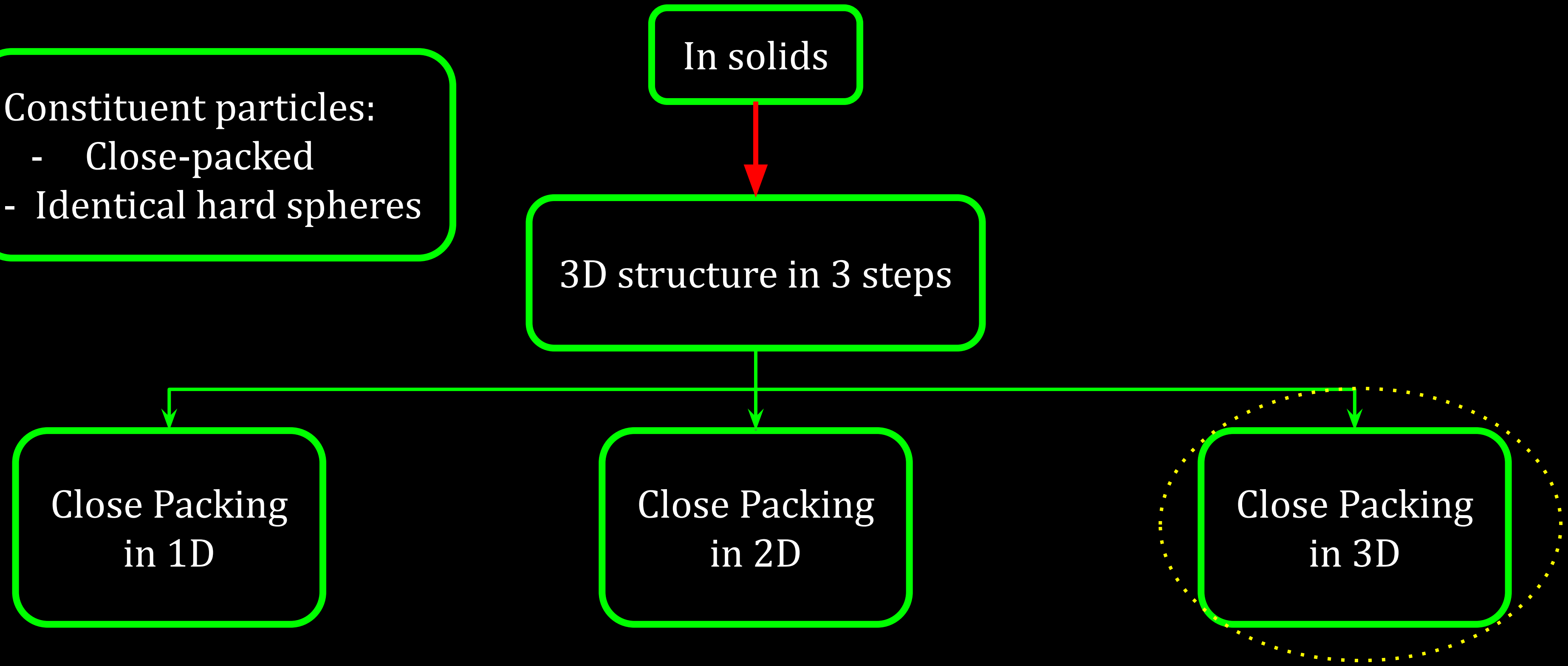
CV 2

Close Packing in 3D, Tetrahedral and Octahedral Voids, Hexagonal and Cubic Close Packing

12C01.3 Close Packed Structures

Close Packed Structures

Constituent particles:
- Close-packed
- Identical hard spheres



12C01.3 Close Packed Structures

Close Packing in 3D

12C01.3 Close Packed Structures

- Close Packing in 3D
  - 3D close packing from 2D S.C.P. layers
  - 3D close packing from 2D H.C.P layers

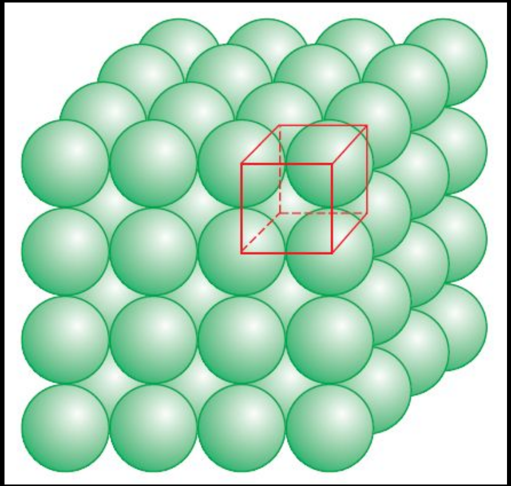
12C01.3 Close Packed Structures

Close Packing in 3D

- 3D close packing from 2D S.C.P. layers
- 3D close packing from 2D H.C.P layers

12C01.3 Close Packed Structures

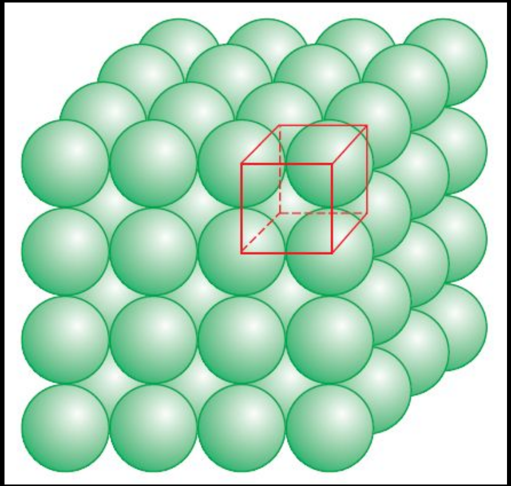
3D close packing from 2D S.C.P. layers



12C01.3 Close Packed Structures

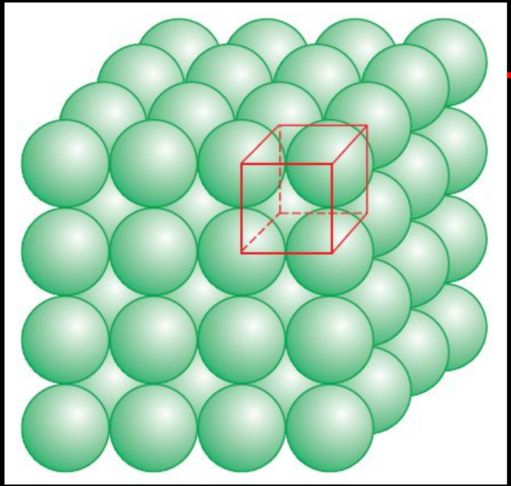
3D close packing from 2D S.C.P. layers

Perfectly aligned Spheres



12C01.3 Close Packed Structures

3D close packing from 2D S.C.P. layers

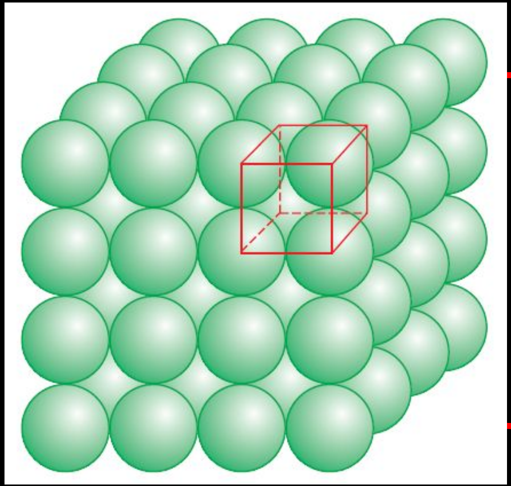


AAA.. type pattern

Perfectly aligned Spheres

12C01.3 Close Packed Structures

3D close packing from 2D S.C.P. layers

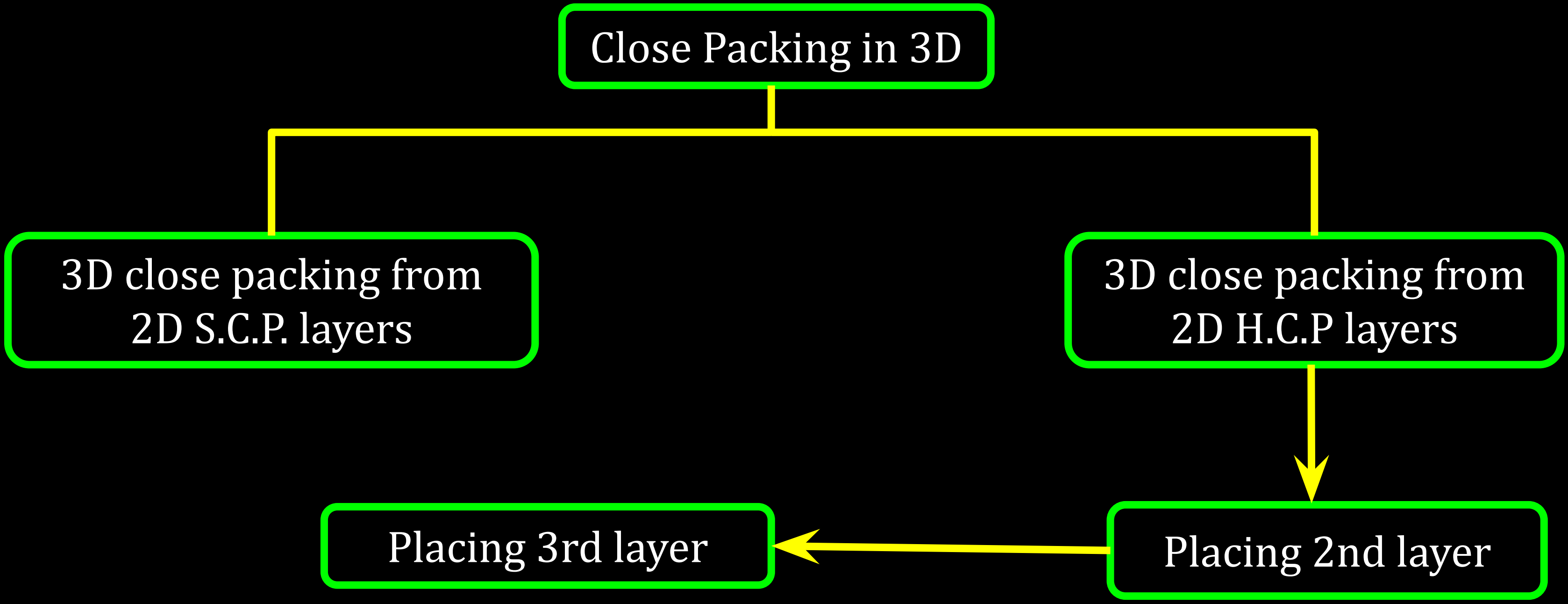


AAA.. type pattern

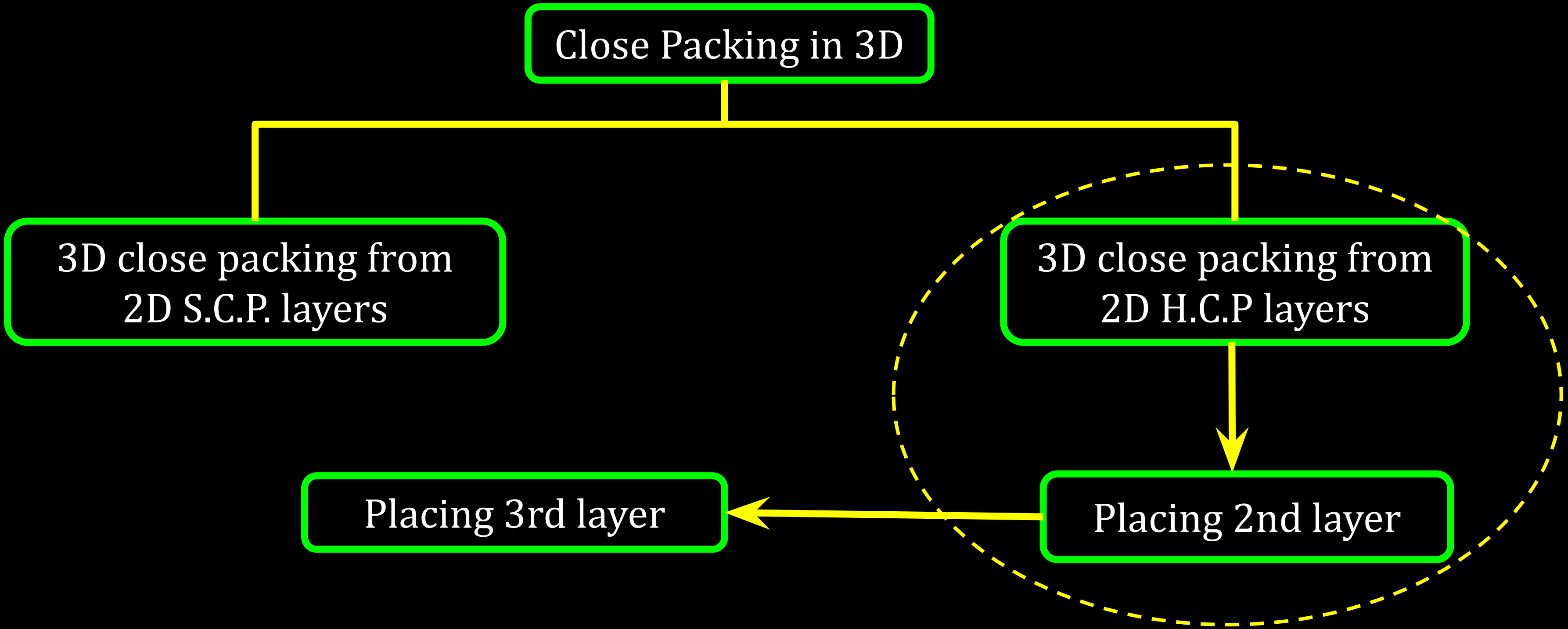
Perfectly aligned Spheres

Simple cubic lattice with primitive cubic unit cell

12C01.3 Close Packed Structures



12C01.3 Close Packed Structures

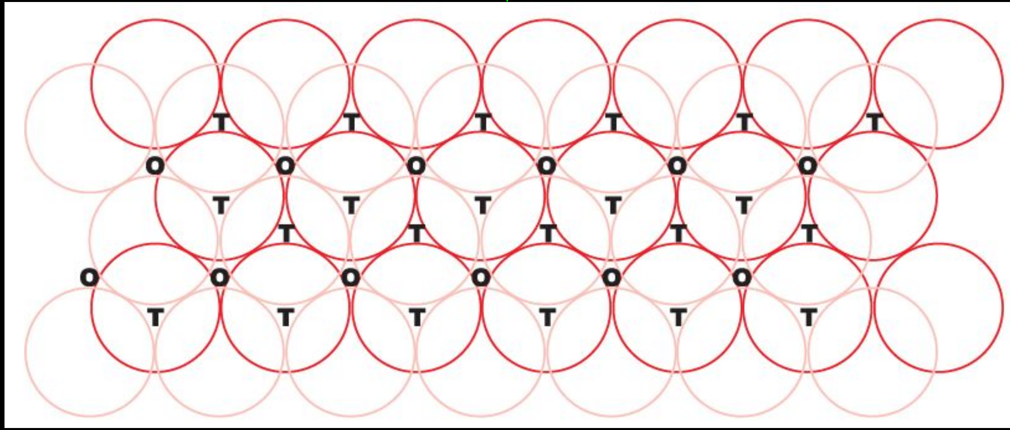


12C01.3 Close Packed Structures

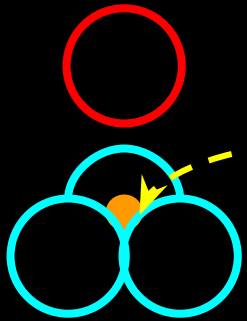
Placing 2nd layer → Tetrahedral and Octahedral voids are formed

12C01.3 Close Packed Structures

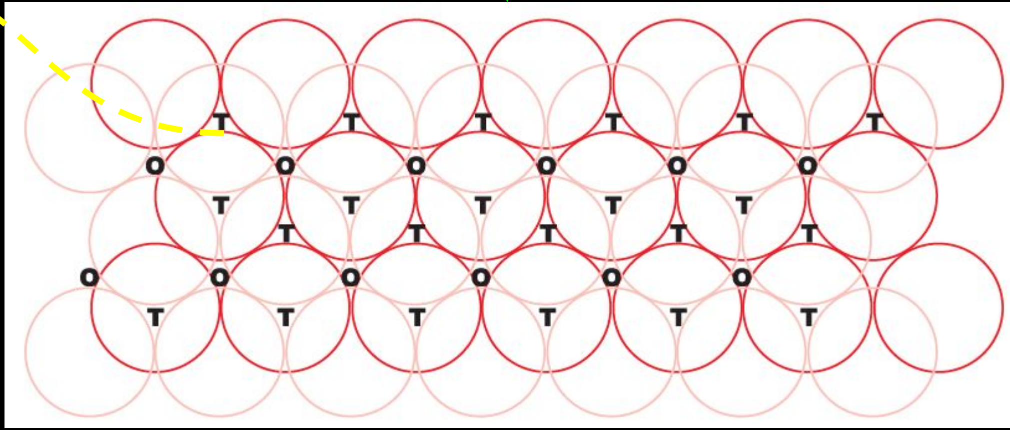
Tetrahedral and Octahedral voids



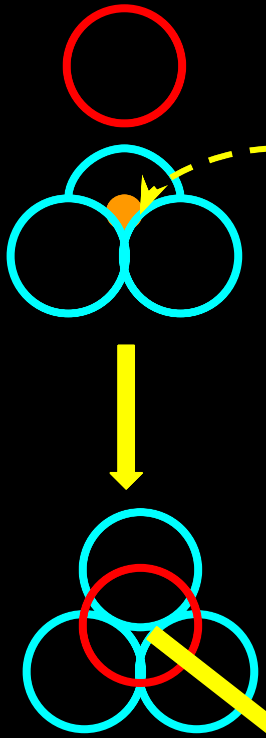
12C01.3 Close Packed Structures



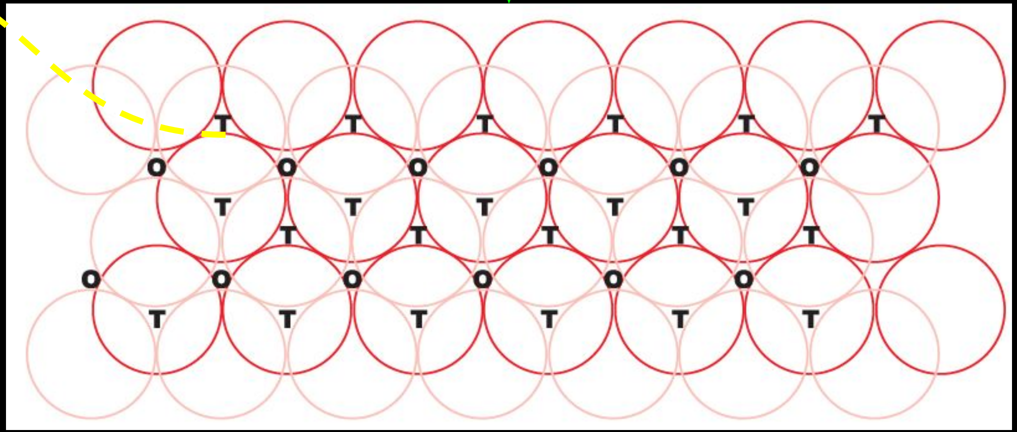
Tetrahedral and Octahedral voids



12C01.3 Close Packed Structures



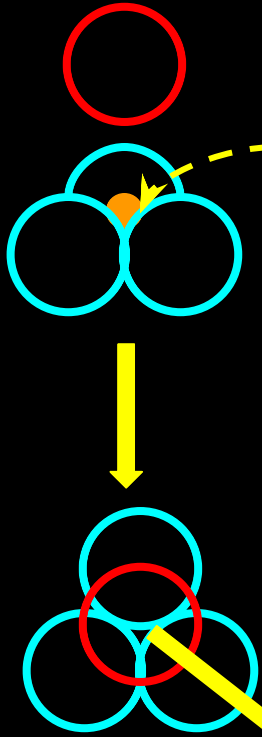
Tetrahedral and Octahedral voids

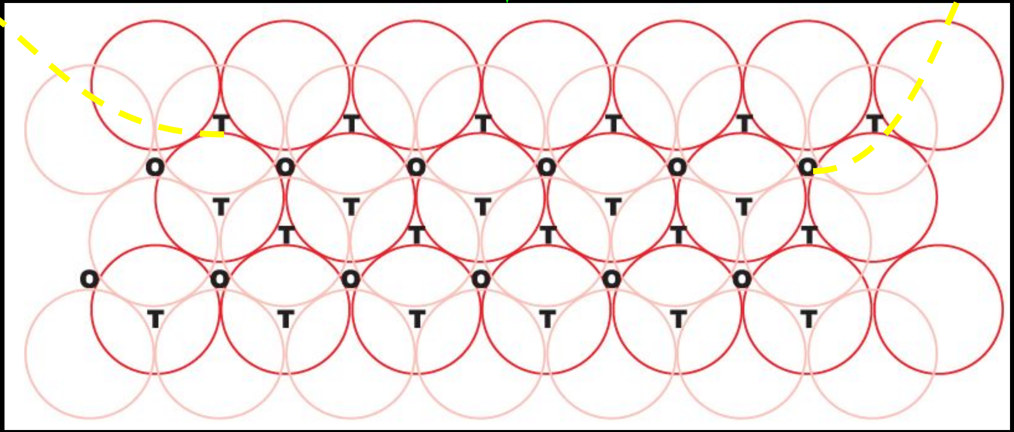


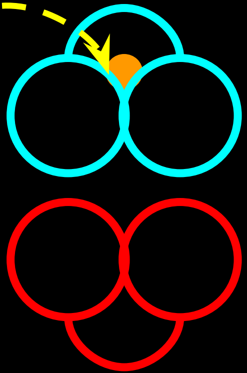
Tetrahedral void

12C01.3 Close Packed Structures

Tetrahedral and Octahedral voids



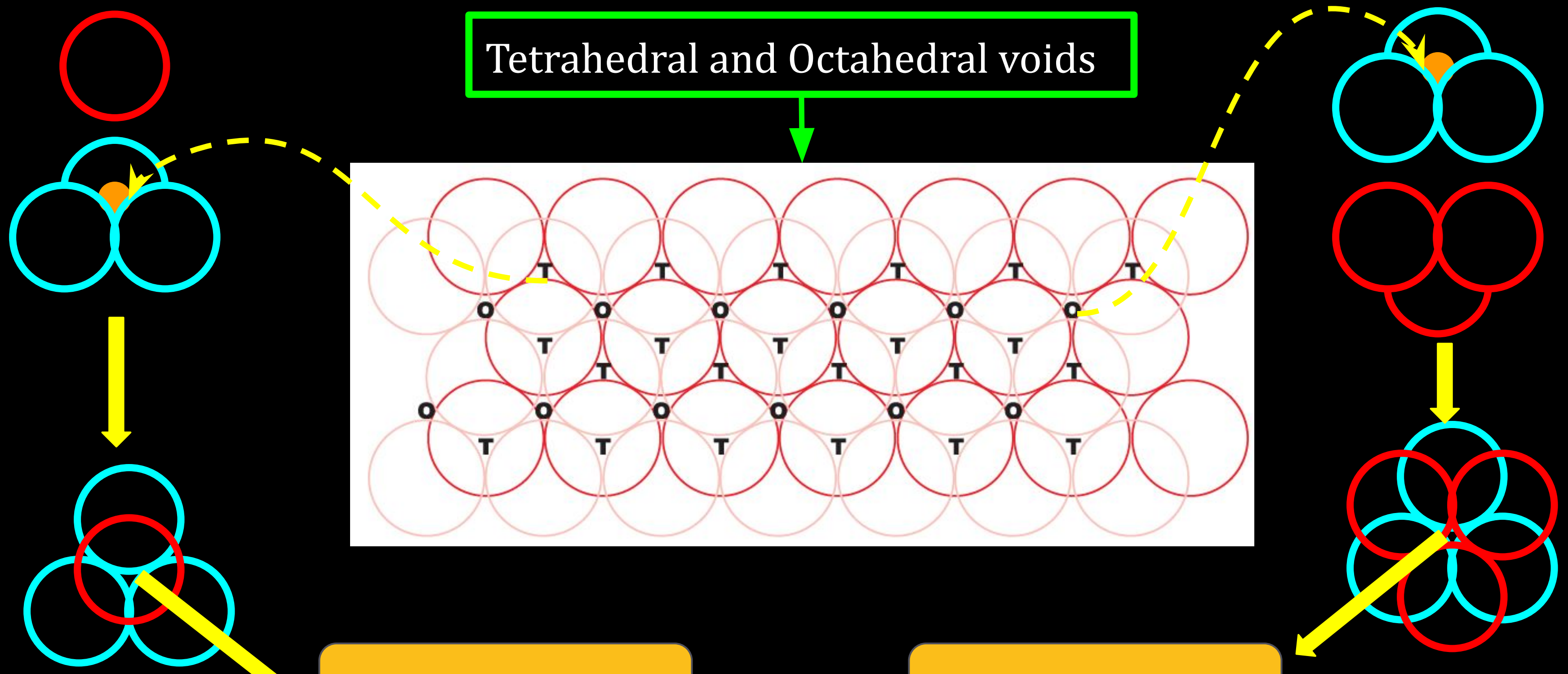




Tetrahedral void

12C01.3 Close Packed Structures

Tetrahedral and Octahedral voids



Tetrahedral void

Octahedral void

12C01.3 Close Packed Structures

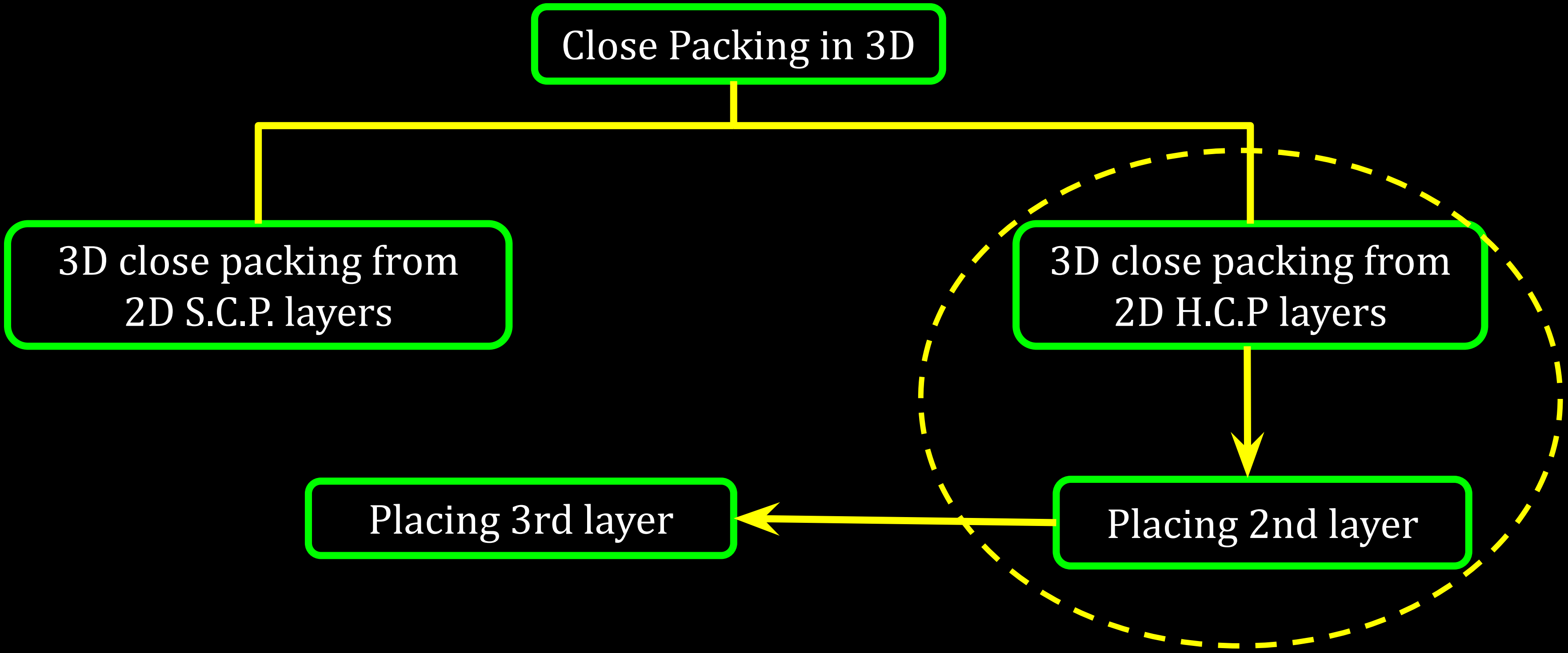
Number of voids

If the number of close packed spheres be N

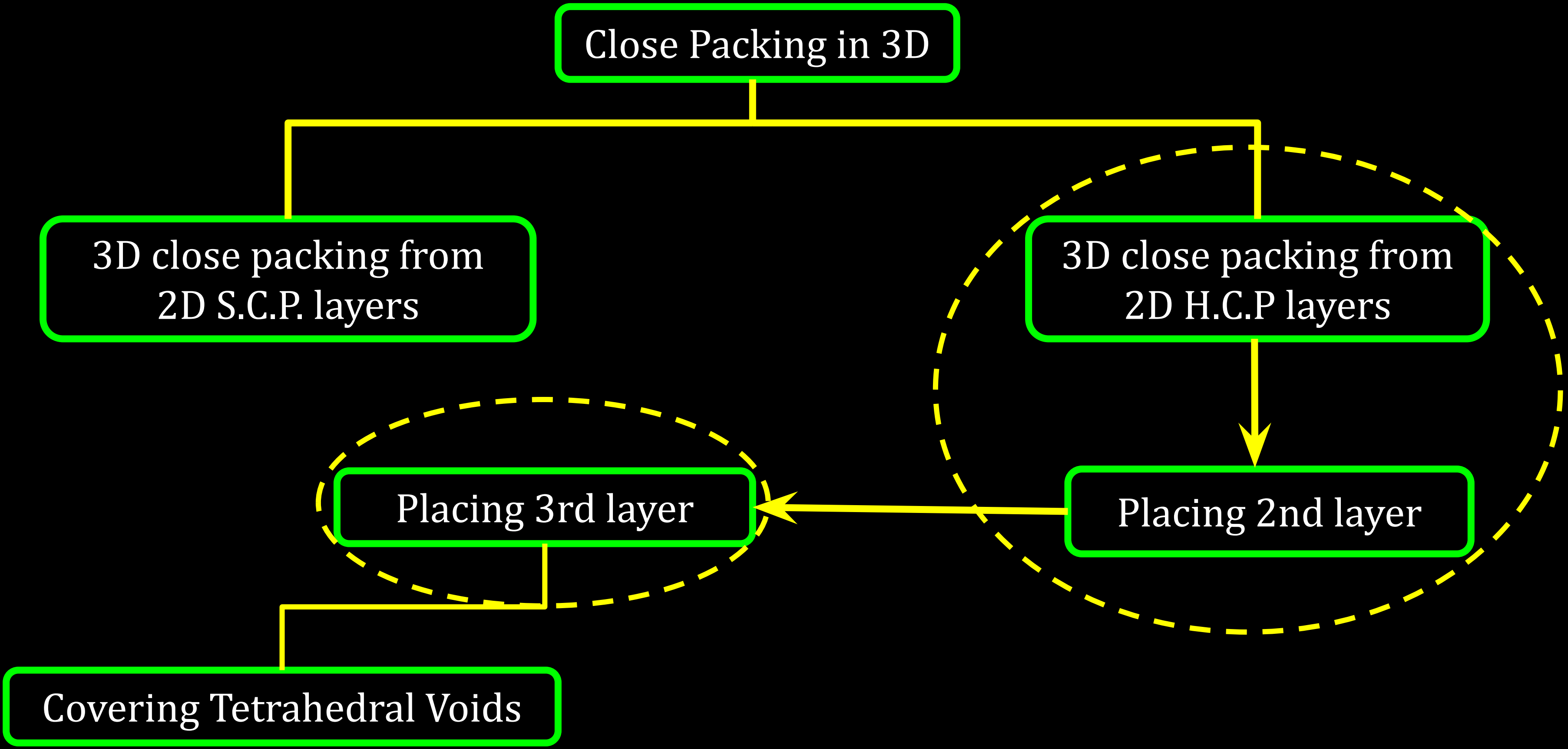
Number of octahedral voids $= N$

Number of tetrahedral voids $= 2N$

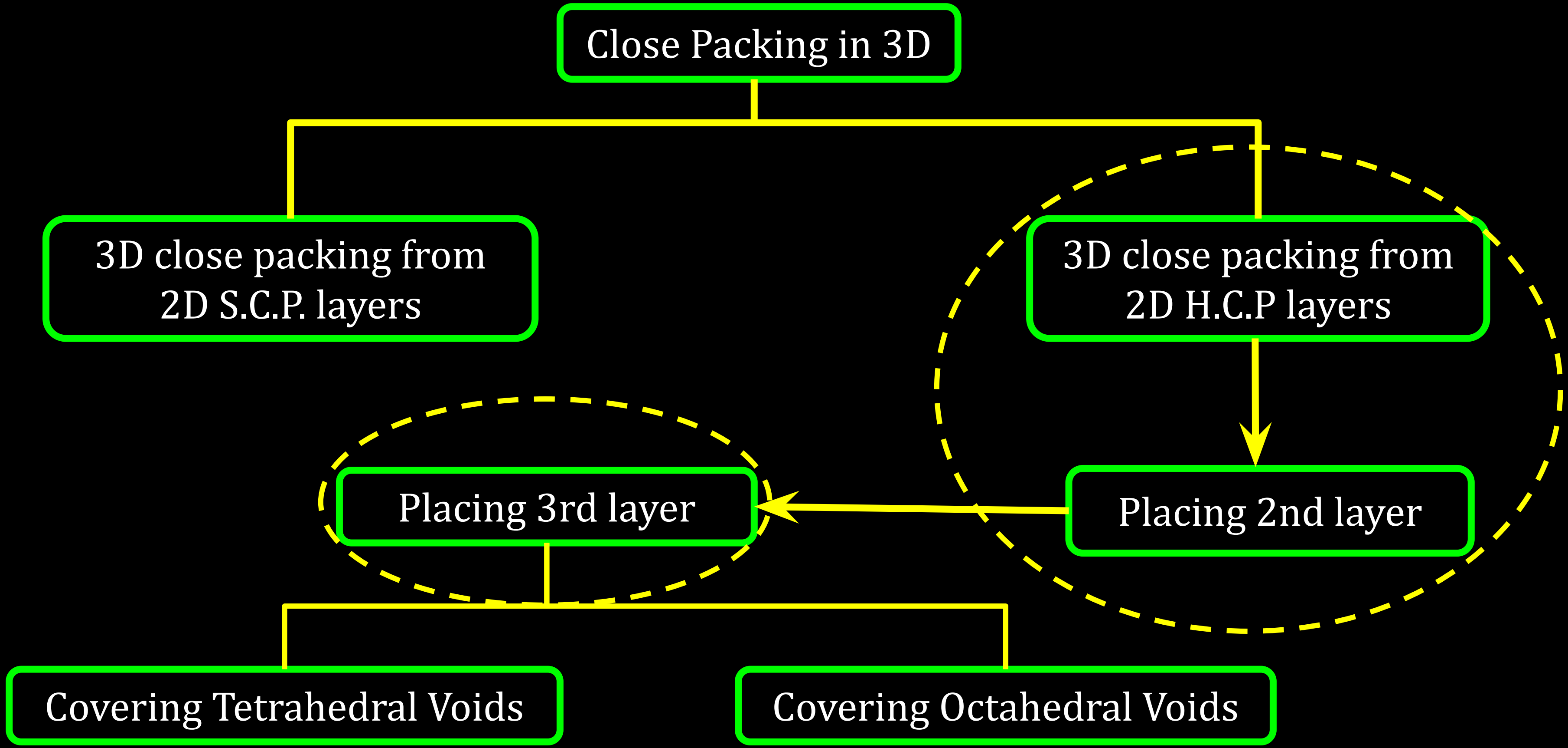
12C01.3 Close Packed Structures



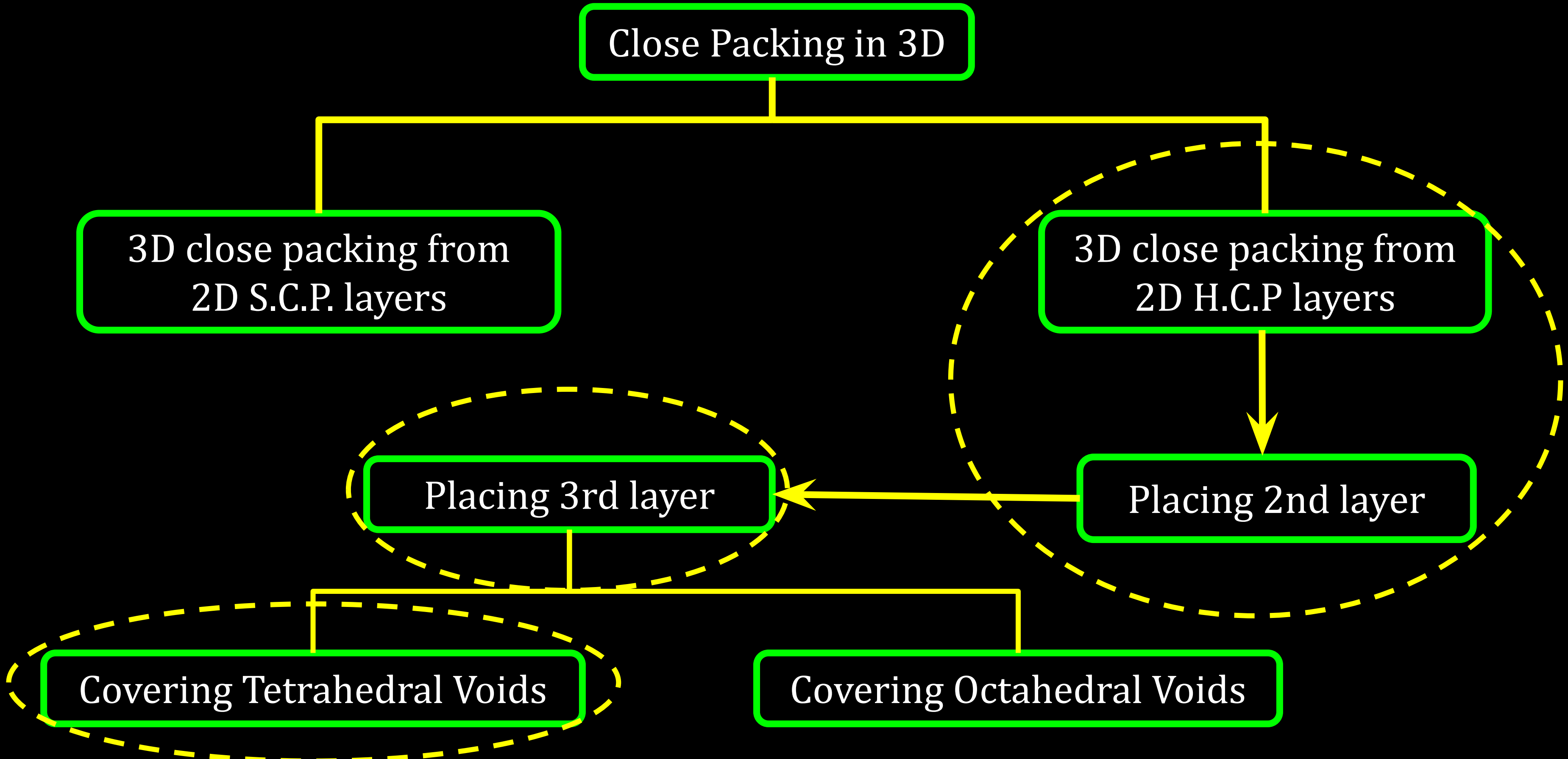
12C01.3 Close Packed Structures



12C01.3 Close Packed Structures



12C01.3 Close Packed Structures



12C01.3 Close Packed Structures

Covering Tetrahedral Voids

12C01.3 Close Packed Structures

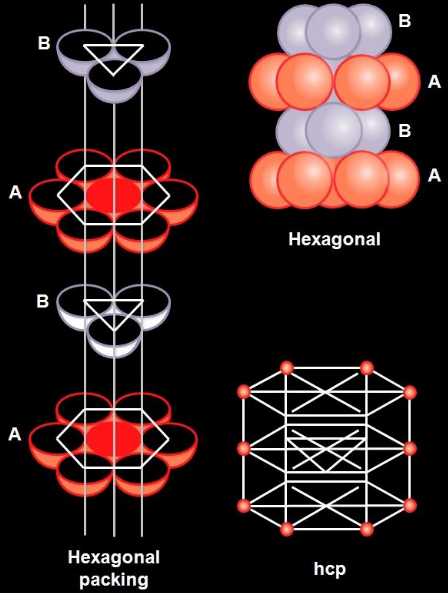
Covering Tetrahedral Voids

Tetrahedral voids of the 2nd layer covered by the spheres of the 3rd layer

12C01.3 Close Packed Structures

Covering Tetrahedral Voids

Tetrahedral voids of the 2nd layer covered by the spheres of the 3rd layer

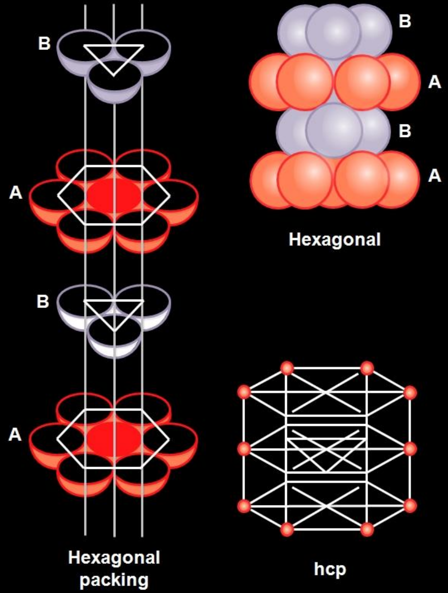


12C01.3 Close Packed Structures

Covering Tetrahedral Voids

Tetrahedral voids of the 2nd layer covered by the spheres of the 3rd layer

ABAB Pattern

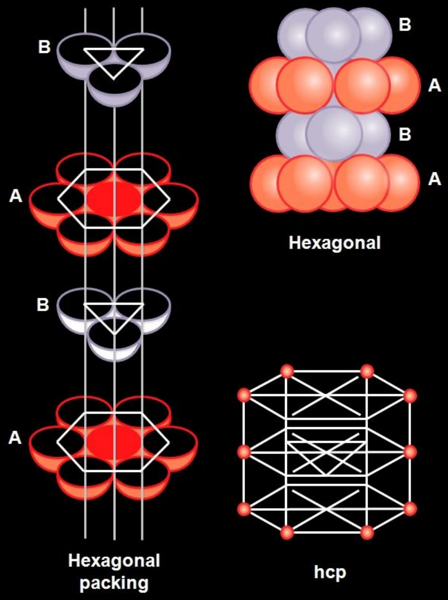


12C01.3 Close Packed Structures

Covering Tetrahedral Voids

Tetrahedral voids of the 2nd layer covered by the spheres of the 3rd layer

ABAB Pattern



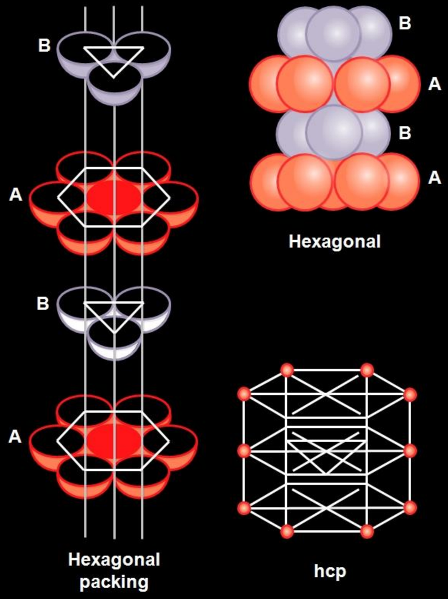
Hexagonal close packed structure (hcp)

12C01.3 Close Packed Structures

Covering Tetrahedral Voids

Tetrahedral voids of the 2nd layer covered by the spheres of the 3rd layer

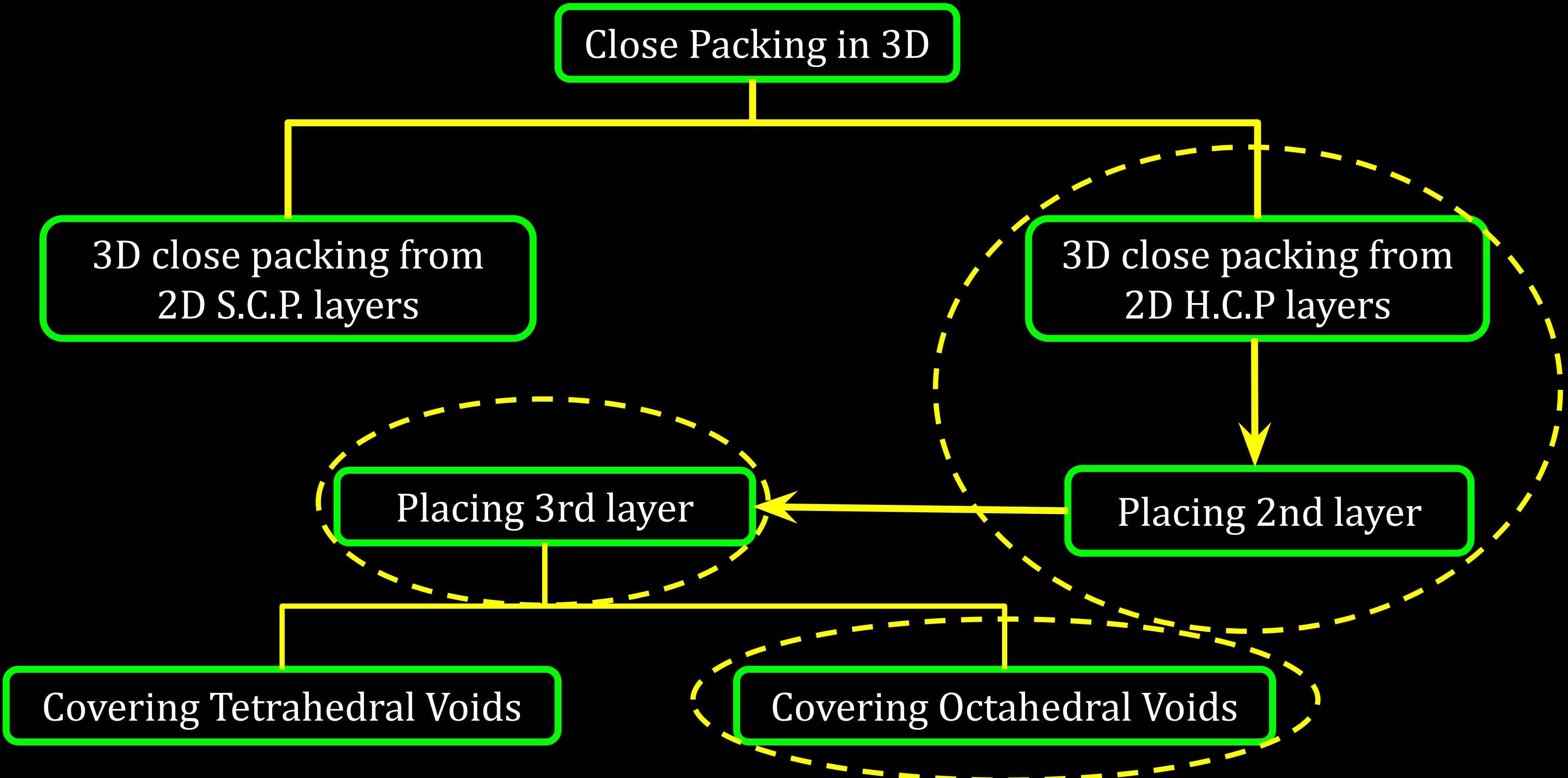
ABAB Pattern



Hexagonal close packed structure (hcp)

Ex. - Mg , Zn

12C01.3 Close Packed Structures



Covering Octahedral Voids

12C01.3 Close Packed Structures

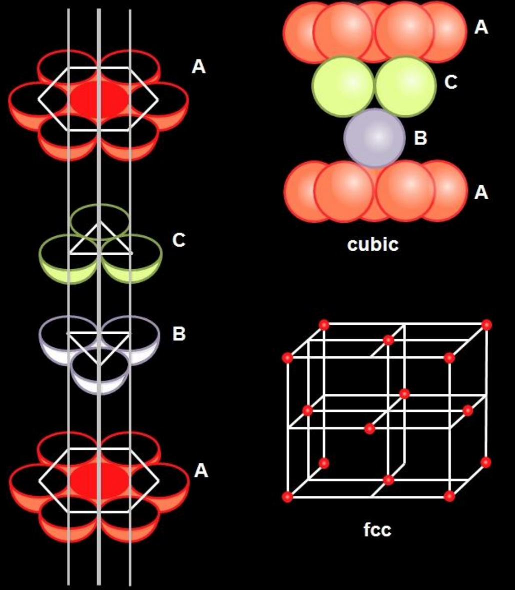
Covering Octahedral Voids

Octahedral voids of the 2nd layer covered by the spheres of the 3rd layer

12C01.3 Close Packed Structures

Covering Octahedral Voids

Octahedral voids of the 2nd layer covered by the spheres of the 3rd layer

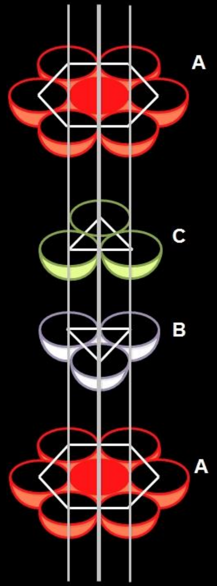


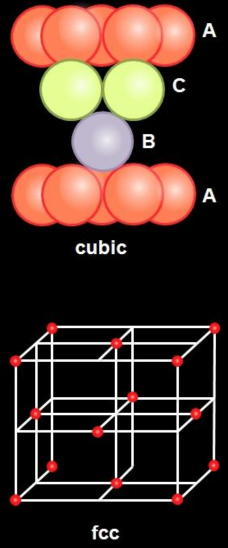
12C01.3 Close Packed Structures

Covering Octahedral Voids

Octahedral voids of the 2nd layer covered by the spheres of the 3rd layer

ABCABC pattern



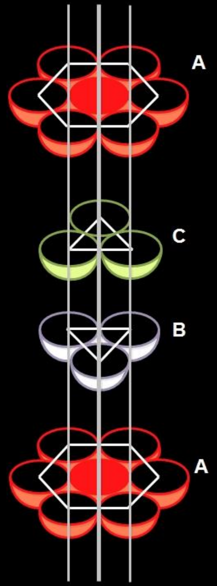


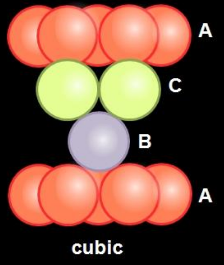
12C01.3 Close Packed Structures

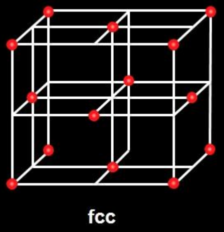
Covering Octahedral Voids

Octahedral voids of the 2nd layer covered by the spheres of the 3rd layer

ABCABC pattern







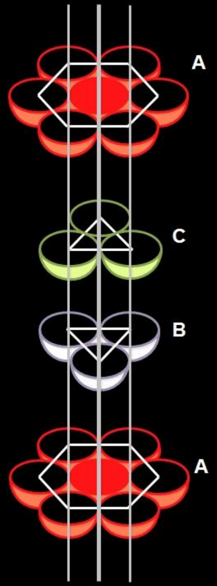
Cubic close packed (ccp) or face centred cubic (fcc) structure

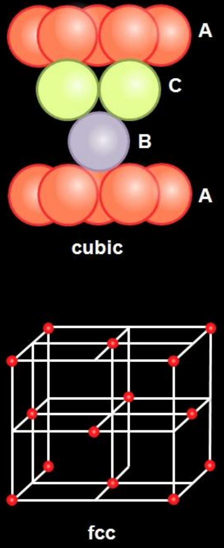
12C01.3 Close Packed Structures

Covering Octahedral Voids

Octahedral voids of the 2nd layer covered by the spheres of the 3rd layer

ABCABC pattern





Cubic close packed (ccp) or face centred cubic (fcc) structure

Ex. - Cu ,Ag

CV 3

Formula of a Compound and Number of Voids Filled, Locating Tetrahedral and Octahedral Voids

12C01.3 Close Packed Structures

Positions of Ions
in Ionic Solids

12C01.3 Close Packed Structures

Bigger ions (usually anions) form the close packed structure and the smaller ions (usually cations) occupy the voids

Positions of Ions in Ionic Solids

12C01.3 Close Packed Structures

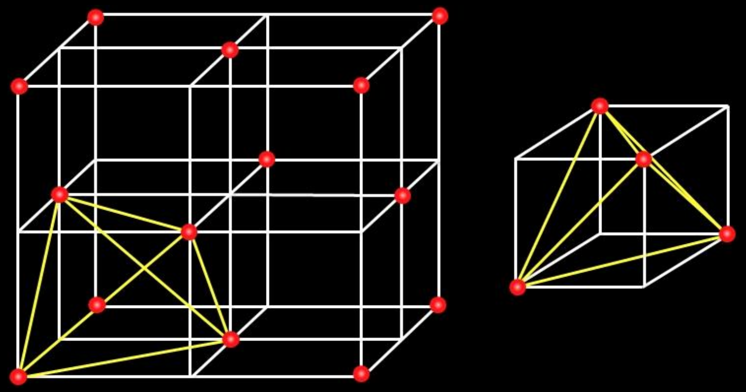
Bigger ions (usually anions) form the close packed structure and the smaller ions (usually cations) occupy the voids

Positions of Ions in Ionic Solids

Fraction of octahedral or tetrahedral voids that are occupied, depends upon the chemical formula of the compound

12C01.3 Close Packed Structures

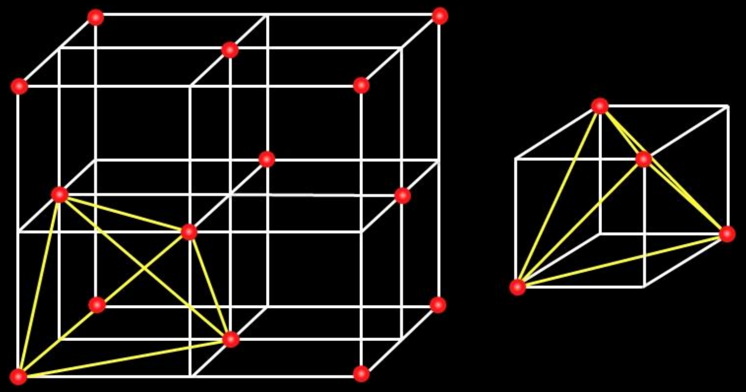
Locating Tetrahedral Voids



ccp or fcc unit cell

12C01.3 Close Packed Structures

Fcc unit cell → 8 small cubes → Each small cube has 4 atoms



ccp or fcc unit cell

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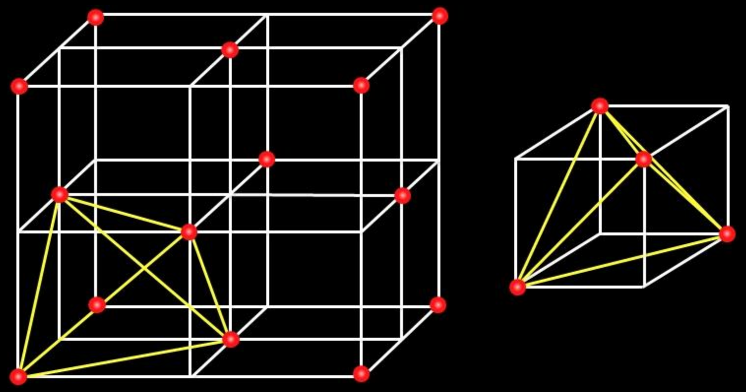
Regular tetrahedron



ccp or fcc unit cell

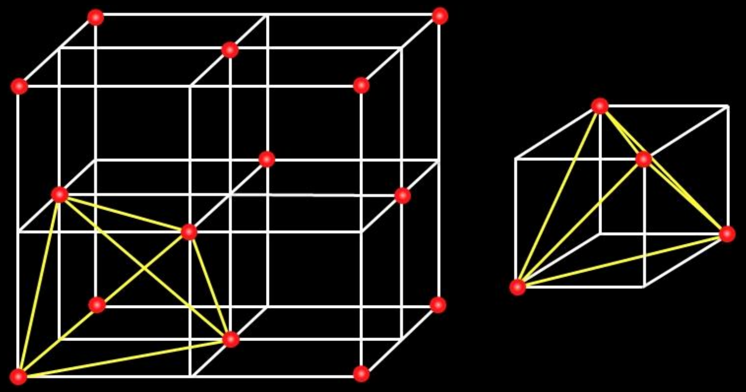
12C01.3 Close Packed Structures

Fcc unit cell → 8 small cubes → Each small cube has 4 atoms

Regular tetrahedron

Tetrahedral void in each small cube = 1

Tetrahedral voids in unit cell = 8



ccp or fcc unit cell

12C01.3 Close Packed Structures

Fcc unit cell → 8 small cubes → Each small cube has 4 atoms

Tetrahedral void in each small cube = 1

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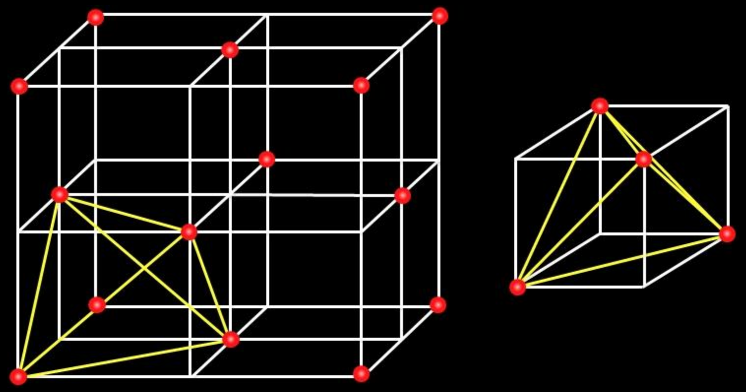


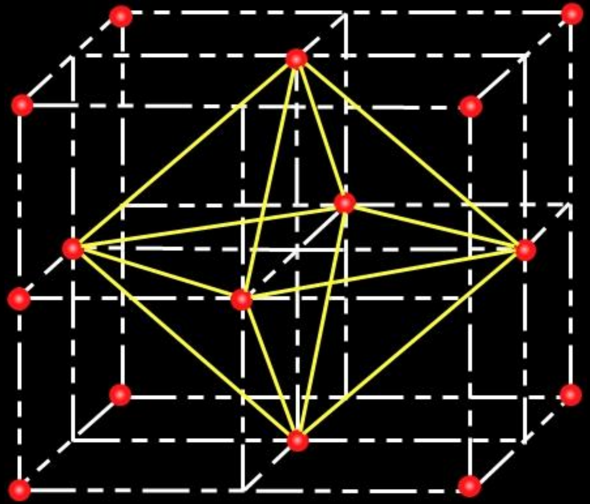
Regular tetrahedron

No. of atoms = 4

No. of Tetrahedral voids = 8

ccp or fcc unit cell

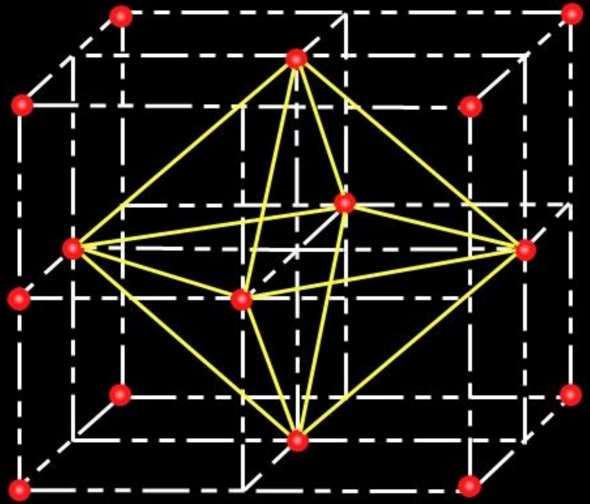
Locating Octahedral Voids



ccp or fcc unit cell

12C01.3 Close Packed Structures

Body centre, C is surrounded by 6 atoms on face centres

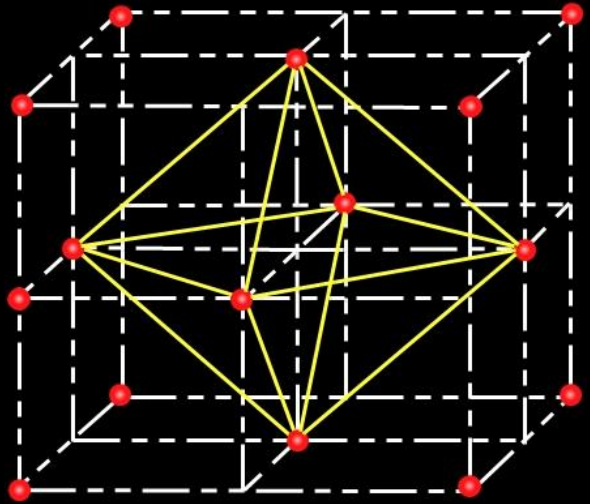


ccp or fcc unit cell

12C01.3 Close Packed Structures

Body centre, C is surrounded by 6 atoms on face centres

Octahedron

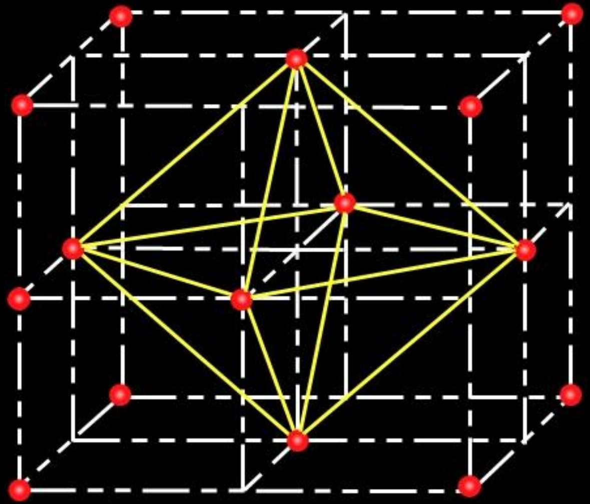


ccp or fcc unit cell

Body centre, C is surrounded by 6 atoms on face centres

Octahedron

Octahedral void at body centre of cube = 1
Octahedral voids at centre of all the 12 edges = 12
Total Voids= 12*¼ + 1 = 4
Total atoms in ccp unit cell = 4 = Total octahedral voids



ccp or fcc unit cell

Conceptest

Ready for Challenge

12C01.3 Close Packed Structures

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution : ? (NCERT Example 1.1 Page no. 18)

12C01.3 Close Packed Structures

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution : ?

(NCERT Example 1.1 Page no. 18)

Pause the Video

Time duration 2 minute

12C01.3 Close Packed Structures

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution :

Compound is formed by elements X and Y

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution :

Compound is formed by elements X and Y

Atoms of the element Y (anions) make ccp

Atoms of Y = 4

12C01.3 Close Packed Structures

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution :

Compound is formed by elements X and Y

Atoms of the element Y (anions) make ccp

Atoms of Y = 4

element X (cations) occupy all octahedral voids

Atoms of X = 4

12C01.3 Close Packed Structures

Question : A compound is formed by two elements X and Y. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids. What is the formula of the compound?

Solution :

Compound is formed by elements X and Y

Atoms of the element Y (anions) make ccp

Atoms of Y = 4

element X (cations) occupy all octahedral voids

Atoms of X = 4

X : Y = 1 : 1

Formula of compound = XY

12C01.3 Close Packed Structures

: Reference Questions :

Intext Question : 1.14, 1.16 (NCERT page no. 23)

Exercise Question : 1.7, 1.19 (NCERT page no. 33)

Work Book Question : 10, 15

12C01.4

Packing Efficiency and Crystal Density

12C01.4 Packing Efficiency and Crystal Density

- Packing Efficiency in hcp and ccp Structures

12C01.4 Packing Efficiency and Crystal Density

- Packing Efficiency in hcp and ccp Structures
- Packing Efficiency in bcc Structures
- Packing Efficiency in Simple Cubic Lattice

12C01.4 Packing Efficiency and Crystal Density

- Packing Efficiency in hcp and ccp Structures
- Packing Efficiency in bcc Structures
- Packing Efficiency in Simple Cubic Lattice
- Crystal Density

CV 1

Packing Efficiency and Coordination Number

12C01.4 Packing Efficiency and Crystal Density

$$\text{Packing Efficiency} = \frac{\text{Volume occupied by atoms in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

12C01.4 Packing Efficiency and Crystal Density

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- P.E. in bcc Structures
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12C01.4 Packing Efficiency and Crystal Density

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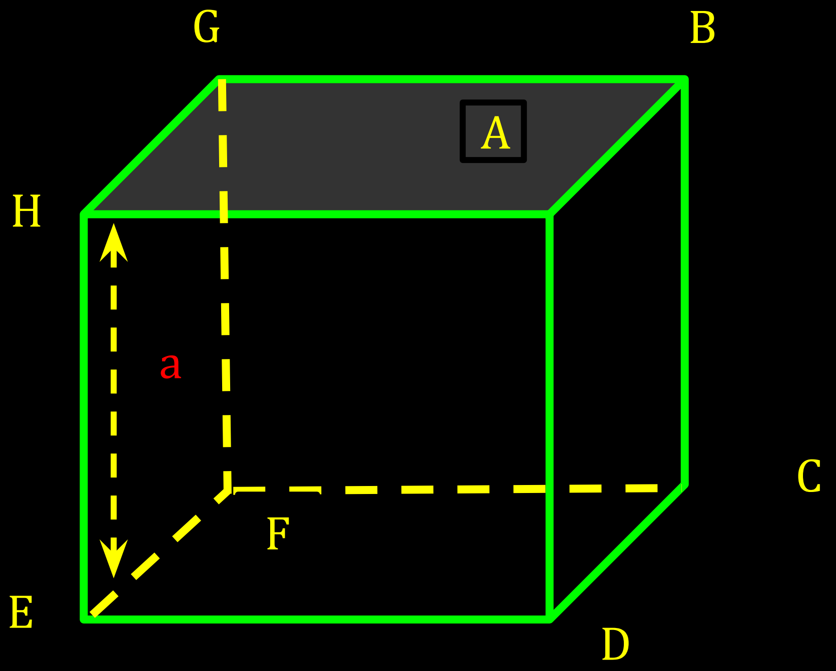
- P.E. in bcc Structures
- P.E. in Simple Cubic Lattice

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

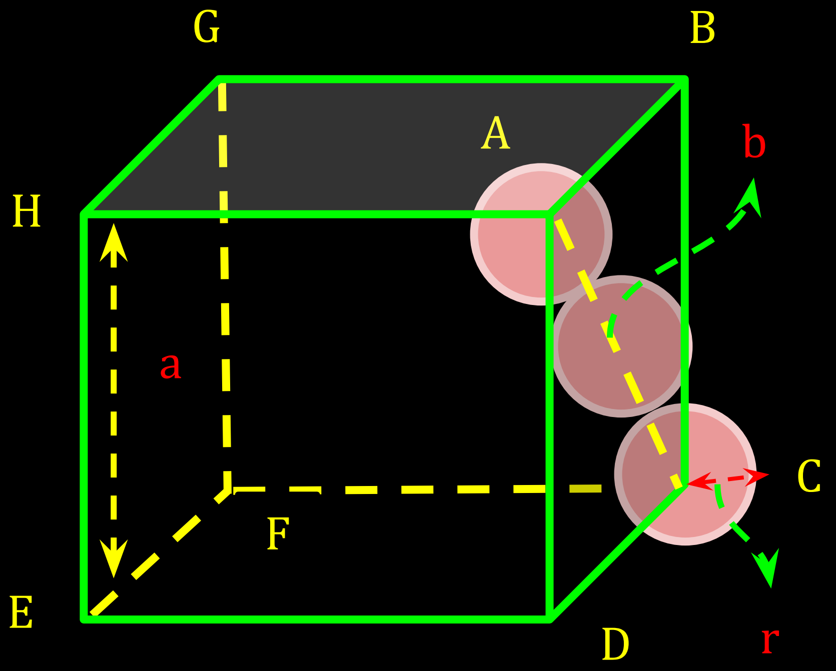
12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures



12C01.4 Packing Efficiency and Crystal Density

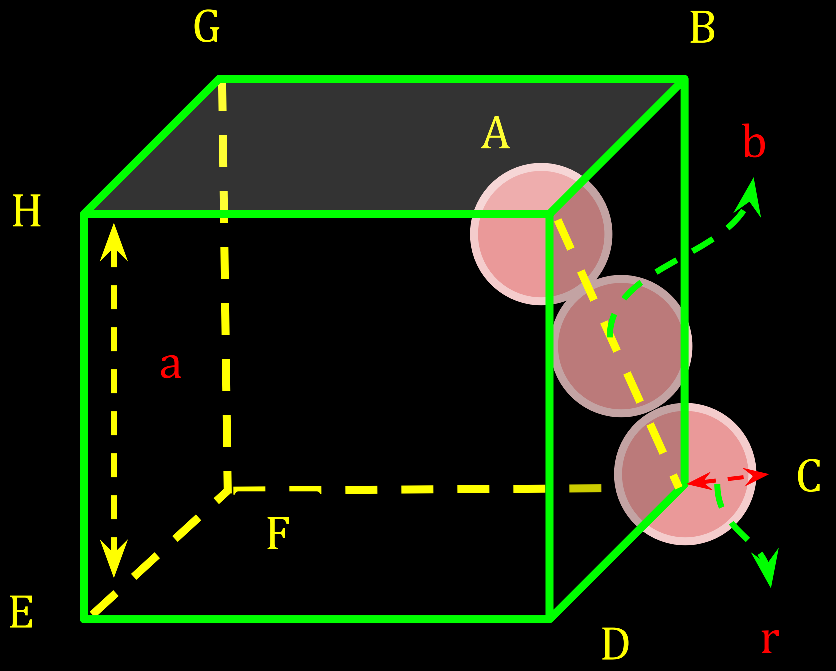
Packing Efficiency in hcp and ccp Structures



12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12

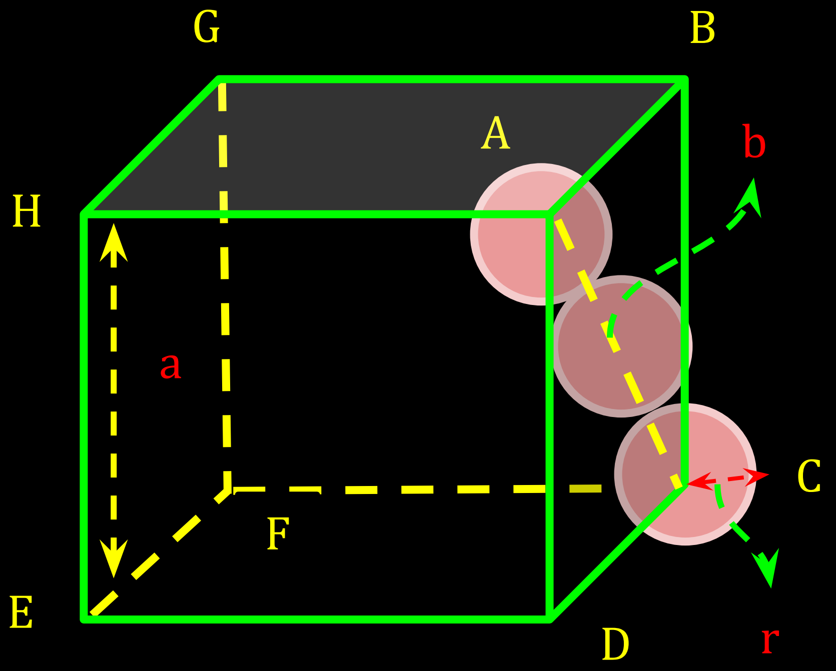


12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12

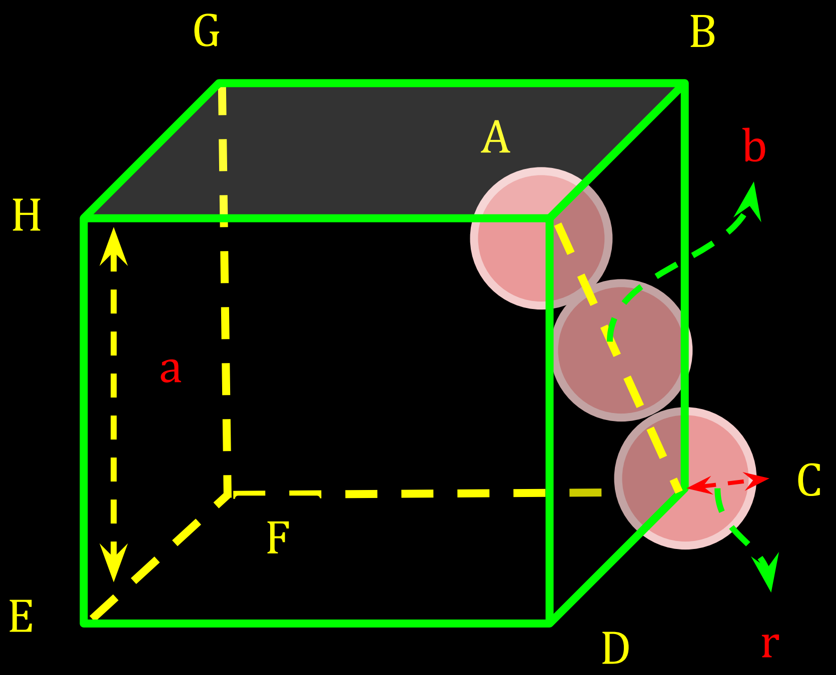
Relation b/w edge length (a) and radius(r)



12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



Relation b/w edge length (a) and radius(r)

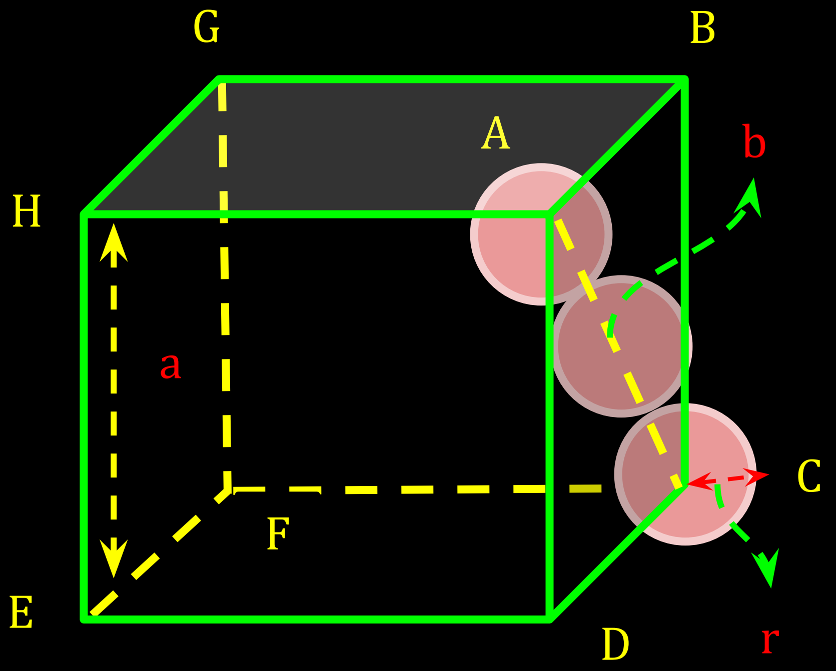
In Δ ABC, $AC^2 = b^2 = BC^2 + AB^2$

$b^2 = a^2 + a^2 = 2a^2$ or $b = \sqrt{2}\,a$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



Relation b/w edge length (a) and radius(r)

In Δ ABC, $AC^2 = b^2 = BC^2 + AB^2$

$b^2 = a^2 + a^2 = 2a^2$ or $b = \sqrt{2}\,a$

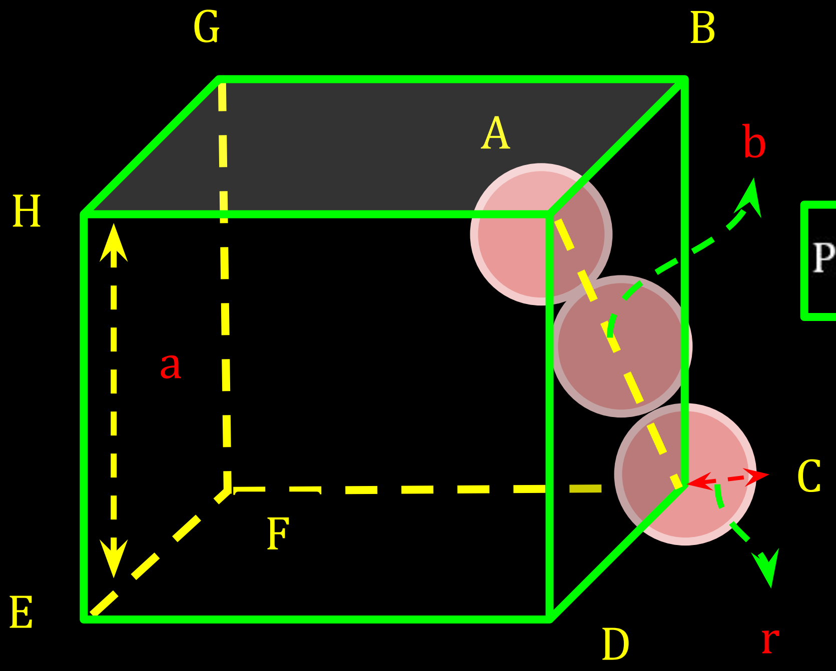
We know, $b = 4r$, thus, $\sqrt{2}\,a = 4r$

$a = 2\sqrt{2}\,r$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



Relation b/w edge length (a) and radius(r)

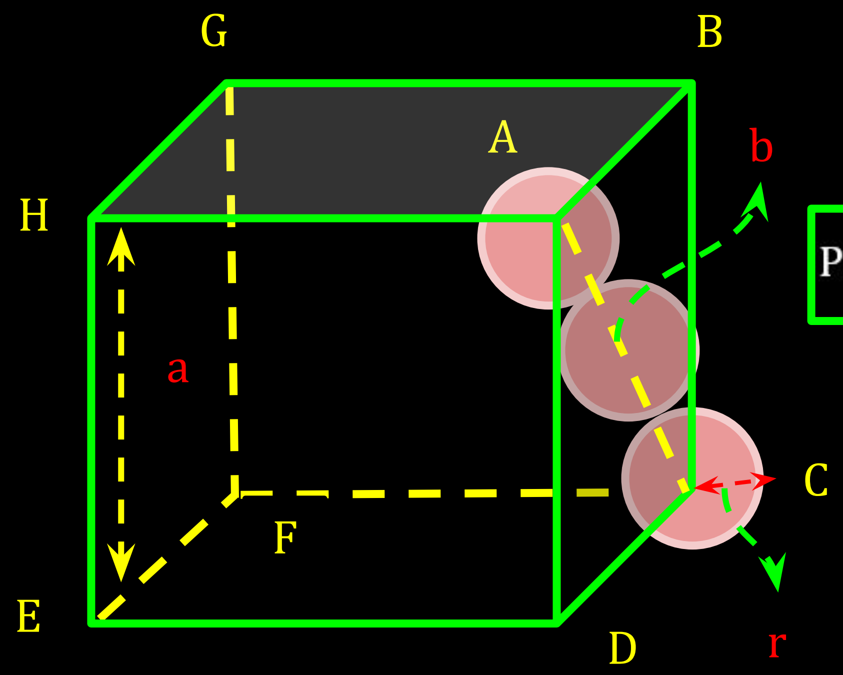
$$a = 2\sqrt{2}\,r$$

$$P.E = \frac{\text{Volume occupied by 4 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



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$$a = 2\sqrt{2}\ r$$

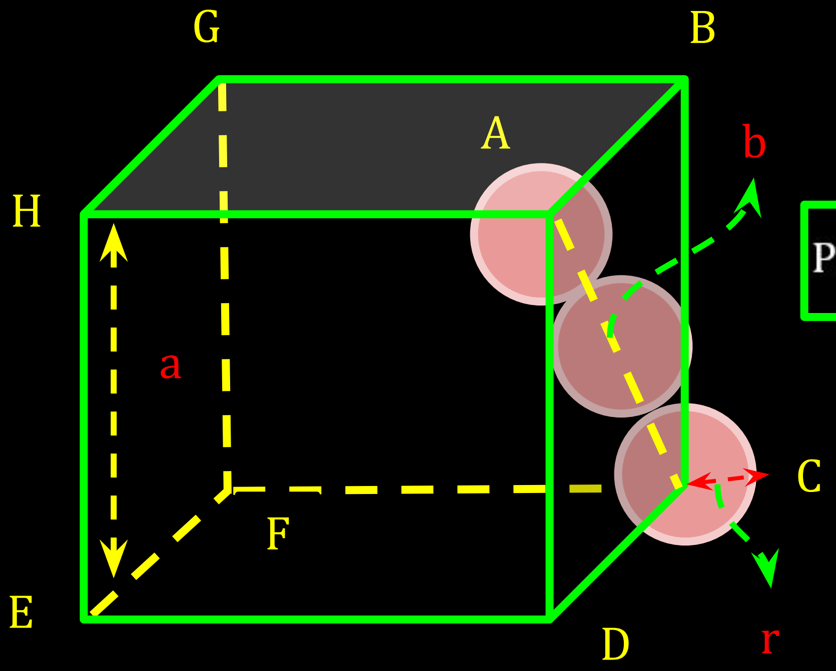
$$P.E = \frac{\text{Volume occupied by 4 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{4 \times (4/3)\pi r^3 \times 100}{(2\sqrt{2}\ r)^3}$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



Relation b/w edge length (a) and radius(r)

$$a = 2\sqrt{2}\, r$$

$$P.E = \frac{\text{Volume occupied by 4 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{4 \times (4/3)\pi r^3 \times 100}{(2\sqrt{2}\, r)^3}$$

$$P.E = 74\%$$

CV 2

Packing Efficiency in bcc Structures

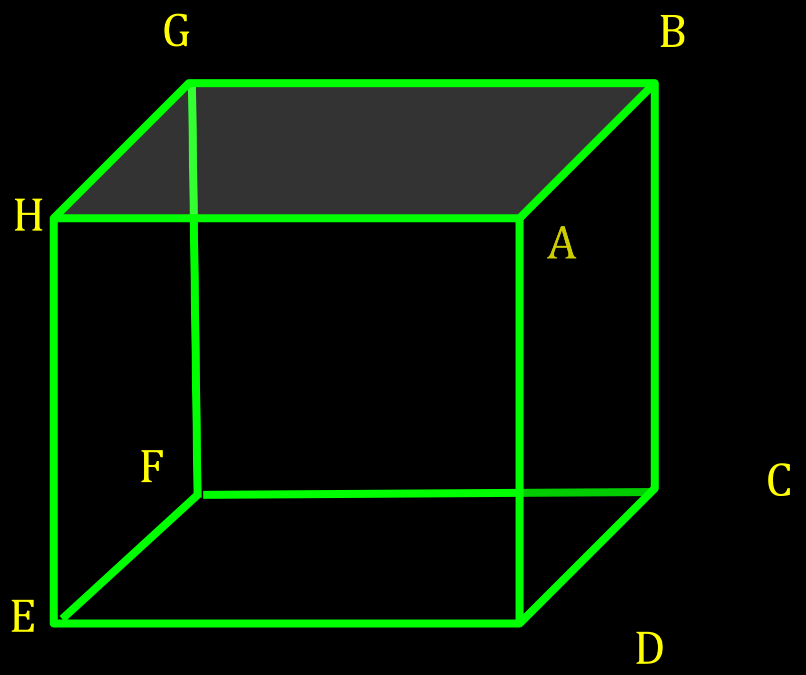
12C01.4 Packing Efficiency and Crystal Density

$$\text{Packing Efficiency} = \frac{\text{Volume occupied by atoms in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

- P.E. in hcp and ccp Structures
- P.E. in bcc Structures
- P.E. in Simple Cubic Lattice

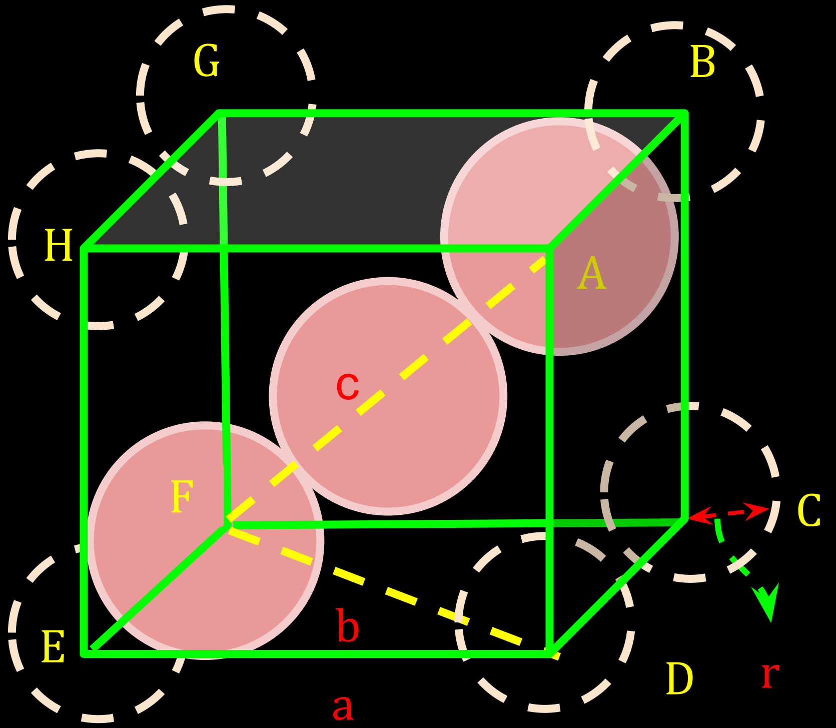
12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures



12C01.4 Packing Efficiency and Crystal Density

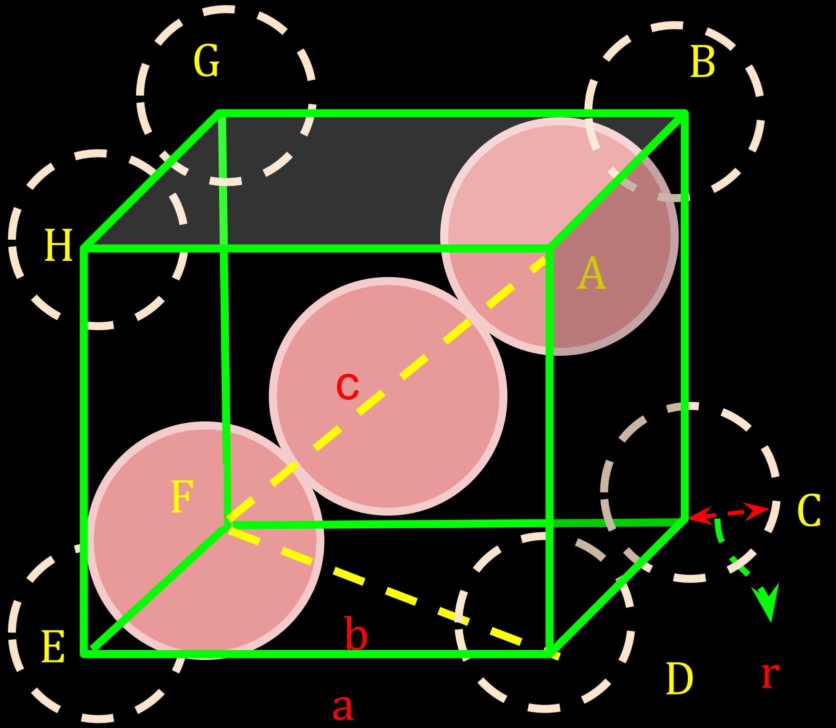
Packing Efficiency in bcc Structures



12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

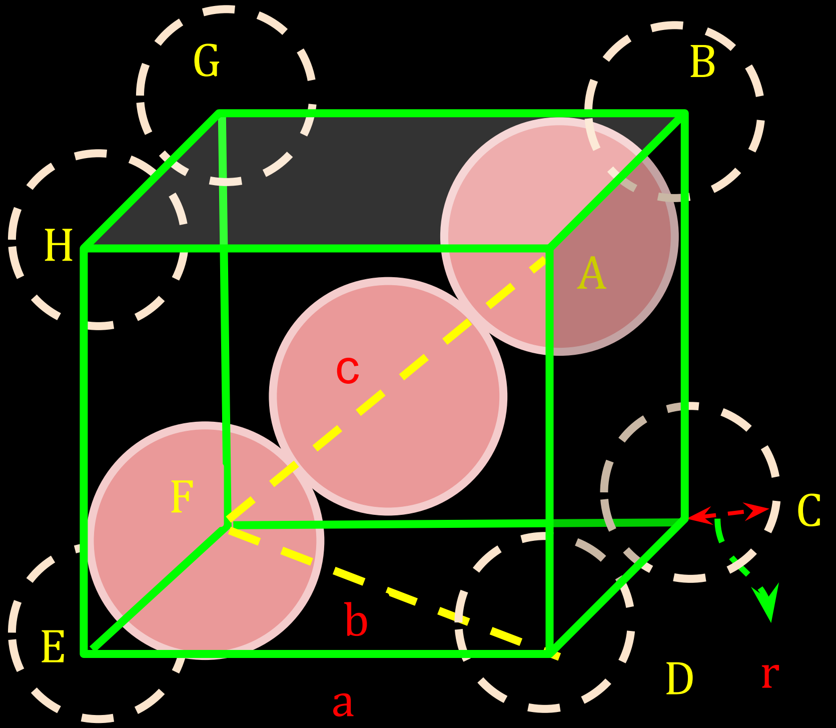
Coordination Number = 8



12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8

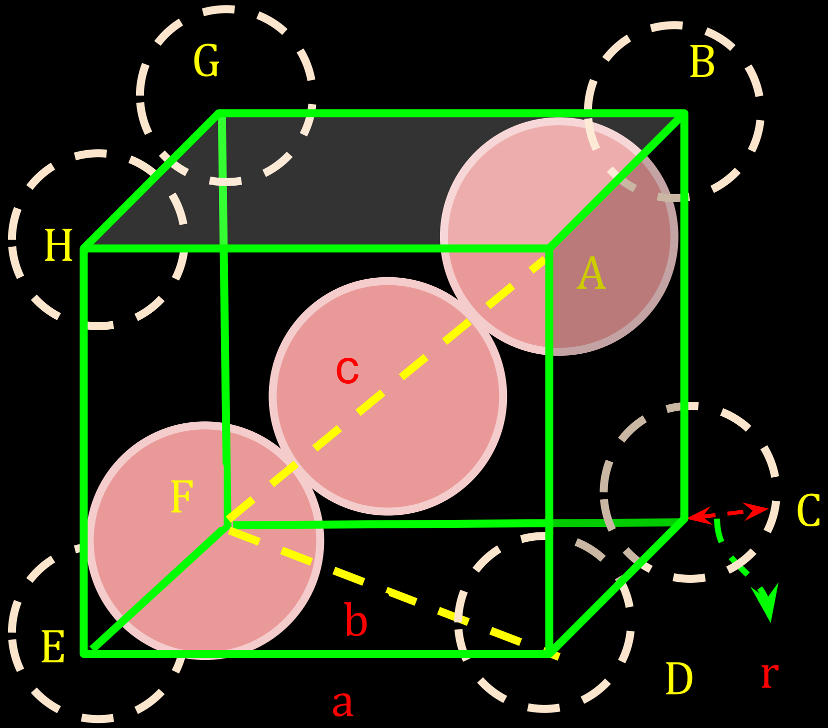


Relation b/w edge length (a) and radius(r)

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8



Relation b/w edge length (a) and radius(r)

In Δ EFD, $b^2 = a^2 + a^2 = 2a^2$, thus, $b = \sqrt{2}\ a$

Now in Δ AFD

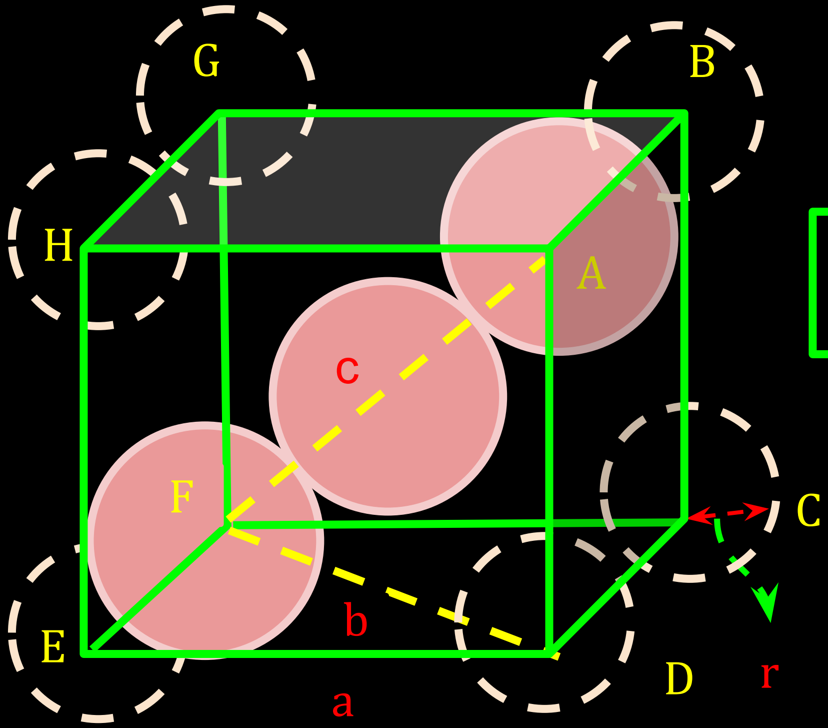
$c^2 = a^2 + b^2 = a^2 + 2a^2 = 3a^2$, thus, $c = \sqrt{3}a$

Now, $\sqrt{3}a = 4r$, thus, $a = 4r/\sqrt{3}$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8



Relation b/w edge length (a) and radius(r)

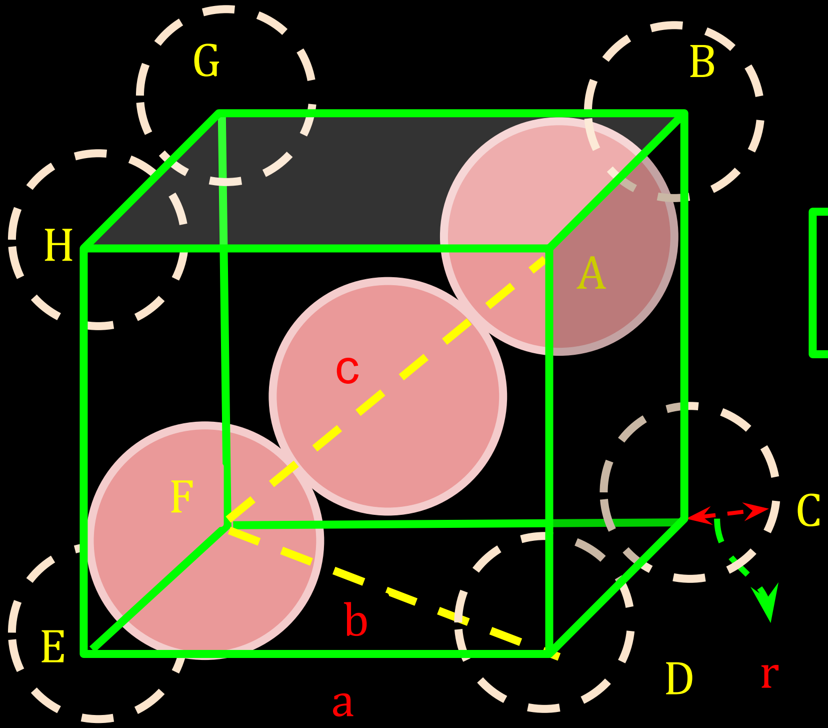
$a = 4r/\sqrt{3}$

$$P.E = \frac{\text{Volume occupied by 2 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8



Relation b/w edge length (a) and radius(r)

$a = 4r/\sqrt{3}$

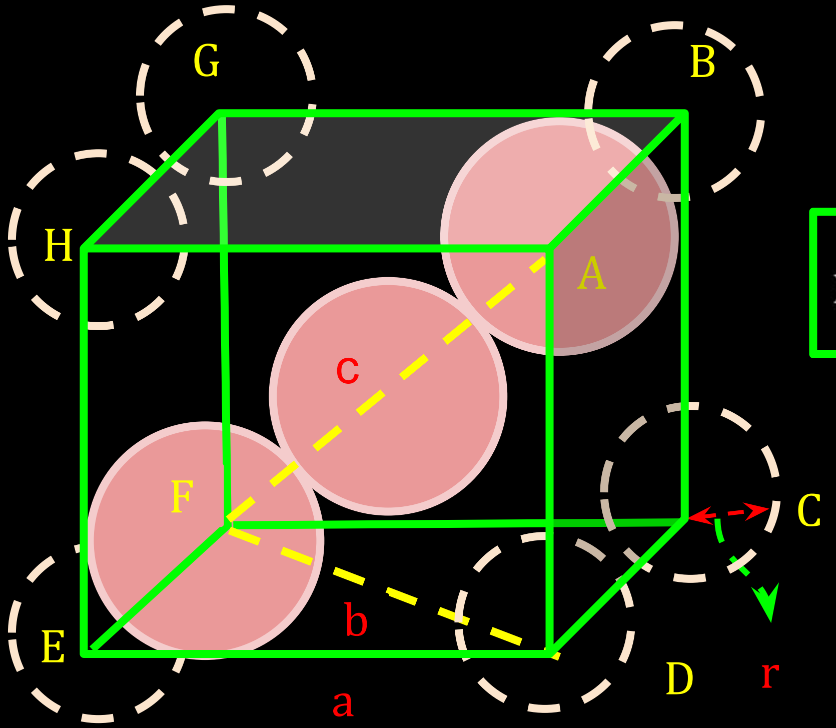
$$P.E = \frac{\text{Volume occupied by 2 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{2 \times (4/3)\,\pi r^3 \times 100}{[(4/\sqrt{3})r]^3}$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8



Relation b/w edge length (a) and radius(r)

$a = 4r/\sqrt{3}$

$$P.E = \frac{\text{Volume occupied by 2 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{2 \times (4/3)\pi r^3 \times 100}{[(4/\sqrt{3})r]^3}$$

$P.E = 68\%$

CV 3

Packing Efficiency in Simple Cubic Lattice

12C01.4 Packing Efficiency and Crystal Density

$$\text{Packing Efficiency} = \frac{\text{Volume occupied by atoms in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

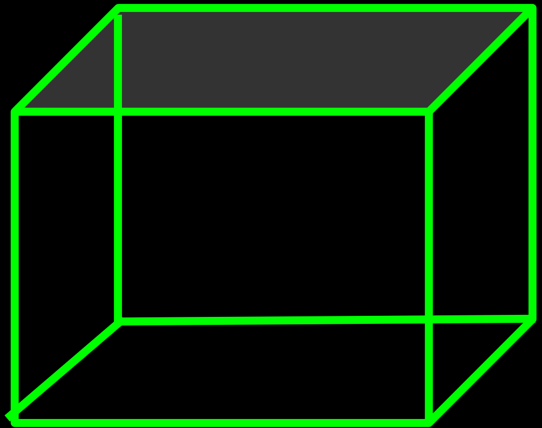
P.E. in hcp and ccp Structures

P.E. in bcc Structures

P.E. in Simple Cubic Lattice

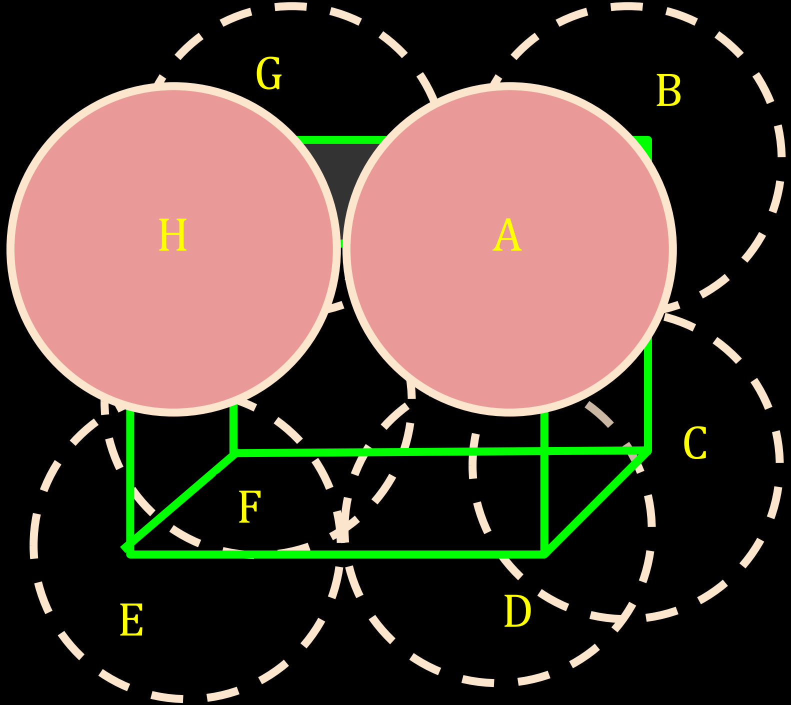
12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in Simple Cubic Lattice



12C01.4 Packing Efficiency and Crystal Density

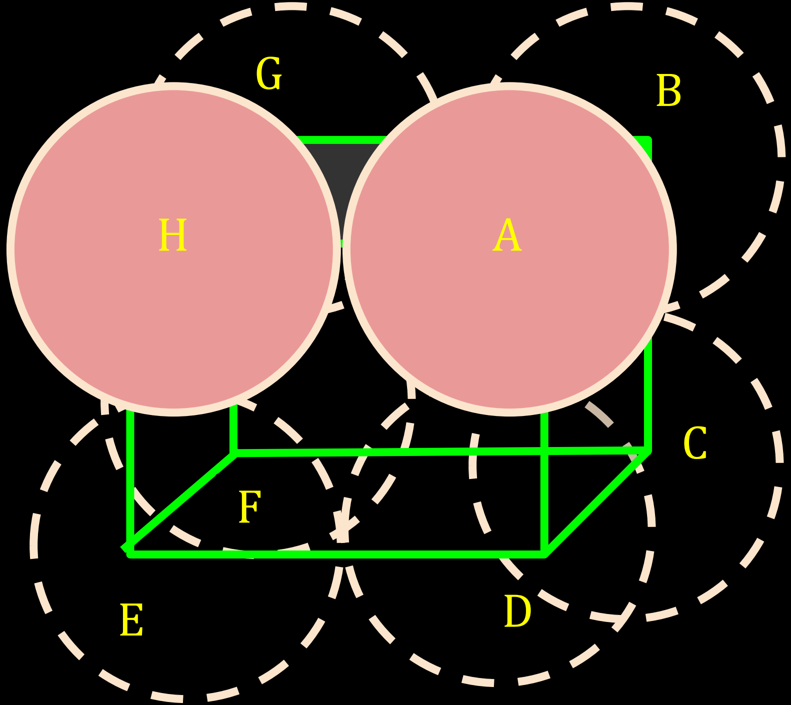
Packing Efficiency in Simple Cubic Lattice



12C01.4 Packing Efficiency and Crystal Density

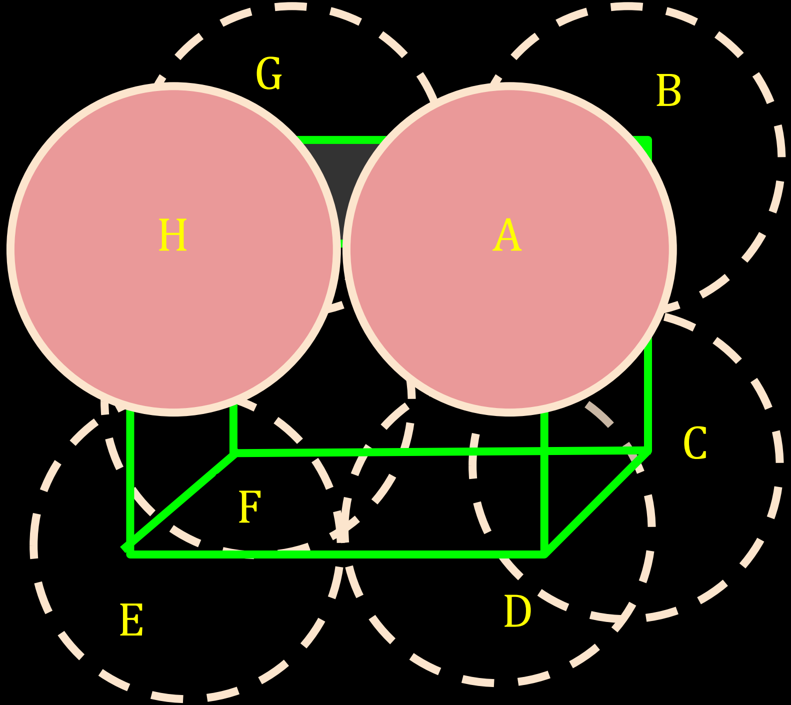
Packing Efficiency in Simple Cubic Lattice

Coordination Number = 6



12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in Simple Cubic Lattice



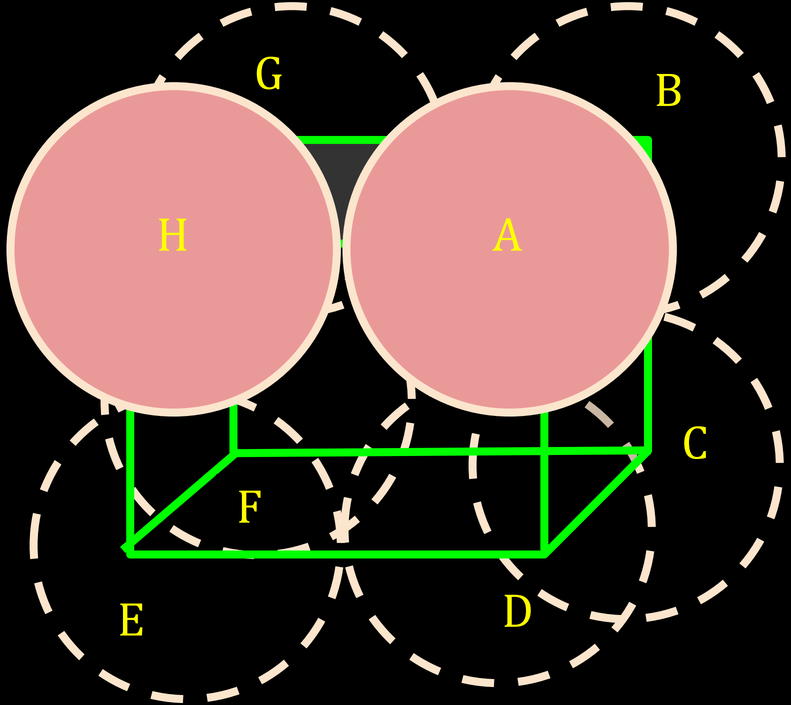
Coordination Number = 6

Relation b/w edge length (a) and radius(r)

$a = 2\,r$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in Simple Cubic Lattice



Coordination Number = 6

Relation b/w edge length (a) and radius(r)

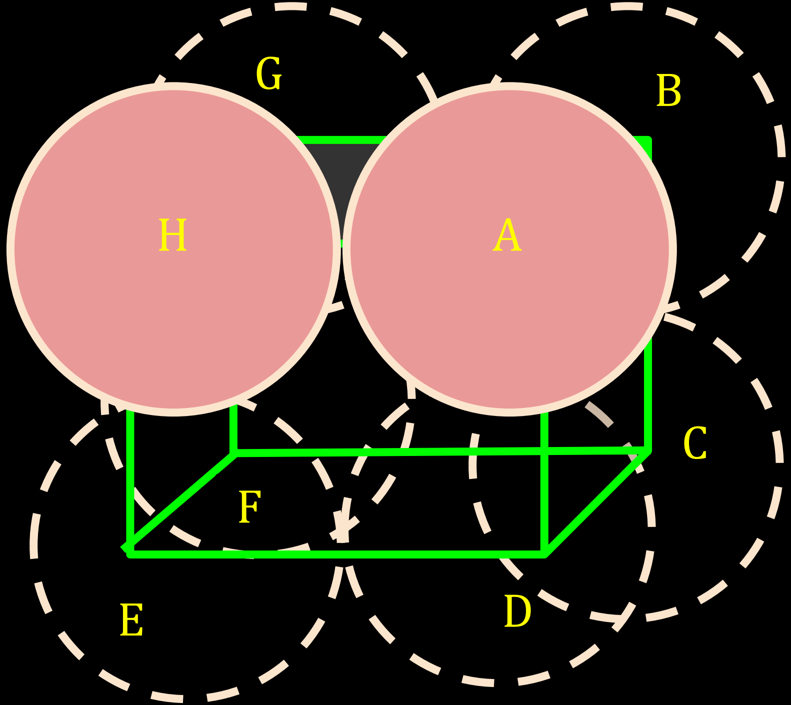
$a = 2\,r$

$$P.E = \frac{\text{Volume occupied by 1 atom} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{(4/3)\,\pi r^3 \times 100}{8r^3}$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in Simple Cubic Lattice



Coordination Number = 6

Relation b/w edge length (a) and radius(r)

$a = 2\,r$

$$P.E = \frac{\text{Volume occupied by 1 atom} \times 100}{\text{Total volume of the unit cell}}$$

$$P.E = \frac{(4/3)\,\pi r^3 \times 100}{8r^3}$$

$$P.E = 52.4\%$$

12C01.4 Packing Efficiency and Crystal Density

Revision Table

Characteristics	fcc	bcc	Simple Cubic
No. of atoms (Z)	4	2	1
Coordination no.	12	8	6
Relation b/w a and r	$a = 2\sqrt{2}\, r$	$a = 4r/\sqrt{3}$	$a = 2\, r$
P.E	74	68	52.4

CV 4

Crystal Density

12C01.4 Packing Efficiency and Crystal Density

$$\frac{\text{Mass of unit cell}}{\text{Volume of unit cell (V)}} = \text{Density of unit cell (d)}$$

12C01.4 Packing Efficiency and Crystal Density

No. of atom

Mass of each atom

$$\boxed{Z \times m}$$

$$\frac{\text{Mass of unit cell}}{\text{Volume of unit cell (V)}} = \text{Density of unit cell (d)}$$

12C01.4 Packing Efficiency and Crystal Density

No. of atom

Mass of each atom

Molar mass/Avogadro no. = M/N_A

$Z \times m$

$$\frac{\text{Mass of unit cell}}{\text{Volume of unit cell (V)}} = \text{Density of unit cell (d)}$$

12C01.4 Packing Efficiency and Crystal Density

No. of atom

Mass of each atom → Molar mass/Avogadro no. = M/N_A

$Z \times m$ → $Z \times M/N_A$

$$\frac{\text{Mass of unit cell}}{\text{Volume of unit cell (V)}} = \text{Density of unit cell (d)}$$

12C01.4 Packing Efficiency and Crystal Density

No. of atom

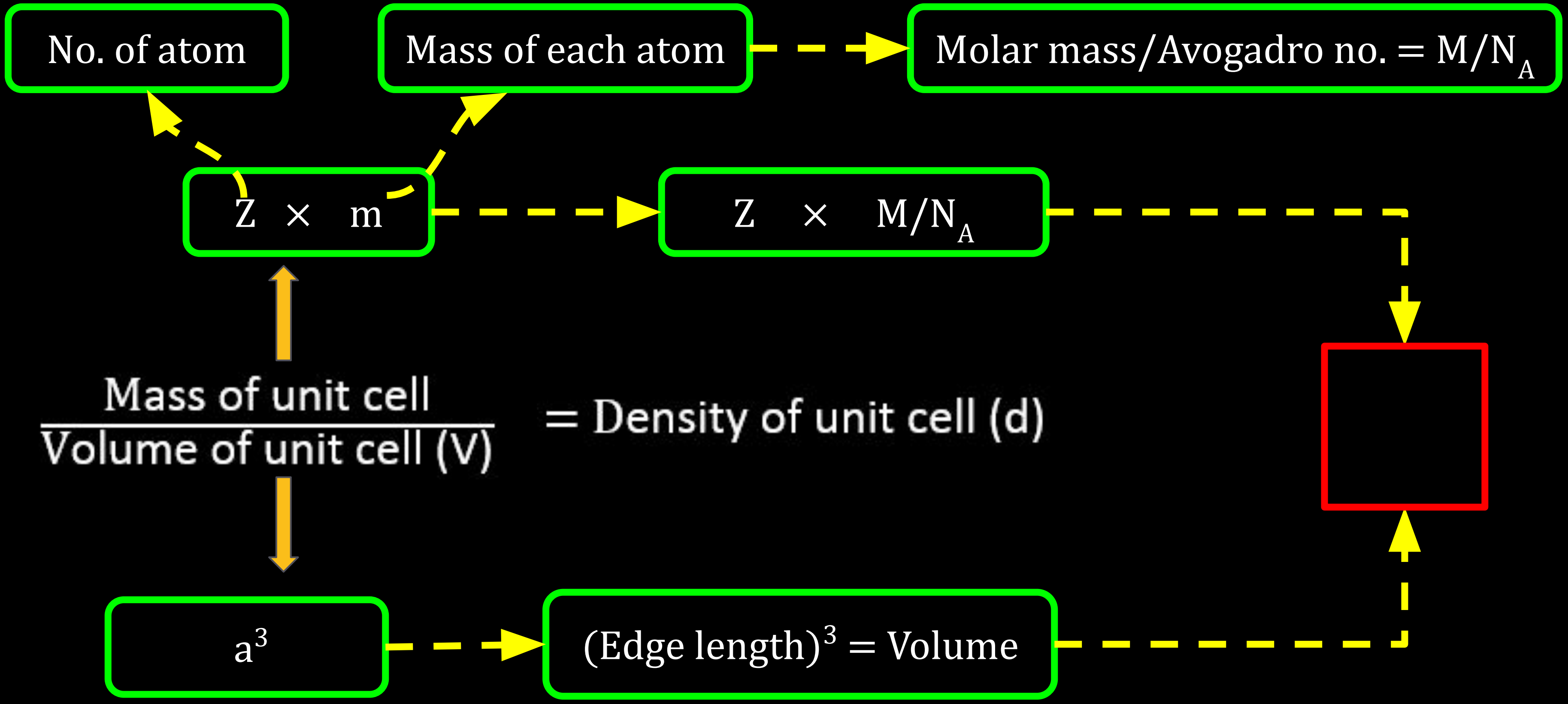
Mass of each atom → Molar mass/Avogadro no. $= M/N_A$

$Z \times m$ → $Z \times M/N_A$

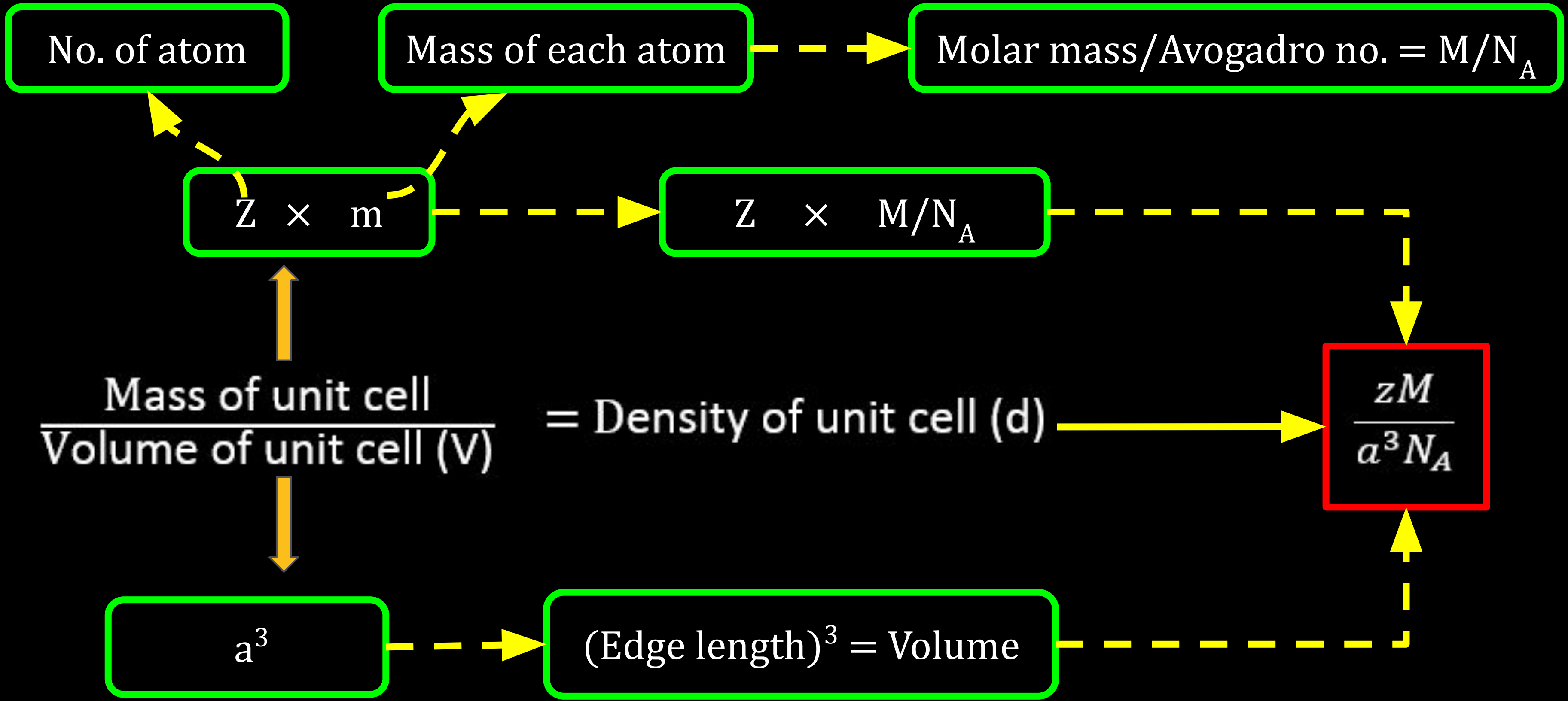
$$\frac{\text{Mass of unit cell}}{\text{Volume of unit cell (V)}} = \text{Density of unit cell (d)}$$

a^3 → $(\text{Edge length})^3 = \text{Volume}$

12C01.4 Packing Efficiency and Crystal Density



12C01.4 Packing Efficiency and Crystal Density



Conceptest

Ready for Challenge

12C01.4 Packing Efficiency and Crystal Density

Question : An element has a bcc structure with a cell edge of 288 pm. The density of the element is 7.2 g/cm^3. How many atoms are present in 208 g of the element?

Solution : ? (NCERT Example 1.3 Page no. 20)

12C01.4 Packing Efficiency and Crystal Density

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Solution : ? (NCERT Example 1.3 Page no. 20)

Pause the Video Time duration 2 minute

12C01.4 Packing Efficiency and Crystal Density

Question : An element has a bcc structure with a cell edge of 288 pm. The density of the element is 7.2 g/cm^3. How many atoms are present in 208 g of the element?

Solution : Volume of unit cell $= (288 \text{ pm})^3 = (288\times10^{-12} \text{ m})^3 = (288\times10^{-10} \text{ cm})^3$

$= 2.39\times10^{-23} \text{ cm}^3$

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$$\boxed{\text{Volume of 208 g of element} = \frac{\text{Mass}}{\text{Density}} = \frac{208 \text{ g}}{7.2 \text{ g cm}^{-3}} = 28.88 \text{ cm}^3}$$

12C01.4 Packing Efficiency and Crystal Density

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12C01.4 Packing Efficiency and Crystal Density

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$$\boxed{\text{Volume of 208 g of element} = \frac{\text{Mass}}{\text{Density}} = \frac{208 \text{ g}}{7.2 \text{ g cm}^{-3}} = 28.88 \text{ cm}^3}$$

$$\boxed{\text{Number of unit cells in this volume} = \frac{28.88}{2.39\times10^{-23}} = 12.08\times10^{23} \text{ unit cells}}$$

$$\boxed{\text{Total no. of atoms in 208 } g = 2\times12.08\times10^{23} = 24.16\times10^{23} \text{ atoms}}$$

12C01.3 Close Packed Structures

: Reference Questions :

Intext Question : 1.17 (NCERT page no. 24)

Exercise Question : 1.5 (NCERT page no. 32), 1.10 (NCERT page no. 33)
1.13 (NCERT page no. 33), 1.15 (NCERT page no. 33)
1.24 (NCERT page no. 34)

Work Book Question : 16, 17, 18 and 20

12C01.5

Imperfections in Solids

12C01.5 Imperfections in Solids

- Imperfections in Solids

12C01.5 Imperfections in Solids

- Imperfections in Solids

- Electrical Properties of Solids

12C01.5 Imperfections in Solids

- Imperfections in Solids

- Electrical Properties of Solids

- Magnetic Properties of Solids

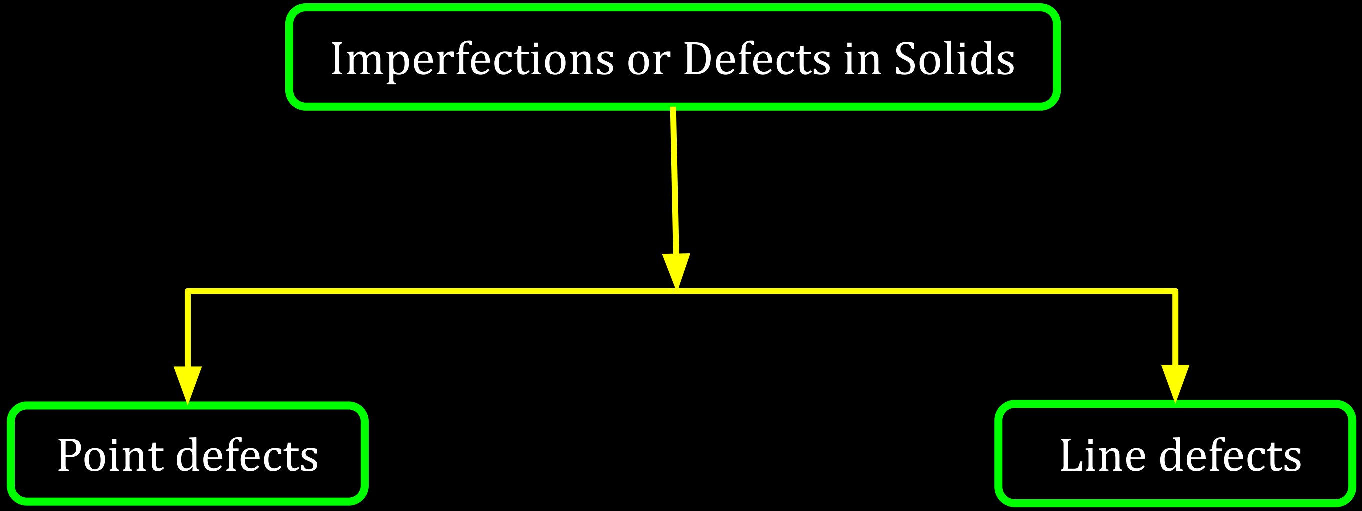
CV 1

Imperfections in Solids, Point defects

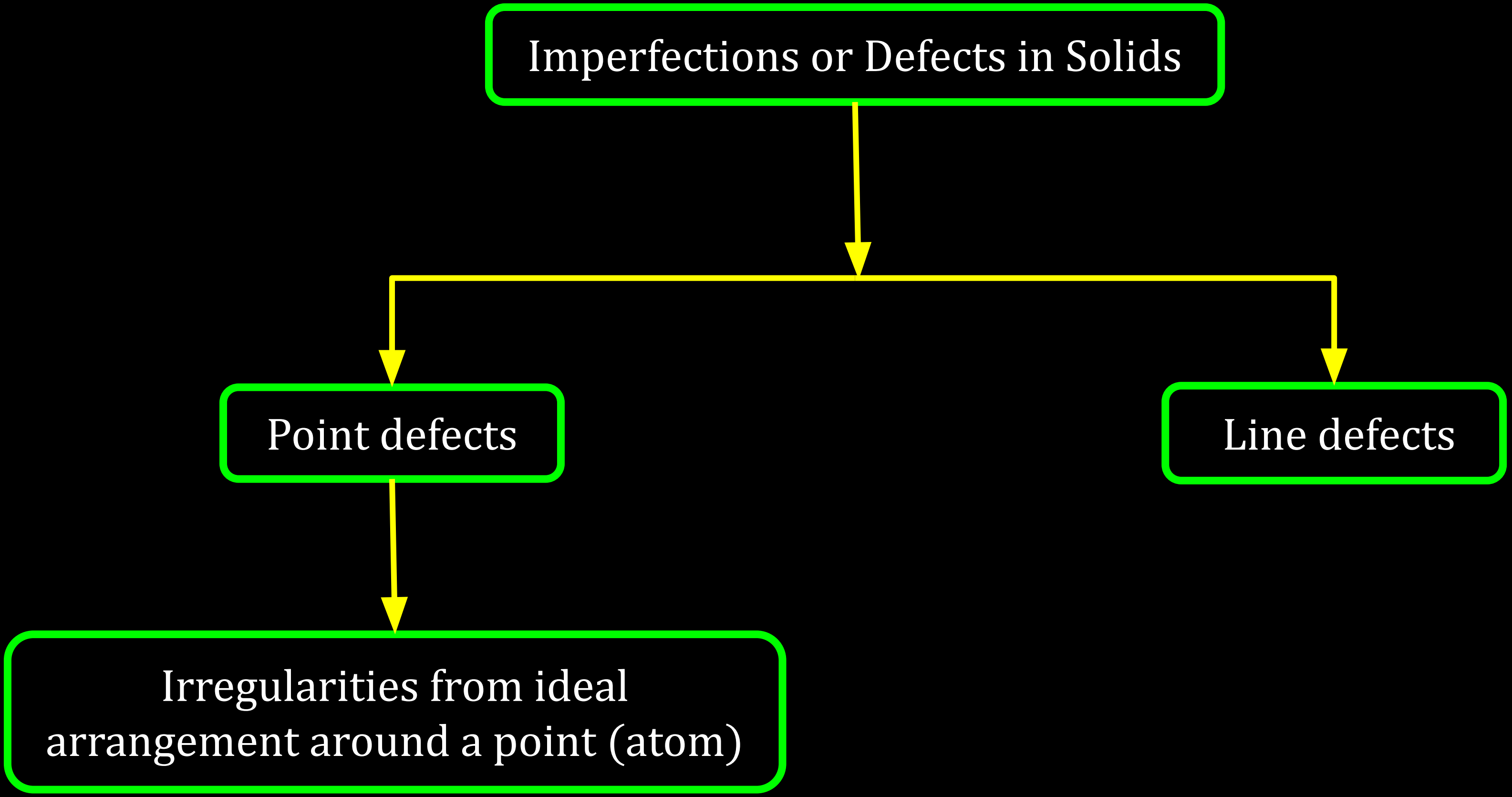
12C01.5 Imperfections in Solids

Imperfections or Defects in Solids

12C01.5 Imperfections in Solids



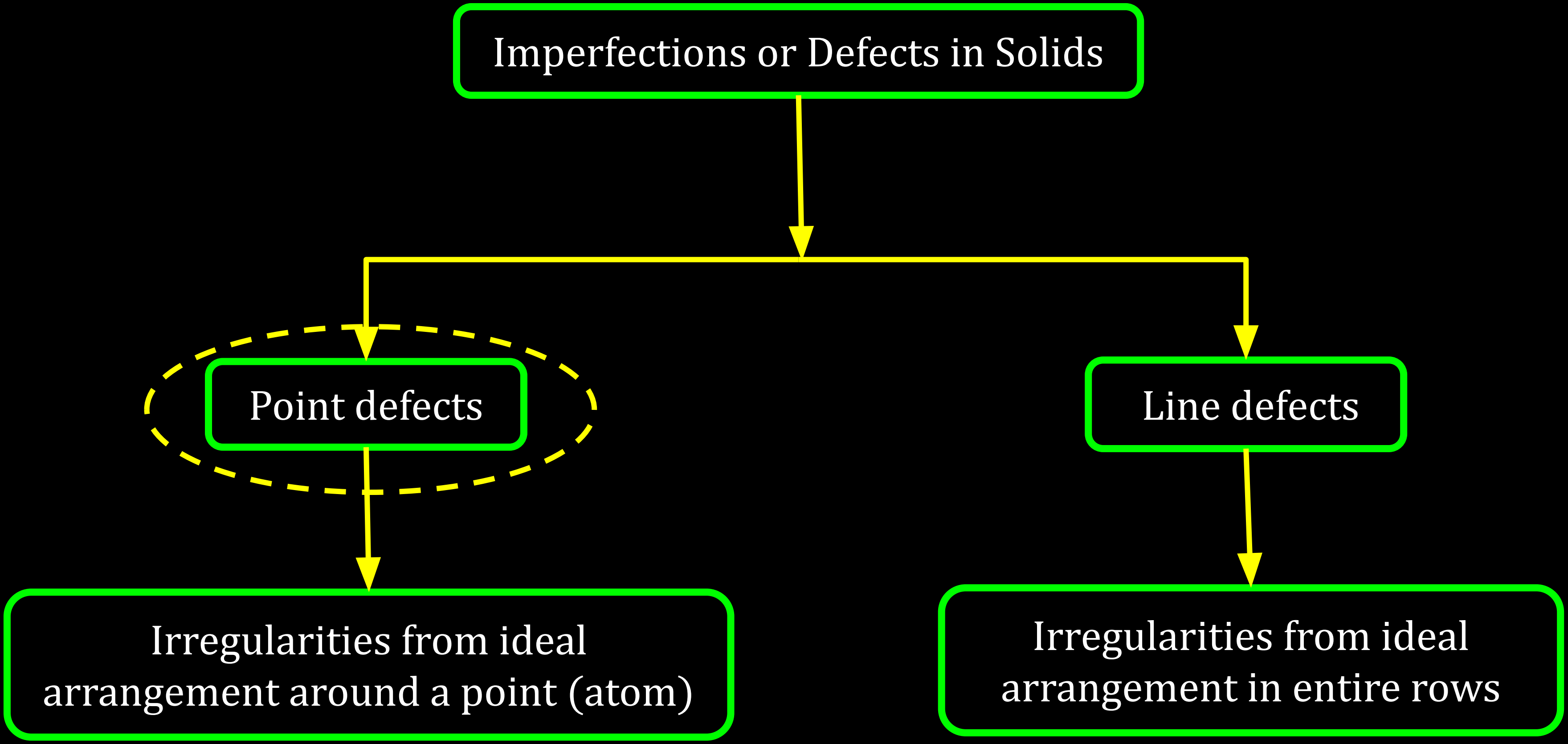
12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

- Imperfections or Defects in Solids
  - Point defects
    - Irregularities from ideal arrangement around a point (atom)
  - Line defects
    - Irregularities from ideal arrangement in entire rows

12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
- Impurity
- Non-stoichiometric

12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
- Impurity
- Non-stoichiometric

- Don’t disturb the stoichiometry of the solid.

12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
- Impurity
- Non-stoichiometric

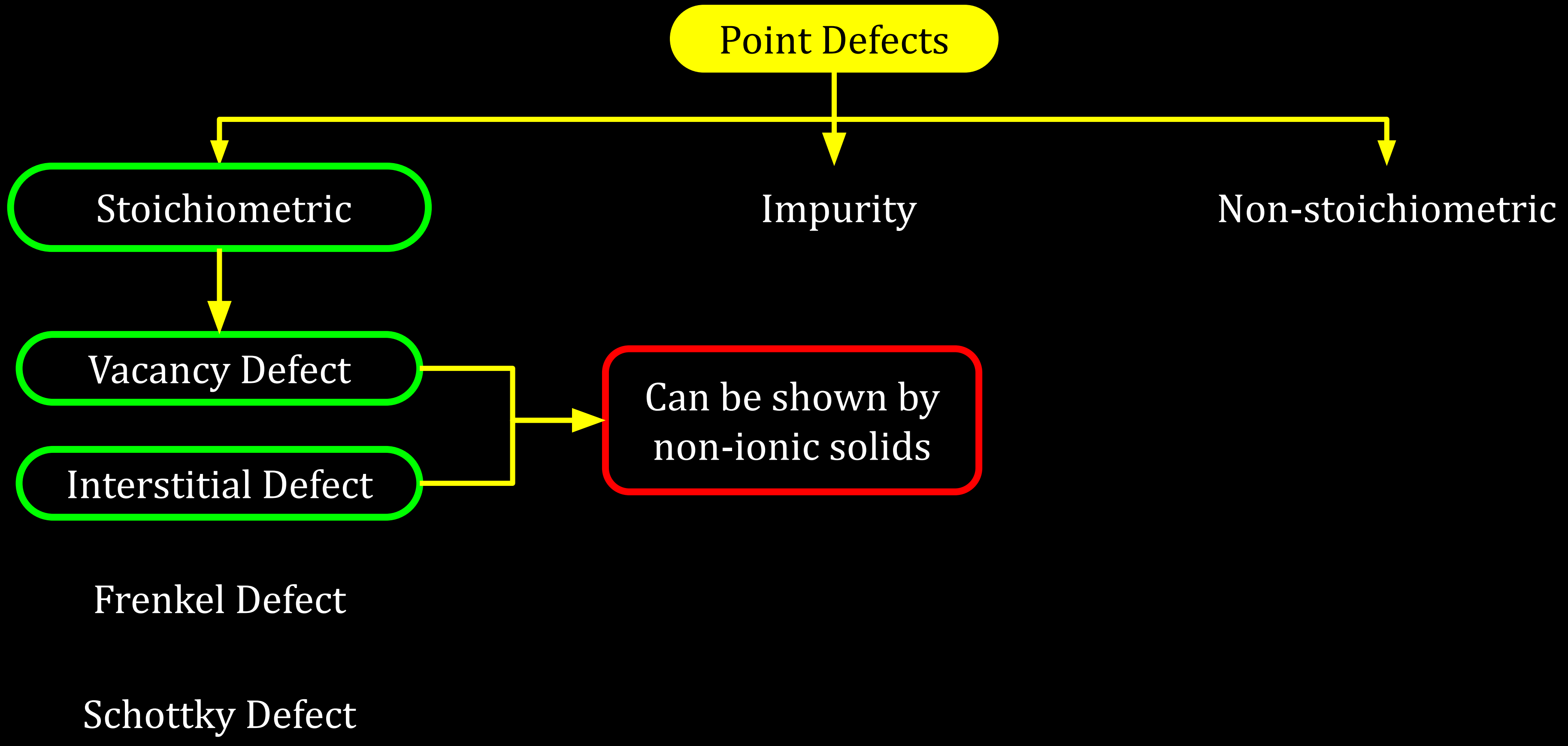
Stoichiometric

- Don't disturb the stoichiometry of the solid.
- Also called intrinsic or thermodynamic defects

12C01.5 Imperfections in Solids

- Point Defects
  - Stoichiometric
    - Vacancy Defect
    - Interstitial Defect
    - Frenkel Defect
    - Schottky Defect
  - Impurity
  - Non-stoichiometric

12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

Shown by ionic solids

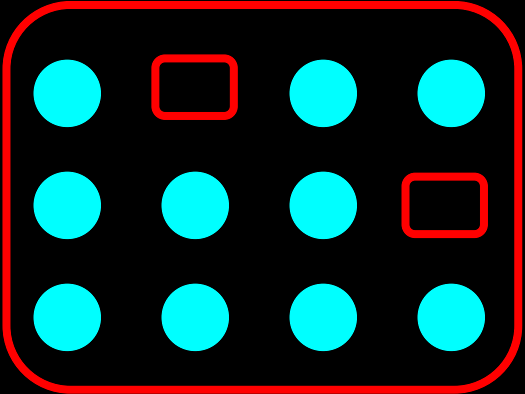
12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

Vacancy Defect:

- Some of the lattice sites are vacant



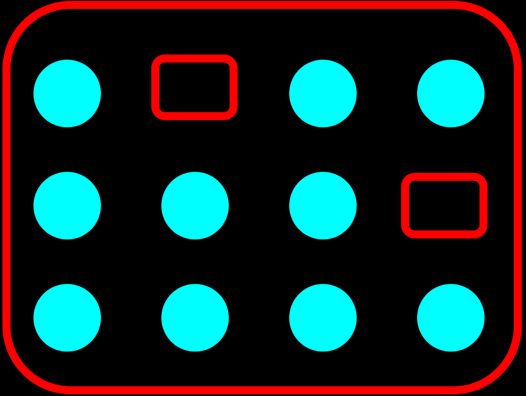
12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

Vacancy Defect:

- Some of the lattice sites are vacant
- Decrement in density.

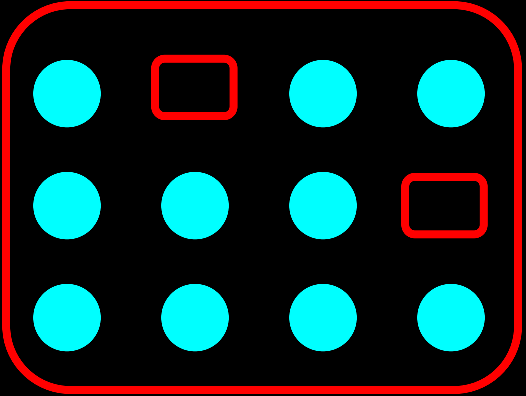


12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

- Some of the lattice sites are vacant
- Decrement in density.
- Also develop on heating



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

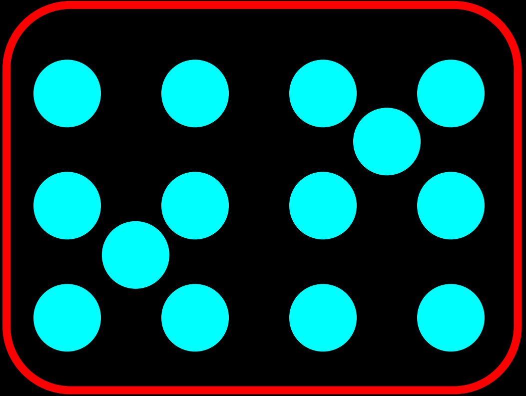
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Some atoms or molecules occupy an interstitial site.



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

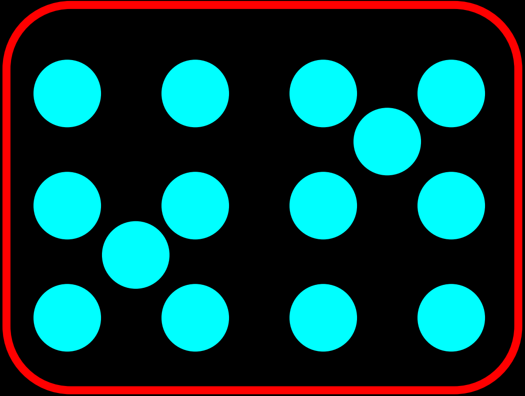
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Some atoms or molecules occupy an interstitial site.
- Increment in density.



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

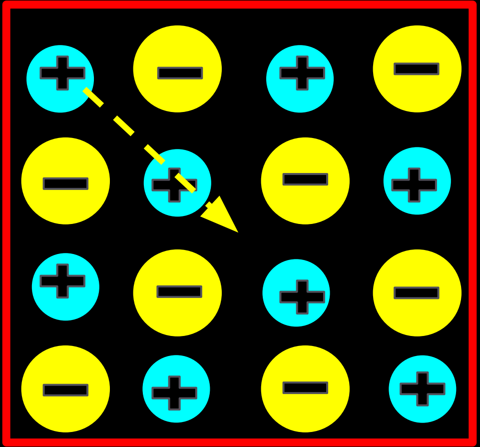
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Smaller ion (cation) moved from normal to interstitial.



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

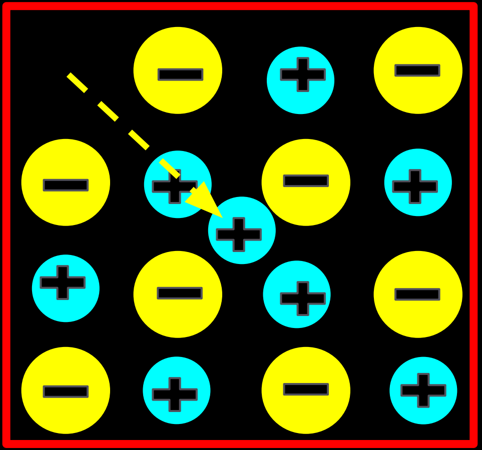
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Smaller ion (cation) moved from normal to interstitial.
- Also called dislocation defect



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

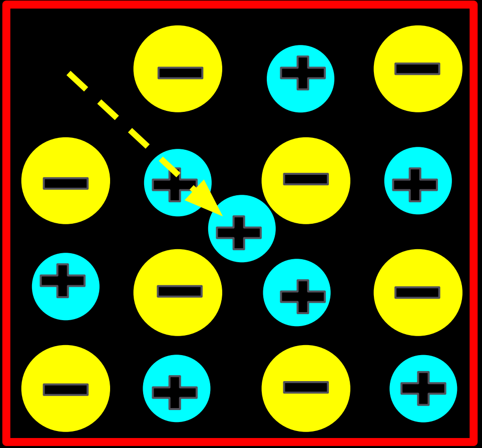
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Smaller ion (cation) moved from normal to interstitial.
- Also called dislocation defect
- Density does not change



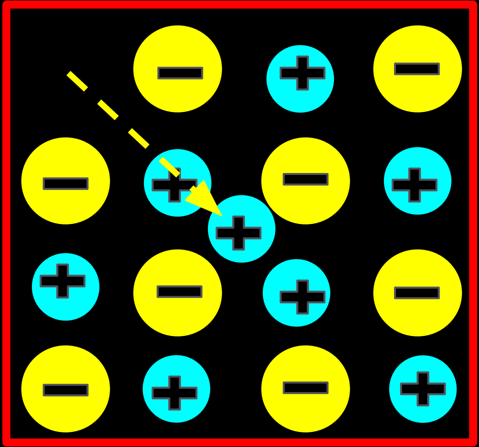
12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

Frenkel Defect:

- Smaller ion (cation) moved from normal to interstitial.
- Also called dislocation defect
- Density does not change
- Ex - ZnS, AgCl, AgBr, AgI



12C01.5 Imperfections in Solids

Point Defects

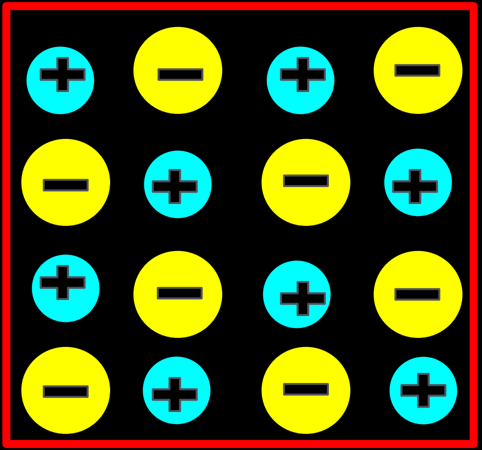
Stoichiometric

- Vacancy Defect
- Interstitial Defect
- Frenkel Defect
- Schottky Defect

Impurity

Non-stoichiometric

- Number of missing cations and anions are equal.



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

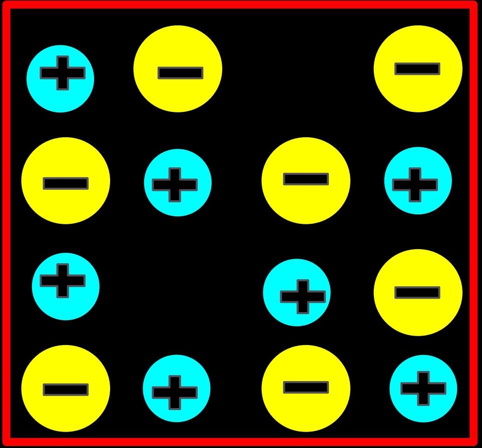
Vacancy Defect

Interstitial Defect

Frenkel Defect

Schottky Defect

- Number of missing cations and anions are equal.

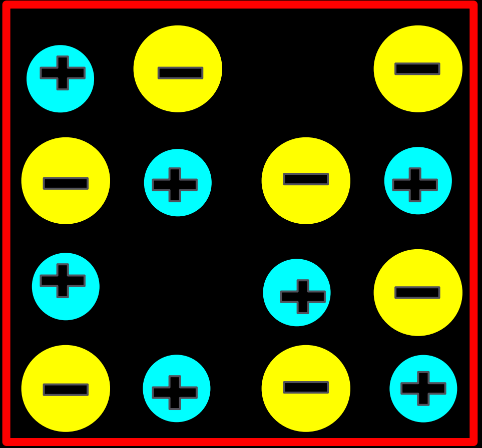


12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

- Number of missing cations and anions are equal.
- Electrical neutral.

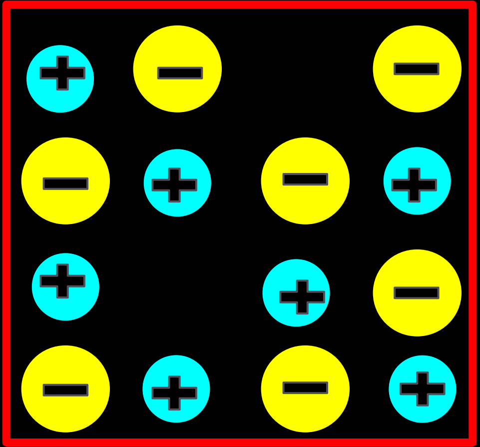


12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
  - Vacancy Defect
  - Interstitial Defect
  - Frenkel Defect
  - Schottky Defect
- Impurity
- Non-stoichiometric

- Number of missing cations and anions are equal.
- Electrical neutral.
- Density of crystal decrease



12C01.5 Imperfections in Solids

Point Defects

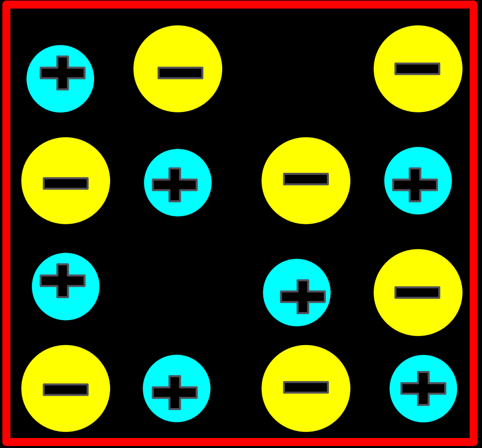
- Stoichiometric
- Impurity
- Non-stoichiometric

Stoichiometric

- Vacancy Defect
- Interstitial Defect
- Frenkel Defect
- Schottky Defect

Schottky Defect:

- Number of missing cations and anions are equal.
- Electrical neutral.
- Density of crystal decrease
- Ex. - NaCl, KCl, CsCl, AgBr



12C01.5 Imperfections in Solids

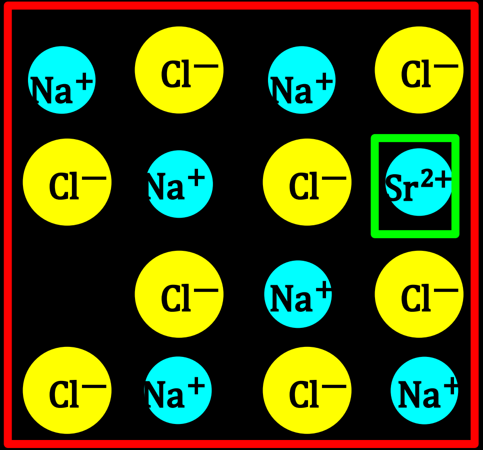
Point Defects

Stoichiometric

Impurity

Non-stoichiometric

- Some cationic sites are occupied by other cations



Solid solution of NaCl and $SrCl_2$

12C01.5 Imperfections in Solids

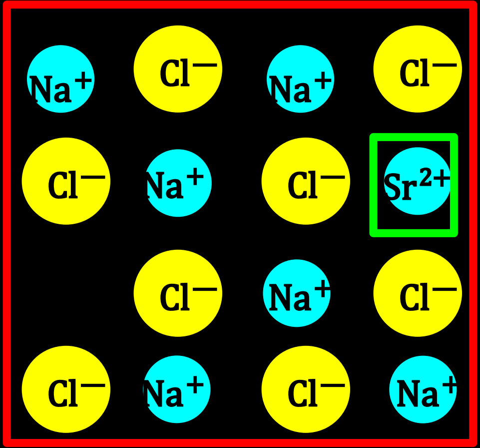
Point Defects

Stoichiometric

Impurity

Non-stoichiometric

- Some cationic sites are occupied by other cations
- Some Na^+ sites are replaced by Sr^{2+} ions.



Solid solution of NaCl and $SrCl_2$

12C01.5 Imperfections in Solids

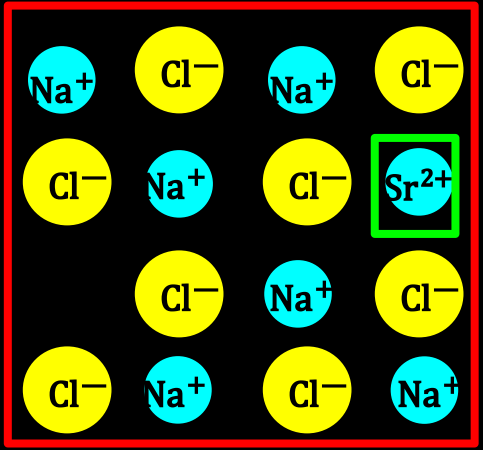
Point Defects

Stoichiometric

Impurity

Non-stoichiometric

- Some cationic sites are occupied by other cations
- Some Na^+ sites are replaced by Sr^{2+} ions.
- Ex. - solid solution of AgCl & $CdCl_2$



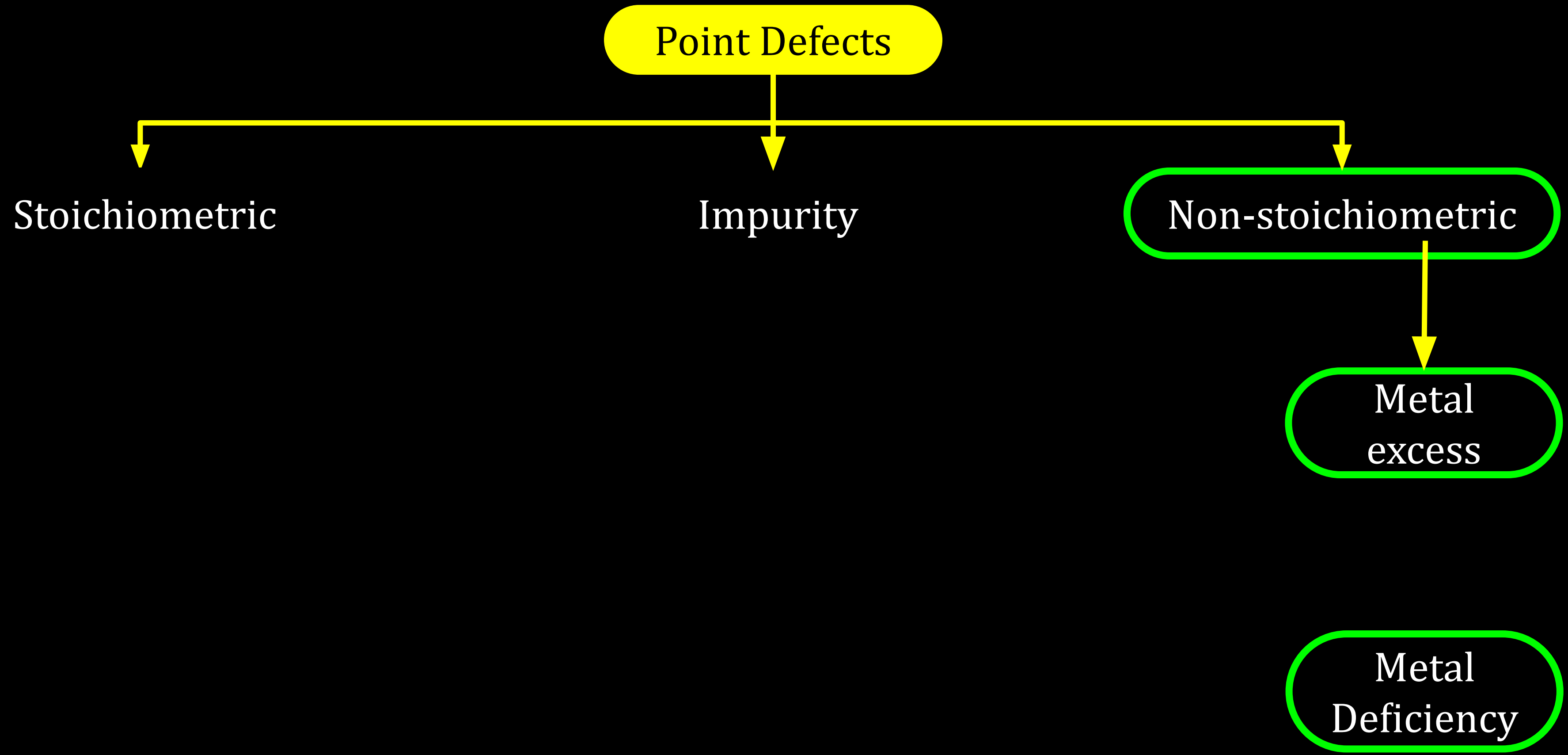
Solid solution of NaCl and $SrCl_2$

12C01.5 Imperfections in Solids

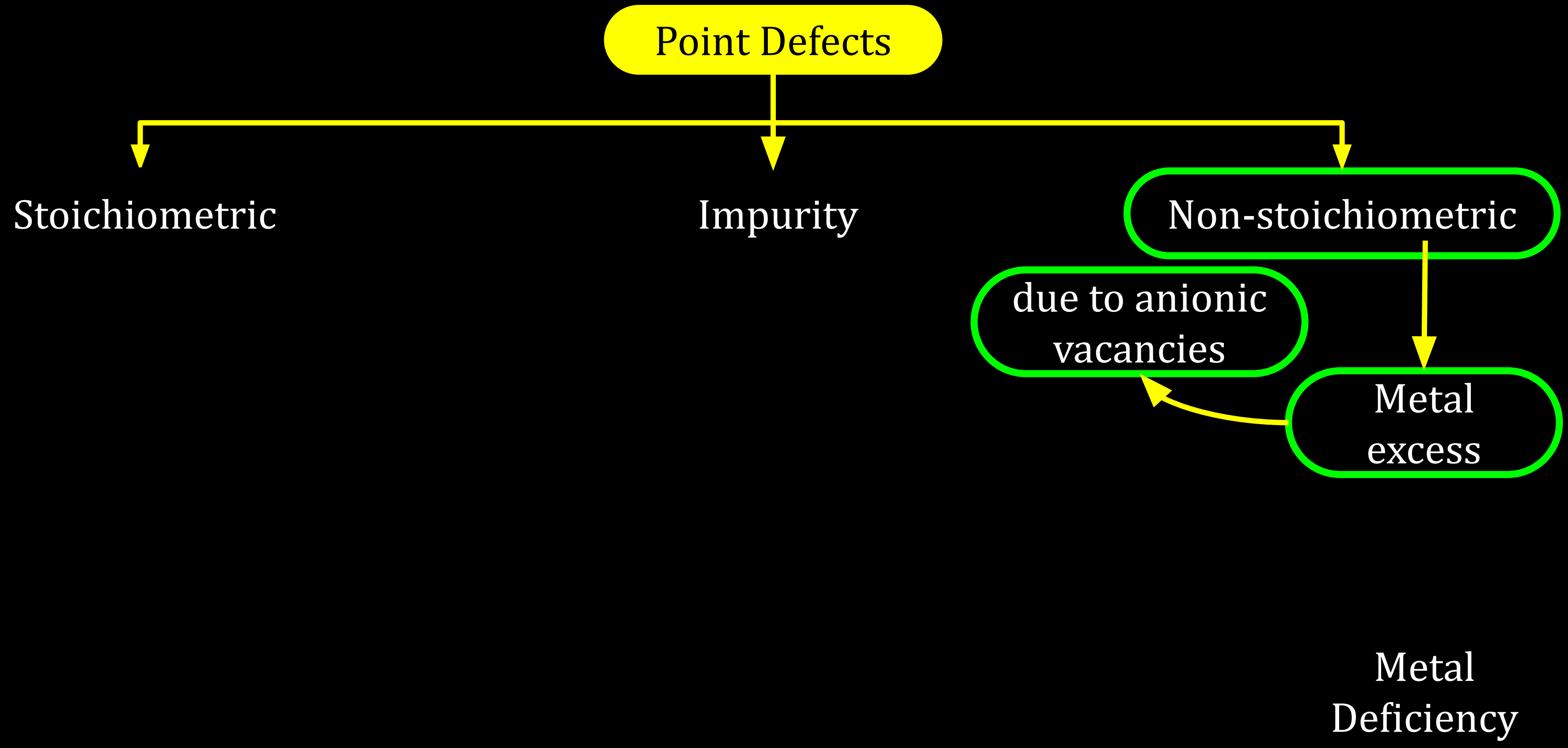
Point Defects

- Stoichiometric
- Impurity
- Non-stoichiometric
  - Contain the constituent elements in non stoichiometric ratio

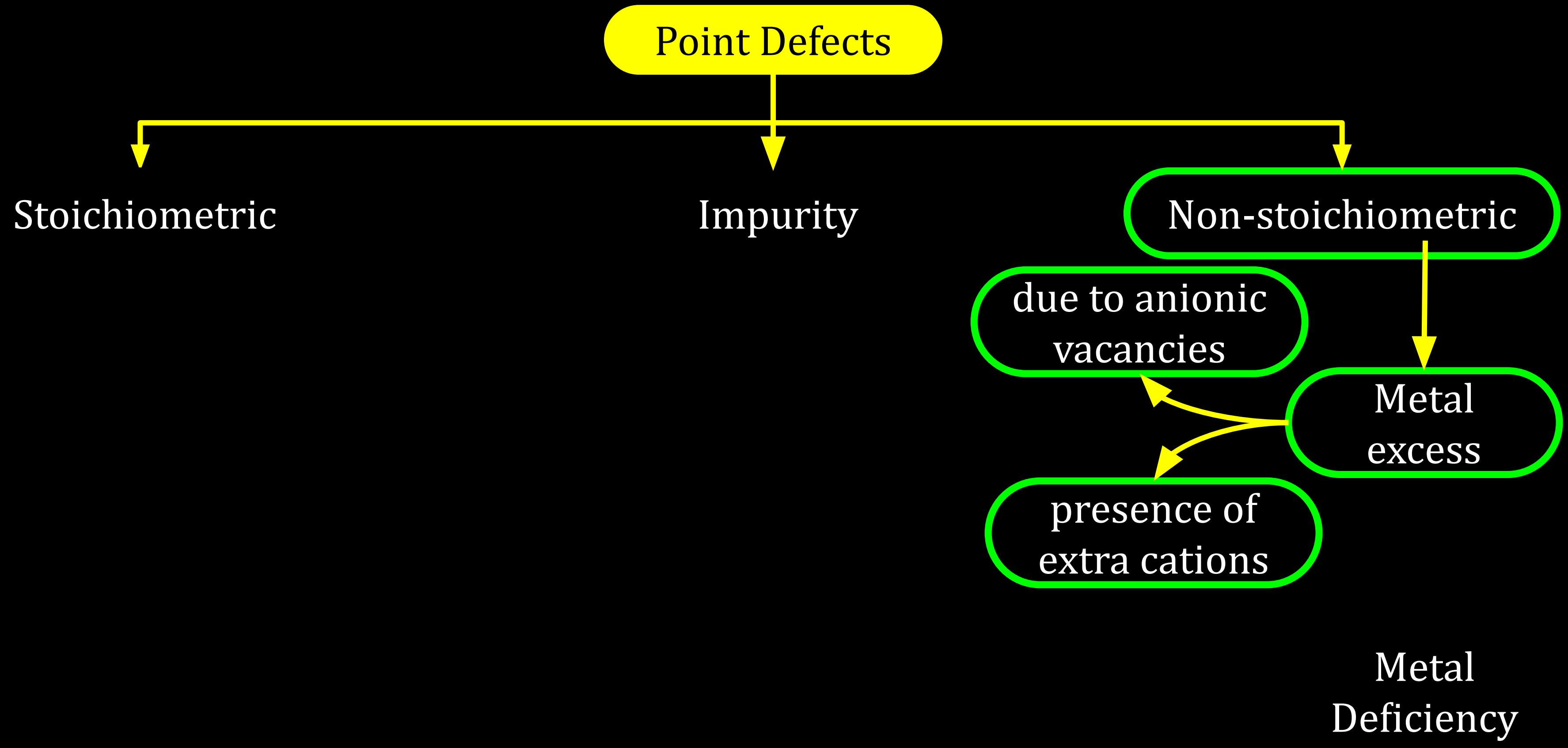
12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids



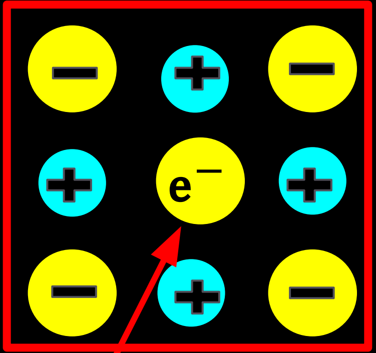
12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Point Defects

- Stoichiometric
- Impurity
- Non-stoichiometric
  - Metal excess
    - due to anionic vacancies
  - Metal Deficiency



F-centres

- Shown by alkali halides like NaCl

12C01.5 Imperfections in Solids

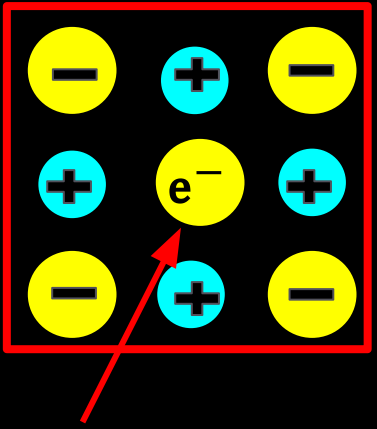
Point Defects

Stoichiometric

Impurity

Non-stoichiometric

- Metal excess
  - due to anionic vacancies
- Metal Deficiency



F-centres

- Shown by alkali halides like NaCl
- Anions diffuse to surface to react with M and e^- diffuse into crystal.

12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

Metal excess

presence of extra cations

- Zinc oxide on heating :-

  White → Yellow

  (Lose oxygen)

Metal Deficiency

12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

Metal excess

presence of extra cations

- Zinc oxide on heating :-

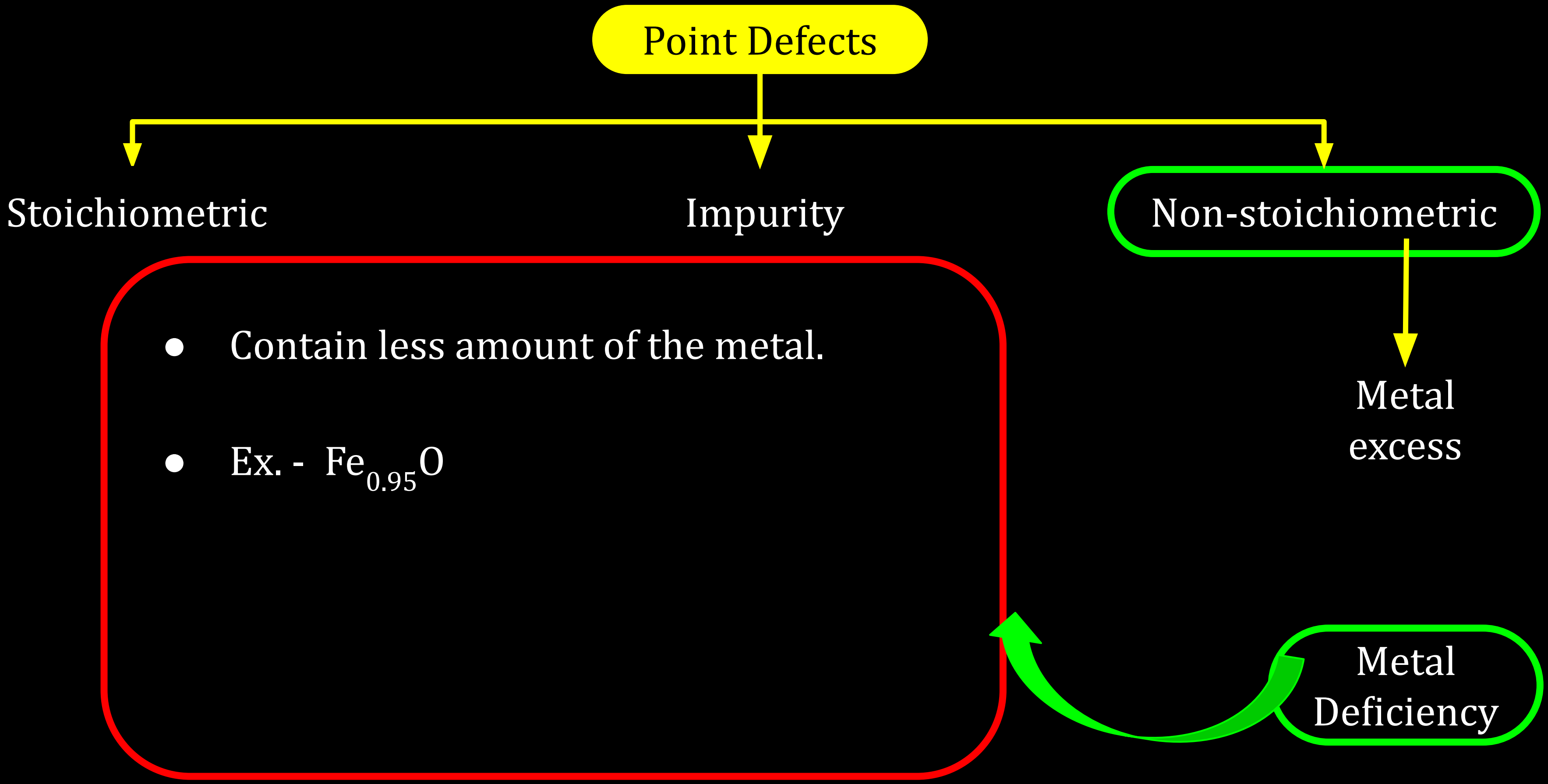
  White $\longrightarrow$ Yellow

  (Lose oxygen)

- Excess Zn^{2+} ions move to interstitial sites and the electrons to neighbouring interstitial sites

Metal Deficiency

12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Point Defects

Stoichiometric

Impurity

Non-stoichiometric

Metal excess

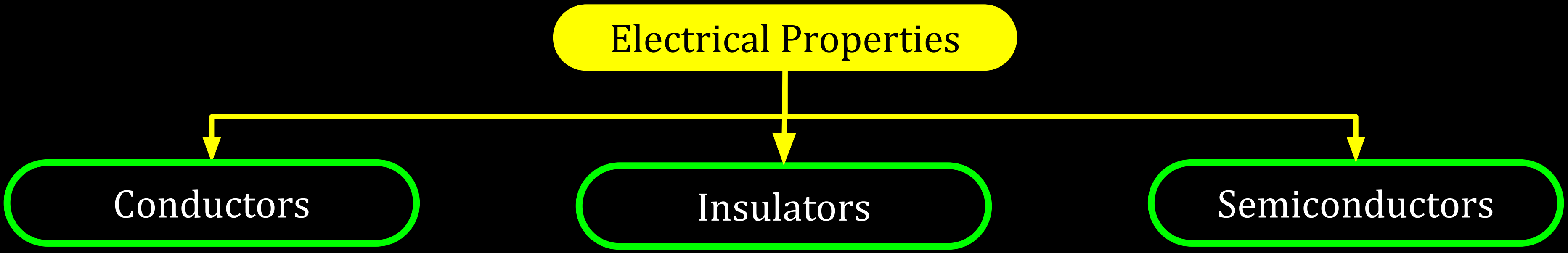
- Contain less amount of the metal.
- Ex. - $Fe_{0.95}O$
- Some Fe^{2+} cations are missing and loss of positive charge is made up by required number of Fe^{3+} ions

Metal Deficiency

CV 2

Electrical Properties of Solids, Band Theory

12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Electrical Properties

- Conductors
- Insulators
- Semiconductors

Conductors:

- Conductivities ranging b/w 10^4 to 10^7 $ohm^{-1}m^{-1}$

12C01.5 Imperfections in Solids

Electrical Properties

Conductors

Insulators

Semiconductors

- Conductivities ranging b/w 10^4 to 10^7 $ohm^{-1}m^{-1}$
- Metals are good conductors

12C01.5 Imperfections in Solids

Electrical Properties

- Conductors
- Insulators
- Semiconductors

Insulators:

- Solids with very low conductivities. (10^{-20} to 10^{-10} $ohm^{-1}m^{-1}$)

12C01.5 Imperfections in Solids

Electrical Properties

Conductors

Insulators

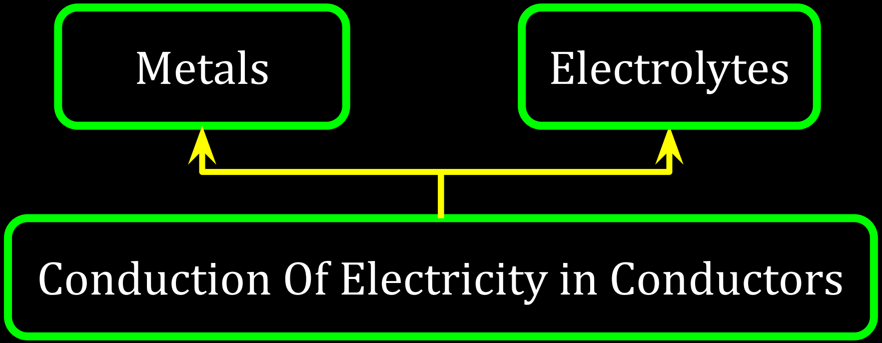
Semiconductors

- Solids with conductivities in the intermediate range.
  (10^{-6} to 10^{4} $ohm^{-1}m^{-1}$)

12C01.5 Imperfections in Solids

Conduction Of Electricity in Conductors

12C01.5 Imperfections in Solids



12C01.5 Imperfections in Solids

Metals

Electrolytes

Aqueous & molten state through movement of ions

Conduction Of Electricity in Conductors

12C01.5 Imperfections in Solids

Conduction Of Electricity in Conductors

- Metals: Solid and molten state through movement of electrons
- Electrolytes: Aqueous & molten state through movement of ions

12C01.5 Imperfections in Solids

Solid and molten state through movement of electrons

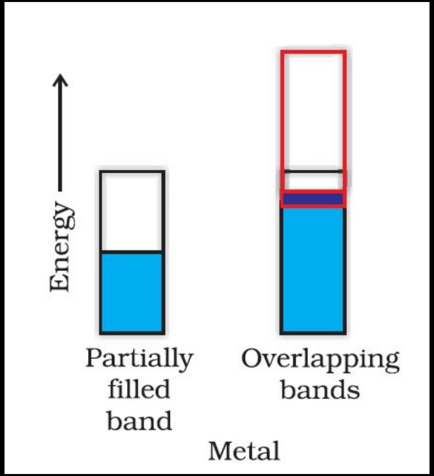
Metals

Electrolytes

Conduction Of Electricity in Conductors

Conductivity depends on valence e^- per atom.

Band formation



12C01.5 Imperfections in Solids

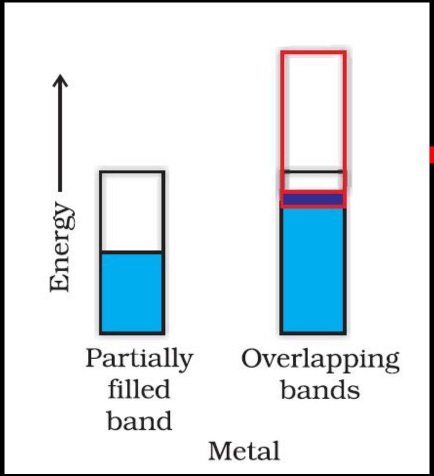
Conduction Of Electricity in Conductors

Metals

Solid and molten state through movement of electrons

Conductivity depends on valence e^- per atom.

Band formation



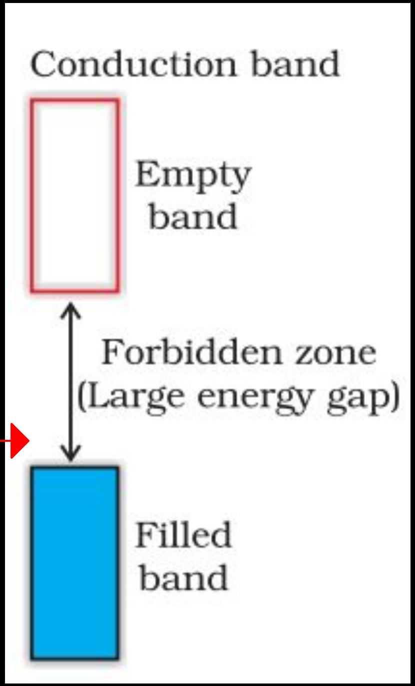
e^- can flow easily under an applied electric field.

Electrolytes

12C01.5 Imperfections in Solids

Conduction of Electricity in Insulators

Gap between filled valence band & conduction band is large

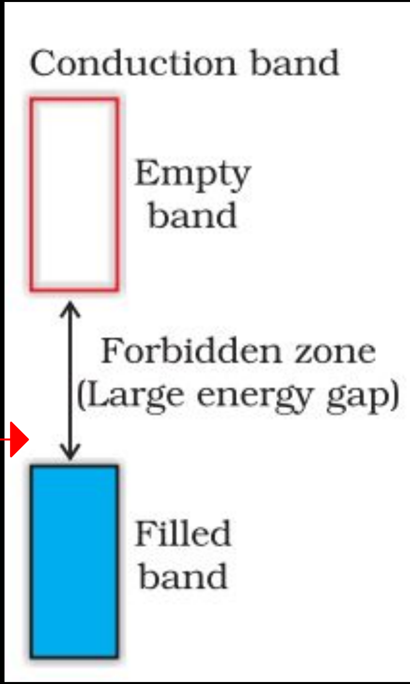


12C01.5 Imperfections in Solids

Conduction of Electricity in Insulators

Gap between filled valence band & conduction band is large

e^- cannot jump.

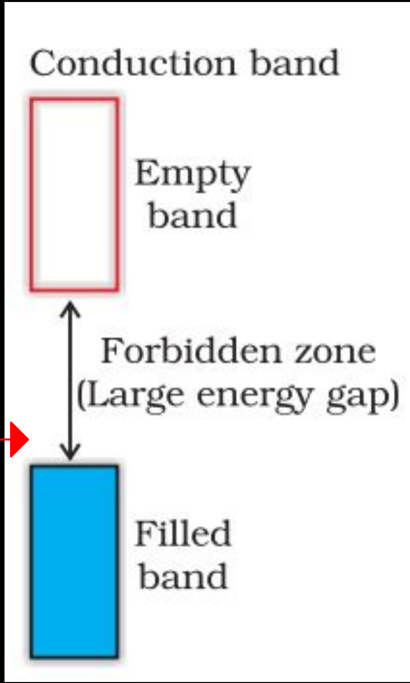


Conduction of Electricity in Insulators

Gap between filled valence band & conduction band is large

e^- cannot jump.

Very small conductivity.

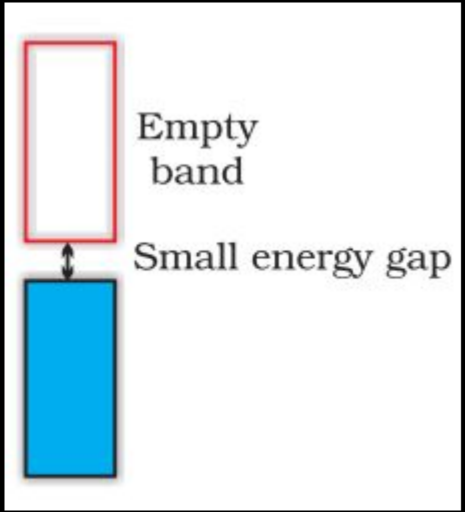


Conduction of Electricity in Semiconductors

12C01.5 Imperfections in Solids

Gap b/w the bands is small

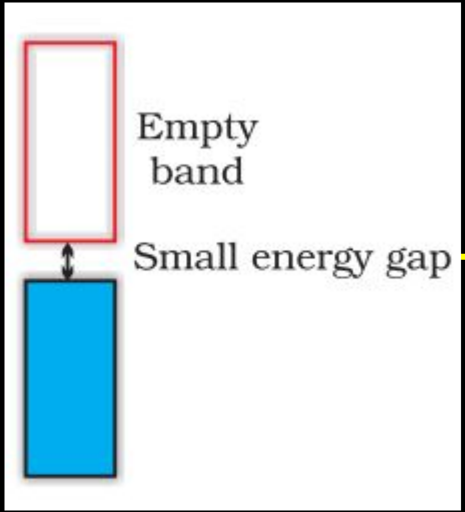
Conduction of Electricity in Semiconductors



12C01.5 Imperfections in Solids

Gap b/w the bands is small

Conduction of Electricity in Semiconductors

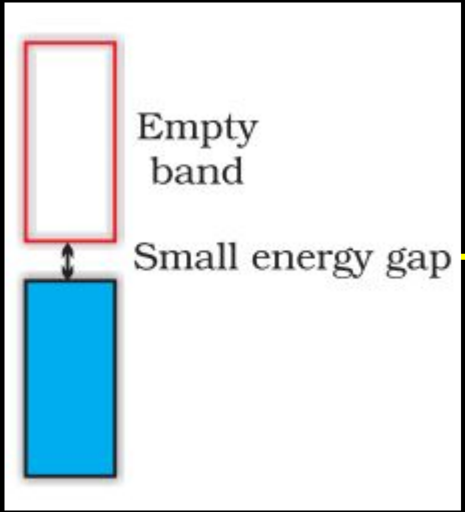


- Conductivity increases with rise in T.
- More electrons can jump.

12C01.5 Imperfections in Solids

Conduction of Electricity in Semiconductors

Gap b/w the bands is small

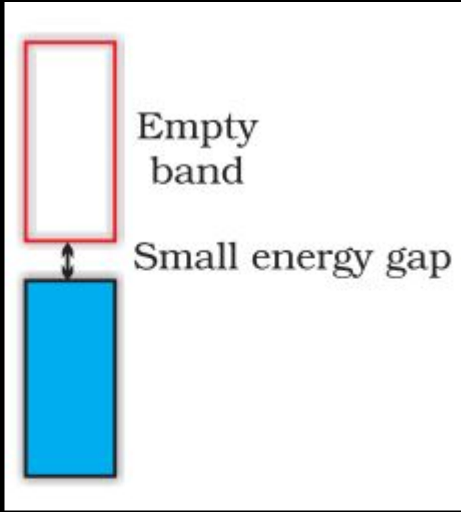


- Conductivity increases with rise in T.
- More electrons can jump.
- These are called intrinsic semiconductors
- Examples are Si and Ge

12C01.5 Imperfections in Solids

Gap b/w the bands is small

Conduction of Electricity in Semiconductors



- Conductivity increases with rise in T.
- More electrons can jump.
- These are called intrinsic semiconductors
- Examples are Si and Ge
- Conductivity of these is too low for practical use,and this is increased by doping.

12C01.5 Imperfections in Solids

Doping

12C01.5 Imperfections in Solids

Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

12C01.5 Imperfections in Solids

Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

Electron deficient impurities

12C01.5 Imperfections in Solids

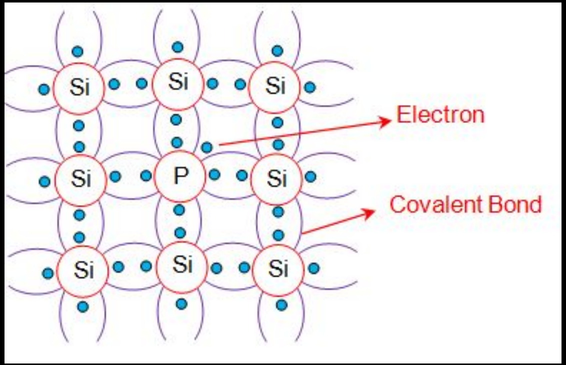
Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

- Group 14 elements like Si and Ge doped with group 15 elements like P

Electron deficient impurities



n-type semiconductor

12C01.5 Imperfections in Solids

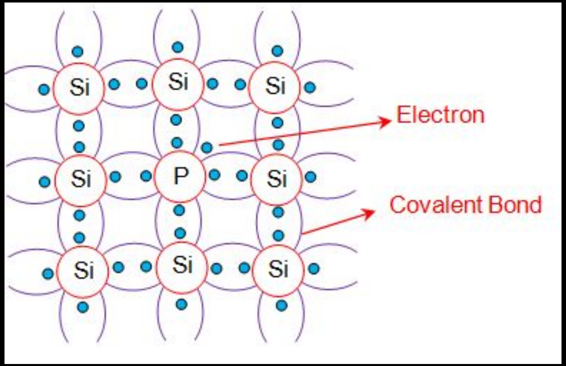
Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

- Group 14 elements like Si and Ge doped with group 15 elements like P
- Delocalised electrons increase the conductivity.

Electron deficient impurities



n-type semiconductor

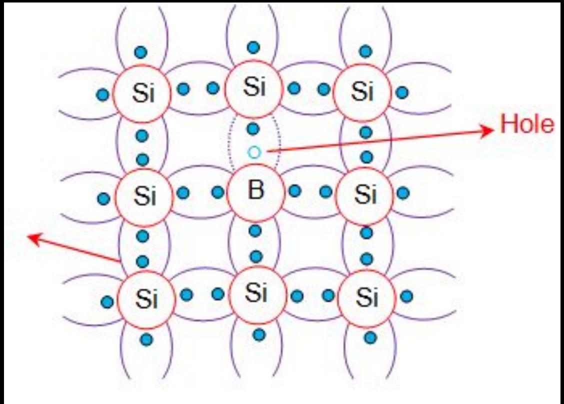
12C01.5 Imperfections in Solids

Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

Electron deficient impurities



p-type semiconductors

- Group 14 elements (Si, Ge) doped with group 13 elements like B

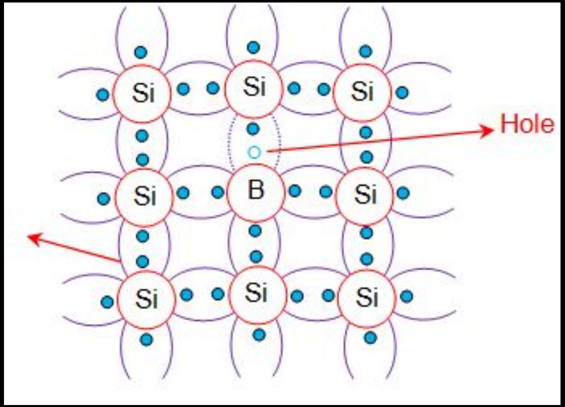
12C01.5 Imperfections in Solids

Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

Electron deficient impurities



p-type semiconductors

- Group 14 elements (Si, Ge) doped with group 13 elements like B

- During the movement of e^- from one place to other, electron hole is created at original position

CV 3

Magnetic Properties of Solids

Origin of Magnetic Behavior of Solids

↓

e^- in atom behaves like tiny magnet

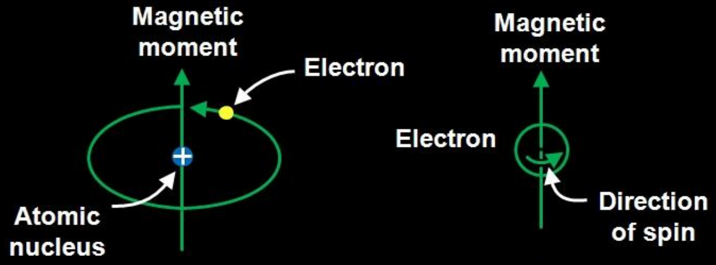
12C01.5 Imperfections in Solids

Origin of Magnetic Behavior of Solids

e^- in atom behaves like tiny magnet

orbital motion around nucleus

spin around its own axis



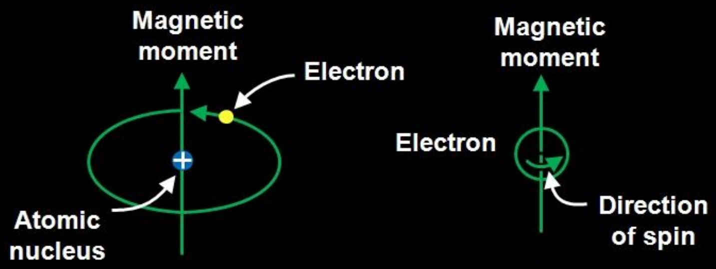
Origin of Magnetic Behavior of Solids

e^- in atom behaves like tiny magnet

orbital motion around nucleus

spin around its own axis

- Moving e^- (charged particle) has permanent spin and an orbital magnetic moment.



12C01.5 Imperfections in Solids

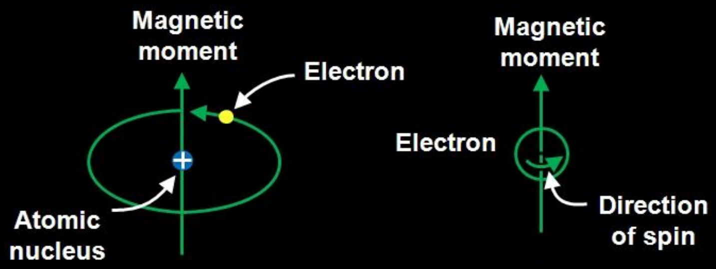
Origin of Magnetic Behavior of Solids

e^- in atom behaves like tiny magnet

orbital motion around nucleus

spin around its own axis

- Moving e^- (charged particle) has permanent spin and an orbital magnetic moment.
- Magnitude of this magnetic moment is very small and it's unit is Bohr magneton(μB).
- $1\ \mu B = 9.27 \times 10^{-24}\ Am^2$



12C01.5 Imperfections in Solids

Types of substances on the basis of their magnetic properties

12C01.5 Imperfections in Solids

Types of substances on the basis of their magnetic properties

diamagnetic → No Unpaired Electrons

12C01.5 Imperfections in Solids

Types of substances on the basis of their magnetic properties

- diamagnetic → No Unpaired Electrons
- paramagnetic
- ferromagnetic
- antiferromagnetic
- ferrimagnetic

paramagnetic, ferromagnetic, antiferromagnetic, ferrimagnetic: Presence of Unpaired Electrons

Domain

Magnetic moment of all the atoms or molecules present in a particular section in the given substance

12C01.5 Imperfections in Solids

Magnetic Properties

Type	Domain Alignment	Magnetic behaviour	Examples

Magnetic Properties

Type	Domain Alignment	Magnetic behaviour	Examples
paramagnetic			

12C01.5 Imperfections in Solids

Magnetic Properties

Type	Domain Alignment	Magnetic behaviour	Examples
paramagnetic	$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$		

Magnetic Properties

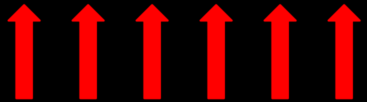
Type	Domain Alignment	Magnetic behaviour	Examples
paramagnetic	↑ ↑ ↑ ↑ ↑ ↑	Weakly attracted	

12C01.5 Imperfections in Solids

Magnetic Properties

Type	Domain Alignment	Magnetic behaviour	Examples
paramagnetic	$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$	Weakly attracted	O_2, Cu^{2+}, Fe^{3+}, Cr^{3+}

Magnetic Properties

Type	Domain Alignment	Magnetic behaviour	Examples
paramagnetic		Weakly attracted	O_2, Cu^{2+}, Fe^{3+}, Cr^{3+}
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12C01.5 Imperfections in Solids

Magnetic Properties

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12C01.5 Imperfections in Solids

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12C01.5 Imperfections in Solids

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12C01.5 Imperfections in Solids

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12C01.5 Imperfections in Solids

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antiferromagnetic	↑ ↓ ↑ ↓ ↑ ↓	No attraction	MnO
ferrimagnetic	↑ ↓ ↑ ↑ ↑ ↓	Weakly attracted	Fe_3O_4, $MgFe_2O_4$

12C01.5 Imperfections in Solids

: Reference Questions :

Intext Question : 1.19, 1.20, 1.21, 1.22,1.23,1.24 (NCERT page no. 31)

Exercise Question : 1.17, 1.20, 1.22, 1.23, 1.26 (NCERT page no. 34)

Work Book Question : 1, 2, 3, 6, 9, 11, 12,13 and 19